

1	1	/	DECMATE II ROM CONTENTS	1
2	2			2
3	3	/	DECODED AND DISASSEMBLED BY CHARLES J. LASNER.	3
4	4			4
5	5	/	LAST EDIT: 02-DEC-1991 02:00:00 CJL	5
6	6			6
7	7	/	MAY BE ASSEMBLED WITH '/J' (PAL8 '/F') SWITCH SET.	7
8	8			8
9	9	/	THIS IS THE CODE USED IN THE DECMATE II PRIMARY CONTROL ROM. IT RESIDES IN	9
10	10	/	THREE 2716 PACKAGES KNOWN AS E113, E114, E115. THE THREE ROMS ARE ENCODED	10
11	11	/	INTO A 12-BIT IMAGE OF AN ENTIRE FIELD IN THE PARTICULAR ORGANIZATION	11
12	12	/	ILLUSTRATED BELOW. FOR ALL ROM CONTENTS: 0.X=BIT X IN THE SPACE 0000-3777,	12
13	13	/	4.X=BIT X IN THE SPACE 4000-7777.	13
14	14			14
15	15	/	E113: 0 1 2 3 4 5 6 7	15
16	16			16
17	17	/	BIT: 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7	17
18	18			18
19	19	/	E114: 0 1 2 3 4 5 6 7	19
20	20			20
21	21	/	BIT: 4.0 4.1 4.2 4.3 4.4 4.5 4.6 4.7	21
22	22			22
23	23	/	E115: 0 1 2 3 4 5 6 7	23
24	24			24
25	25	/	BIT: 0.8 4.8 0.9 4.9 0.10 4.10 0.11 4.11	25
26	26			26
27	27	/	THIS IS THE ASSEMBLY OF THE RELEASED ROM SET KNOWN AS 358E2, 359E2, 360E2.	27
28	28			28
29	29	/	WHEN THE DECMATE II IS FIRST POWERED UP, ROM CONTROL IS ENABLED FOR ALL DIRECT	29
30	30	/	(IF) AND INDIRECT (DF) REFERENCES TO CP MEMORY, THUS THE CPU STARTS AT	30
31	31	/	ROM-BASED ADDRESS CP07777. THE ROM CODE MOVES A LOADING PROGRAM TO PAGE ZERO	31
32	32	/	OF THE CORRESPONDING RAM IN FIELD ZERO OF CP MEMORY, WHICH IS THEN STARTED.	32
33	33	/	TO ACCOMPLISH THE INITIAL MOVE OF THE LOADER, THE DF CONTROL IS CHANGED TO RAM	33
34	34	/	WHILE RETAINING IF CONTROL IN THE ROM.	34
35	35			35
36	36	/	THE LOADER MOVES THE BULK OF THE ROM CODE TO CP FIELD SEVEN AND THEN STARTS IT	36
37	37	/	THERE. NO CODE IS MOVED TO ANY ADDRESS LOWER THAN 0100, SINCE THE CODE IMAGE	37
38	38	/	IS STORED OFFSET BY 100. THIS MEANS THAT ROM LOCATION 0000 WINDS UP IN FIELD	38
39	39	/	7 LOCATION 0100 AND SO ON. DURING THE LOAD OPERATION, THE IF CONTROL IS	39
40	40	/	ALWAYS SET TO RAM BECAUSE AT THIS POINT THE LOADER IS RUNNING. THE DF IS	40
41	41	/	SWITCHED BETWEEN ROM AND RAM AS NECESSARY TO ACCOMPLISH THE LOAD OPERATION.	41
42	42	/	AFTER LOADING IS COMPLETE, THE ROM IS TOTALLY DISABLED.	42
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1	44	/	DEFINITIONS.	1
2	45			2
3	46	/	OPERATE INSTRUCTIONS.	3
4	47			4
5	48	7014	R3L= 7014 /ROTATE AC LEFT THREE WITHOUT AFFECTING LINK	5
6	49			6
7	50	/	COMBINED OPERATE INSTRUCTIONS.	7
8	51			8
9	52	7201	NL0001= CLA IAC /LOAD AC WITH 0001	9
10	53	7326	NL0002= CLA CLL CML RTL /LOAD AC WITH 0002	10
11	54	7305	NL002A= CLA CLL IAC RAL /LOAD AC WITH 0002 ON 8/I OR NEWER	11
12	55	7325	NL0003= CLA CLL CML IAC RAL /LOAD AC WITH 0003	12
13	56	7307	NL0004= CLA CLL IAC RTL /LOAD AC WITH 0004	13
14	57	7327	NL0006= CLA CLL CML IAC RTL /LOAD AC WITH 0006	14
15	58	7215	NL0010= CLA IAC R3L /LOAD AC WITH 0010	15
16	59	7203	NL0100= CLA IAC BSW /LOAD AC WITH 0100	16
17	60	7332	NL2000= CLA CLL CML RTR /LOAD AC WITH 2000	17
18	61	7350	NL3777= CLA CLL CMA RAR /LOAD AC WITH 3777	18
19	62	7330	NL4000= CLA CLL CML RAR /LOAD AC WITH 4000	19
20	63	7352	NL5777= CLA CLL CMA RTR /LOAD AC WITH 5777	20
21	64	7333	NL6000= CLA CLL CML IAC RTR /LOAD AC WITH 6000	21
22	65	7346	NL7775= CLA CLL CMA RTL /LOAD AC WITH 7775	22
23	66	7344	NL7776= CLA CLL CMA RAL /LOAD AC WITH 7776	23
24	67	7240	NL7777= CLA CMA /LOAD AC WITH 7777	24
25	68			25
26	69	/	PROCESSOR I/O INSTRUCTIONS.	26
27	70			27
28	71	6266	CPD= 6266 /FORCE DF TO REFER TO MAIN MEMORY	28
29	72	6256	GCF= 6256 /GET CURRENT FIELDS	29
30	73	6004	PEX= 6004 /EXIT FROM CP TO MAIN MEMORY	30
31	74	6003	PG0= 6003 /RESET HLTFLG FLIP-FLOP	31
32	75	6236	PRQ3= 6236 /TYPE 3 PANEL REQUEST	32
33	76	6000	PRS= 6000 /READ PANEL STATUS	33
34	77	6207	RSP1= 6207 /READ FIRST STACK POINTER	34
35	78	6227	RSP2= 6227 /READ SECOND STACK POINTER	35
36	79	6276	SPD= 6276 /FORCE DF TO REFER TO CP MEMORY	36
37	80	6246	WSR= 6246 /WRITE SWITCH REGISTER (ROM CONTROL), CLEAR AC	37
38	81			38
39	82	/	INTERNAL I/O INSTRUCTIONS.	39
40	83			40
41	84	0003	KEY= 03 /KEYBOARD DEVICE CODE	41
42	85	0004	TTY= 04 /SCREEN DEVICE CODE	42
43	86			43
44	87	/	VIDEO INTERRUPT INSTRUCTIONS.	44
45	88			45
46	89	0006	VIDINT= 06 /VIDEO INTERRUPT DEVICE CODE	46
47	90			47
48	91	6060	VFL= VIDINT^10+6000 /SET VIDEO INTERRUPT FLAG	48
49	92	6061	VSF= VIDINT^10+6001 /SKIP ON, CLEAR VIDEO INTERRUPT FLAG	49
50	93	6062	VCL= VIDINT^10+6002 /NOP (CLEAR THE AC?)	50
51	94	6063	VNOP1= VIDINT^10+6003 /(NOP?)	51
52	95	6064	VDUMM1= VIDINT^10+6004 /(NOP?)	52
53	96	6065	VIE= VIDINT^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	53
54	97	6066	VDUMM2= VIDINT^10+6006 /(NOP?)	54
55	98	6067	VDUMM3= VIDINT^10+6007 /(NOP?)	55
56				56
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1	100	/	VIDEO CONTROLLER INSTRUCTION.	1
2	101			2
3	102	0012	VIDDEV= 12 /VIDEO CONTROLLER DEVICE CODE	3
4	103			4
5	104	6120	APTFL= VIDDEV^10+6000 /SET APT INTERRUPT FLAG	5
6	105	6121	APTSKP= VIDDEV^10+6001 /SKIP ON, CLEAR APT INTERRUPT FLAG	6
7	106	6122	LSCREG= VIDDEV^10+6002 /LOAD VIDEO REGISTER SELECT	7
8	107	6123	CGLOAD= VIDDEV^10+6003 /LOAD CHARACTER GENERATOR ADDRESS/DATA	8
9	108	6124	VLOAD= VIDDEV^10+6004 /LOAD SELECTED REGISTER FROM AC[4-11]	9
10	109	6125	APTIE= VIDDEV^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	10
11	110	6126	VIDCON= VIDDEV^10+6006 /AC TO VIDEO CONTROL	11
12	111	6127	VREAD= VIDDEV^10+6007 /READ SELECTED REGISTER	12
13	112			13
14	113	/	PRINTER INTERFACE INSTRUCTIONS.	14
15	114			15
16	115	0032	PRICON= 32 /PRINTER INPUT DEVICE CODE	16
17	116			17
18	117	6320	PRIFL= PRICON^10+6000 /SET PRINTER INPUT FLAG	18
19	118	6321	PRISKP= PRICON^10+6001 /SKIP ON, CLEAR PRINTER INPUT FLAG	19
20	119	6322	PRICLR= PRICON^10+6002 /CLEAR THE AC	20
21	120	6323	PRINO1= PRICON^10+6003 /NOP	21
22	121	6324	PRIRS= PRICON^10+6004 /OR INPUT BUFFER WITH AC	22
23	122	6325	PRIIE= PRICON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	23
24	123	6326	PRIRB= PRICON^10+6006 /LOAD INPUT BUFFER INTO AC	24
25	124	6327	PRINO2= PRICON^10+6007 /NOP	25
26	125			26
27	126	0033	PROCON= 33 /PRINTER OUTPUT DEVICE CODE	27
28	127			28
29	128	6330	PROFL= PROCON^10+6000 /SET PRINTER OUTPUT FLAG	29
30	129	6331	PROSKP= PROCON^10+6001 /SKIP ON, CLEAR PRINTER OUTPUT FLAG	30
31	130	6332	PRON01= PROCON^10+6002 /NOP	31
32	131	6333	PRSB= PROCON^10+6003 /LOAD BAUD RATE PER AC[8-11]	32
33	132	6334	PROPC= PROCON^10+6004 /OUTPUT CHARACTER FROM AC[4-11]	33
34	133	6335	PROIE= PROCON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	34
35	134	6336	PROLS= PROCON^10+6006 /OUTPUT CHARACTER FROM AC[4-11], CLEAR AC	35
36	135	6337	PRON02= PROCON^10+6007 /NOP	36
37	136			37
38	137	/	RD51D I/O INSTRUCTIONS.	38
39	138			39
40	139	6700	RDNOP= 6700 /RESERVED INSTRUCTION	40
41	140	6701	RDSR= 6701 /SKIP ON, CLEAR DATA REQUEST FLAG	41
42	141	6702	RDSC= 6702 /SEND COMMAND PER AC	42
43	142	6703	RDSD= 6703 /SKIP ON, CLEAR DONE FLAG	43
44	143	6704	RDTD= 6704 /TRANSFER DATA TO/FROM AC	44
45	144	6705	RDWE= 6705 /WRITE INTERRUPT ENABLE PER AC[11]	45
46	145	6706	RDSE= 6706 /SKIP ON, CLEAR ERROR FLAG	46
47	146	6707	RDTEST= 6707 /RESERVED TEST INSTRUCTION THAT CLEARS THE AC	47
48	147		/ONLY IF THE RD51D IS INSTALLED.	48
49				49
50				50
51				51
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1	149	/	COMMUNICATIONS PORT DEFINITIONS.		1
2	150				2
3	151	0030	PORTIN= 30	/COMMUNICATIONS PORT INPUT DEVICE CODE	3
4	152				4
5	153	6300	IFL= PORTIN^10+6000	/SET COMMUNICATIONS PORT INPUT/OUTPUT FLAG	5
6	154	6301	ISF= PORTIN^10+6001	/SKIP ON, CLEAR PORT INPUT/OUTPUT FLAG	6
7	155	6302	ICF= PORTIN^10+6002	/NOP (CLEAR THE AC?)	7
8	156	6303	INOP1= PORTIN^10+6003	/(NOP?)	8
9	157	6304	IRS= PORTIN^10+6004	/READ COMMUNICATIONS PORT RECEIVE BUFFER	9
10	158	6305	IIE= PORTIN^10+6005	/PORT I/O INTERRUPT ENABLE PER AC[11]	10
11	159	6306	IRB= PORTIN^10+6006	/READ COMMUNICATIONS PORT RECEIVE BUFFER	11
12	160	6307	INOP2= PORTIN^10+6007	/(NOP?)	12
13	161				13
14	162	0031	PORTOUT=31	/COMMUNICATIONS PORT OUTPUT DEVICE CODE	14
15	163				15
16	164	6310	DUMBFL= PORTOUT^10+6000	/SET COMMUNICATIONS PORT DUMMY FLAG	16
17	165	6311	DUMBSF= PORTOUT^10+6001	/SKIP ON, CLEAR COMMUNICATIONS PORT DUMMY FLAG	17
18	166	6312	DUMBCF= PORTOUT^10+6002	/NOP (CLEAR THE AC?)	18
19	167	6313	ONOP1= PORTOUT^10+6003	/(NOP?)	19
20	168	6314	OPC= PORTOUT^10+6004	/LOAD COMMUNICATIONS PORT TRANSMIT BUFFER	20
21	169	6315	DUMBIE= PORTOUT^10+6005	/PORT DUMMY INTERRUPT ENABLE PER AC[11]	21
22	170	6316	OLS= PORTOUT^10+6006	/LOAD COMMUNICATIONS PORT TRANSMIT BUFFER	22
23	171	6317	ONOP2= PORTOUT^10+6007	/(NOP?)	23
24	172				24
25	173	0036	PORTCON=36	/COMMUNICATIONS PORT CONTROL DEVICE CODE	25
26	174				26
27	175	6360	MFL= PORTCON^10+6000	/SET MODEM CHANGE FLAG	27
28	176	6361	MSF= PORTCON^10+6001	/SKIP ON, CLEAR MODEM CHANGE FLAG	28
29	177	6362	MLC= PORTCON^10+6002	/LOAD MODEM CONTROL REGISTER	29
30	178	6363	MSB= PORTCON^10+6003	/LOAD BAUD RATE REGISTER	30
31	179	6364	MRS= PORTCON^10+6004	/READ MODEM STATUS REGISTER	31
32	180	6365	MIE= PORTCON^10+6005	/MODEM CHANGE INTERRUPT ENABLE PER AC[11]	32
33	181	6366	MPSCC= PORTCON^10+6006	/ACCESS MULTIPROTOCOL SERIAL CONTROLLER	33
34	182	6367	MPRESET=PORTCON^10+6007	/RESET MULTIPROTOCOL SERIAL CONTROLLER	34
35					35
36					36
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38					38
39					39
40					40
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1	184	/	TTY: TRAP DEFINITIONS.	1
2	185			2
3	186	0007	TTYCON= 07 /DEVICE CODE FOR TTY: TRAP	3
4	187			4
5	188	6070	TTYFL= TTYCON^10+6000 /SET TTY: TRAP FLAG	5
6	189	6071	TTYSF= TTYCON^10+6001 /SKIP ON, CLEAR TTY: TRAP FLAG	6
7	190	6072	TTYCLR= TTYCON^10+6002 /CLEAR THE AC	7
8	191	6073	TTNOP1= TTYCON^10+6003 /NOP	8
9	192	6074	TTNOP2= TTYCON^10+6004 /NOP	9
10	193	6075	TTYIE= TTYCON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	10
11	194	6076	TTNOP3= TTYCON^10+6006 /NOP	11
12	195	6077	TTNOP4= TTYCON^10+6007 /NOP	12
13	196			13
14	197	/	CLOCK DEFINITIONS.	14
15	198			15
16	199	0013	CLKCON= 13 /REAL-TIME CLOCK DEVICE CODE	16
17	200			17
18	201	6130	CLFL= CLKCON^10+6000 /SET CLOCK FLAG	18
19	202	6131	CLSK= CLKCON^10+6001 /SKIP ON, CLEAR CLOCK FLAG	19
20	203	6132	CLNOP1= CLKCON^10+6002 /NOP	20
21	204	6133	CLNOP2= CLKCON^10+6003 /NOP	21
22	205	6134	CLNOP3= CLKCON^10+6004 /NOP	22
23	206	6135	CLIE= CLKCON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	23
24	207	6136	CLNOP4= CLKCON^10+6006 /NOP	24
25	208	6137	CLNOP5= CLKCON^10+6007 /NOP	25
26	209			26
27	210	/	KEYBOARD DEFINITIONS.	27
28	211			28
29	212	0011	KBICON= 11 /KEYBOARD INPUT DEVICE CODE	29
30	213			30
31	214	6110	KBIFL= KBICON^10+6000 /SET KEYBOARD INPUT FLAG	31
32	215	6111	KBISF= KBICON^10+6001 /SKIP ON, CLEAR KEYBOARD INPUT FLAG	32
33	216	6112	KBICLR= KBICON^10+6002 /CLEAR THE AC	33
34	217	6113	KINOP1= KBICON^10+6003 /NOP	34
35	218	6114	KBIRS= KBICON^10+6004 /OR AC WITH KEYBOARD INPUT DATA	35
36	219	6115	KBIIE= KBICON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	36
37	220	6116	KBIRB= KBICON^10+6006 /READ KEYBOARD INPUT DATA INTO AC	37
38	221	6117	KINOP2= KBICON^10+6007 /NOP	38
39	222			39
40	223	0005	KBOCON= 05 /KEYBOARD OUTPUT DEVICE CODE	40
41	224			41
42	225	6050	KBOFL= KBOCON^10+6000 /SET KEYBOARD OUTPUT FLAG	42
43	226	6051	KBOSF= KBOCON^10+6001 /SKIP ON, CLEAR KEYBOARD OUTPUT FLAG	43
44	227	6052	KONOP1= KBOCON^10+6002 /NOP	44
45	228	6053	KONOP2= KBOCON^10+6003 /NOP	45
46	229	6054	KBOPC= KBOCON^10+6004 /OUTPUT AC TO KEYBOARD	46
47	230	6055	KBOIE= KBOCON^10+6005 /WRITE INTERRUPT ENABLE PER AC[11]	47
48	231	6056	KBOLS= KBOCON^10+6006 /OUTPUT AC TO KEYBOARD, CLEAR AC	48
49	232	6057	KONOP3= KBOCON^10+6007 /NOP	49
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1	234		/	RX50/RX01/RX02 DEFINITIONS.	1
2	235				2
3	236	0075		RXCON= 75 /RX50 DEVICE CODE	3
4	237				4
5	238	6750	SEL=	RXCON^10+6000 /SELECT DISKETTE PAIR PER AC[0] AND AC[11]	5
6	239	6751	LCD=	RXCON^10+6001 /LOAD COMMAND REGISTER, CLEAR AC	6
7	240	6752	XDR=	RXCON^10+6002 /TRANSFER DATA	7
8	241	6753	STR=	RXCON^10+6003 /SKIP ON TRANSFER FLAG, CLEAR TRANSFER FLAG	8
9	242	6754	SER=	RXCON^10+6004 /SKIP ON ERROR FLAG, CLEAR ERROR FLAG	9
10	243	6755	SDN=	RXCON^10+6005 /SKIP ON DONE FLAG, CLEAR DONE FLAG	10
11	244	6756	INTR=	RXCON^10+6006 /WRITE INTERRUPT ENABLE PER AC[11]	11
12	245	6757	RXINIT=	RXCON^10+6007 /INITIALIZE CONTROLLER AND DRIVES	12
13	246				13
14	247		/	MRI DEFINITIONS.	14
15	248				15
16	249	5600	JMPIC= JMP I .	/CURRENT PAGE JMP I	16
17	250				17
18	251		/	OTHER DEFINITIONS.	18
19	252				19
20	253	0000	A= 0	/EVEN HALF OF COMMUNICATIONS CHIP	20
21	254	0014	APUCON= 14	/APU/XPU DEVICE CODE	21
22	255	0001	B= 1	/ODD HALF OF COMMUNICATIONS CHIP	22
23	256	0400	DTIME= 400	/DISK WAIT TIME-OUT FACTOR	23
24	257	3660	ROWADR= 3660	/ROWTABLE ADDRESS IN FIELD 1	24
25	258	5350	T0S3BUF=5350	/TRACK 0, SECTOR 3 BUFFER IN FIELD 1	25
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1	260	0007	FIELD 7	/MOST OF THE CODE EXECUTES IN FIELD 7	1
2	261				2
3	262	*7600	PAGE 37	/STARTUP PAGE	3
4	263				4
5	264	/	THE ROM STARTS UP HERE (JUMPED TO FROM THE END OF THE PAGE). SINCE WE ARE		5
6	265	/	STARTING FROM POWERUP, BOTH CONTROL BITS ARE CLEARED. THE ROM IS REFERENCED		6
7	266	/	WHEN MEMORY IS READ VIA BOTH THE INSTRUCTION AND DATA FIELDS INITIALLY.		7
8	267	/	FURTHER, FORCE-ZERO MODE IS IN EFFECT WHICH MUST BE CLEARED TO USE EXTENDED		8
9	268	/	MEMORY.		9
10	269				10
11	270	7600	ROMGO= .	/ROM STARTS HERE	11
12	271				12
13	272	077600 6202	ROMGO, CIF 00	/SET OUR OWN FIELD; START CLEARING FZ MODE	13
14	273	077601 5202	JMP .+1	/THIS JUST LOADS OUR FIELD, BUT CLEARS FZ MODE	14
15	274	077602 7330	NL4000	/SETUP PATTERN FOR RAM DATA FIELD	15
16	275	077603 6246	WSR	/ROM IS ONLY IF NOW, DF IS RAM	16
17	276				17
18	277	/	APPARENTLY THE RAM MAY NOT YET BE WORKING, SO VARIOUS TESTS ARE DONE UNTIL IT		18
19	278	/	RESPONDS CORRECTLY.		19
20	279				20
21	280	077604 7320	MEMTST, CLA STL	/CLEAN UP AND SET A STOP BIT	21
22	281	077605 2745	WATLUP, ISZ I ATABLE+0/(L4001)	/BUMP UP TEST LOCATION	22
23	282	077606 7000	NOP	/IN CASE IT SKIPS	23
24	283	077607 7004	RAL	/MOVE PATTERN OVER	24
25	284	077610 7500	SMA	/DONE ENOUGH?	25
26	285	077611 5205	JMP WATLUP	/NO, KEEP GOING	26
27	286	077612 7200	CLA	/CLEAN UP	27
28	287	077613 6276	SPD	/FORCE INDIRECT REFERENCES TO CP MEMORY	28
29	288				29
30	289	/	TRY TO STORE ALL ZEROES INTO THE RAM TEST LOCATION.		30
31	290				31
32	291	077614 3745	DCA I ATABLE+0/(L4001)	/ATTEMPT TO STORE CLEAR PATTERN	32
33	292	077615 1745	TAD I ATABLE+0/(L4001)	/GET IT BACK	33
34	293	077616 7640	SZA CLA	/SKIP IF ACTUALLY CLEAR	34
35	294	077617 5204	JMP MEMTST	/GO BACK IF IT FLUNKS	35
36	295				36
37	296	/	TRY TO STORE ALTERNATING ZEROES AND ONES INTO THE RAM TEST LOCATION.		37
38	297				38
39	298	077620 1317	TAD L2525/(2525)	/GET A DATA PATTERN	39
40	299	077621 3745	DCA I ATABLE+0/(L4001)	/ATTEMPT TO STORE IT	40
41	300	077622 1745	TAD I ATABLE+0/(L4001)	/GET IT BACK	41
42	301	077623 1320	TAD L5252/(5252)	/ADD ON NEGATED PATTERN	42
43	302	077624 7001	IAC	/ADD ONE TO FINISH THE COMPARISON	43
44	303	077625 7640	SZA CLA	/SKIP IF IT MATCHES	44
45	304	077626 5204	JMP MEMTST	/GO DO IT AGAIN IF IT FLUNKS	45
46	305				46
47	306	/	TRY TO STORE ALTERNATING ONES AND ZEROES INTO THE RAM TEST LOCATION.		47
48	307				48
49	308	077627 1320	TAD L5252/(5252)	/GET A DATA PATTERN	49
50	309	077630 3745	DCA I ATABLE+0/(L4001)	/ATTEMPT TO STORE IT	50
51	310	077631 1745	TAD I ATABLE+0/(L4001)	/GET IT BACK	51
52	311	077632 1317	TAD L2525/(2525)	/ADD ON NEGATED PATTERN	52
53	312	077633 7001	IAC	/ADD ONE TO FINISH THE COMPARISON	53
54	313	077634 7640	SZA CLA	/SKIP IF IT MATCHES	54
55	314	077635 5204	JMP MEMTST	/GO DO IT AGAIN IF IT FLUNKS	55
56					56
57					57
58					58

1	316	/	TRY TO STORE ALL ONES INTO THE RAM TEST LOCATION.	1
2	317			2
3	318	077636 7240	NL7777 /SET DATA PATTERN	3
4	319	077637 3745	DCA I ATABLE+0/(L4001) /ATTEMPT TO STORE IT	4
5	320	077640 7301	NL0001 CLL /SET NEGATED PATTERN FOR COMPARISON	5
6	321	077641 1745	TAD I ATABLE+0/(L4001) /ADD ON TEST VALUE	6
7	322	077642 7640	SZA CLA /SKIP IF IT MATCHES	7
8	323	077643 5204	JMP MEMTST /GO DO IT AGAIN IF IT FLUNKS	8
9	324			9
10	325	/	MOVE THE IMAGE OF THE LOADING PROGRAM TO PAGE ZERO OF RAM.	10
11	326			11
12	327	077644 1321	TAD LPROGM+0/(4001) /GET A WORD	12
13	328	077645 3745	DCA I ATABLE+0/(L4001) /PUT A WORD	13
14	329	077646 1322	TAD LPROGM+1/(CODEXC-1) /GET A WORD	14
15	330	077647 3746	DCA I ATABLE+1/(XR6) /PUT A WORD	15
16	331	077650 1323	TAD LPROGM+2/(-ROMGO+1) /GET A WORD	16
17	332	077651 3747	DCA I ATABLE+2/(MOVCNT) /PUT A WORD	17
18	333	077652 1324	TAD LPROGM+3/(NL7777 CLL) /GET A WORD	18
19	334	077653 3750	DCA I ATABLE+3/(LOADIT+0) /PUT A WORD	19
20	335	077654 1325	TAD LPROGM+4/(DCA XR1) /GET A WORD	20
21	336	077655 3751	DCA I ATABLE+4/(LOADIT+1) /PUT A WORD	21
22	337	077656 1326	TAD LPROGM+5/(CDF 00) /GET A WORD	22
23	338	077657 3752	DCA I ATABLE+5/(LOADLP+0) /PUT A WORD	23
24	339	077660 1327	TAD LPROGM+6/(NL0001) /GET A WORD	24
25	340	077661 3753	DCA I ATABLE+6/(LOADLP+1) /PUT A WORD	25
26	341	077662 1330	TAD LPROGM+7/(WSR) /GET A WORD	26
27	342	077663 3754	DCA I ATABLE+7/(LOADLP+2) /PUT A WORD	27
28	343	077664 1331	TAD LPROGM+10/(TAD I XR1) /GET A WORD	28
29	344	077665 3755	DCA I ATABLE+10/(LOADLP+3) /PUT A WORD	29
30	345	077666 1332	TAD LPROGM+11/(MQL) /GET A WORD	30
31	346	077667 3756	DCA I ATABLE+11/(LOADLP+4) /PUT A WORD	31
32	347	077670 1333	TAD LPROGM+12/(TAD L4001) /GET A WORD	32
33	348	077671 3757	DCA I ATABLE+12/(LOADLP+5) /PUT A WORD	33
34	349	077672 1334	TAD LPROGM+13/(WSR) /GET A WORD	34
35	350	077673 3760	DCA I ATABLE+13/(LOADLP+6) /PUT A WORD	35
36	351	077674 1335	TAD LPROGM+14/(CLA MQA) /GET A WORD	36
37	352	077675 3761	DCA I ATABLE+14/(LOADLP+7) /PUT A WORD	37
38	353	077676 1336	TAD LPROGM+15/(CDF 70) /GET A WORD	38
39	354	077677 3762	DCA I ATABLE+15/(LOADLP+10) /PUT A WORD	39
40	355	077700 1337	TAD LPROGM+16/(DCA I XR6) /GET A WORD	40
41	356	077701 3763	DCA I ATABLE+16/(LOADLP+11) /PUT A WORD	41
42	357	077702 1340	TAD LPROGM+17/(ISZ MOVCNT) /GET A WORD	42
43	358	077703 3764	DCA I ATABLE+17/(LOADLP+12) /PUT A WORD	43
44	359	077704 1341	TAD LPROGM+20/(JMP LOADLP) /GET A WORD	44
45	360	077705 3765	DCA I ATABLE+20/(LOADLP+13) /PUT A WORD	45
46	361	077706 1342	TAD LPROGM+21/(CIF 70) /GET A WORD	46
47	362	077707 3766	DCA I ATABLE+21/(LOADLP+14) /PUT A WORD	47
48	363	077710 1343	TAD LPROGM+22/(JMP I .+1) /GET A WORD	48
49	364	077711 3767	DCA I ATABLE+22/(LOADLP+15) /PUT A WORD	49
50	365	077712 1344	TAD LPROGM+23/(CODESTART) /GET A WORD	50
51	366	077713 3770	DCA I ATABLE+23/(LOADLP+16) /PUT A WORD	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	368		/			NOW STARTUP THE LOADER TO MOVE THE CODE TO RAM.	1
2	369						2
3	370	077714	7201		NL0001	/SETUP FOR	3
4	371	077715	6246		WSR	/RAM IF AND ROM DF	4
5	372	077716	5020		JMP LOADIT	/GO START IT UP	5
6	373						6
7	374	077717	2525	L2525,	2525	/DATA	7
8	375	077720	5252	L5252,	5252	/PATTERNS	8
9	376						9
10	377		7721		LPROGM= .	/IMAGE OF LOADING PROGRAM HERE	10
11	378						11
12	379		*0011		RELOC 11	/FOOL THE ASSEMBLER	12
13	380						13
14	381			XR1,		/AUTO-INDEX USED FOR MOVING THE ROM	14
15	382						15
16	383		*0015		RELOC 15	/FOOL THE ASSEMBLER SOME MORE	16
17	384						17
18	385		0015		LOADER= .	/LOADER EXECUTES FROM HERE	18
19	386						19
20	387	070015*	4001	L4001,	4001	/ROM DISABLE CONSTANT	20
21	388	070016*	0077	XR6,	CODEXC-1	/RAM STORE POINTER	21
22	389	070017*	0201	MOVCNT,	-ROMGO+1	/MOVE COUNTER	22
23	390						23
24	391			/		THE LOADER STARTS HERE. ALL IF REFERENCES ARE NOW IN RAM. THE CODE IN THE	24
25	392			/		ROM WILL BE MOVED TO FIELD 7.	25
26	393						26
27	394	070020*	7340	LOADIT,	NL7777 CLL	/SET (AUTO-INCREMENTED) POINTER	27
28	395	070021*	3011		DCA XR1	/TO LOWEST ROM LOCATION	28
29	396	070022*	6201	LOADLP,	CDF 00	/ROM CODE IS IN FIELD ZERO	29
30	397	070023*	7201		NL0001	/SETUP FOR	30
31	398	070024*	6246		WSR	/RAM IF AND ROM DF	31
32	399	070025*	1411		TAD I XR1	/GET A ROM WORD	32
33	400	070026*	7421		MQL	/SAVE IT FOR NOW	33
34	401	070027*	1015		TAD L4001/(4001)	/SETUP FOR	34
35	402	070030*	6246		WSR	/RAM IF AND RAM DF	35
36	403	070031*	7701		CLA MQA	/GET THE ROM VALUE	36
37	404	070032*	6271		CDF 70	/CODE LOADS INTO FIELD SEVEN	37
38	405	070033*	3416		DCA I XR6	/STORE INTO RAM	38
39	406	070034*	2017		ISZ MOVCNT	/DONE YET?	39
40	407	070035*	5022		JMP LOADLP	/NO, KEEP GOING	40
41	408	070036*	6272		CIF 70	/GOTO NEW EXECUTION FIELD	41
42	409	070037*	5440		JMP I .+1/(CODESTART)	/GO START IT UP	42
43	410	070040*	0200		CODESTART	/THROUGH HERE	43
44	411						44
45	412		*7745		RELOC	/UN-FOOL THE ASSEMBLER	45
46							46
47							47
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1	414	077745	0015	ATABLE,	LOADER+00	/TABLE OF LOADING ADDRESSES FOR MOVED LOADER	1
2	415	077746	0016		LOADER+01	/	2
3	416	077747	0017		LOADER+02	/	3
4	417	077750	0020		LOADER+03	/	4
5	418	077751	0021		LOADER+04	/	5
6	419	077752	0022		LOADER+05	/	6
7	420	077753	0023		LOADER+06	/	7
8	421	077754	0024		LOADER+07	/	8
9	422	077755	0025		LOADER+10	/	9
10	423	077756	0026		LOADER+11	/	10
11	424	077757	0027		LOADER+12	/	11
12	425	077760	0030		LOADER+13	/	12
13	426	077761	0031		LOADER+14	/	13
14	427	077762	0032		LOADER+15	/	14
15	428	077763	0033		LOADER+16	/	15
16	429	077764	0034		LOADER+17	/	16
17	430	077765	0035		LOADER+20	/	17
18	431	077766	0036		LOADER+21	/	18
19	432	077767	0037		LOADER+22	/	19
20	433	077770	0040		LOADER+23	/LAST ADDRESS OF MOVED LOADER CODE	20
21	434						21
22	435	077771	0000	ZBLOCK	7776-.	/EMPTY SPACE	22
23	436						23
24	437			/	WHEN THE CODE IS IN RAM, THESE LOCATIONS IN CP FIELD 0 ARE USED AS THE START		24
25	438			/	OF THE CP INTERRUPT HANDLER. THE MAIN MEMORY INTERRUPTED PC IS STORED IN CP		25
26	439			/	00000 AND CONTROL IS REGAINED AT CP 07777. INVARIABLY, THE INSTRUCTION USED		26
27	440			/	THERE IS JMP I .-1. THE POINTER ADDRESS IS THEREFORE STORED HERE AT CP 07776.		27
28	441						28
29	442	077776	0000	ROMSP,	..	/USED LATER AS CP INTERRUPT POINTER	29
30	443						30
31	444			/	THE ROM PROGRAM GAINS CONTROL HERE. ALSO USED LATER FOR CP INTERRUPTS.		31
32	445						32
33	446	077777	5200	ROMST,	JMP ROMGO	/CONTINUE THERE	33
34							34
35							35
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57							57
58							58

1	448	/	START OF PAGE ZERO WHILE RUNNING IN FIELD 7. THESE LOCATIONS ARE NOT LOADED	1
2	449	/	BY THE ROM'S LOADING PROGRAM BUT ARE USED BY THE MOVED CODE AS TEMPORARIES.	2
3	450			3
4	451	*0000	*0 /START AT THE BEGINNING	4
5	452			5
6	453	NOPUNCH	/DON'T GENERATE BINARY	6
7	454			7
8	455	RIMADR,	/APT RIM-LOADER TEMPORARY	8
9	456	INTADR,	/CP-INTERRUPTS PC STORED HERE	9
10	457			10
11	458	/	THE FOLLOWING ARE DEFINITIONS FOR MAIN MEMORY INTERRUPTS IN FIELD 0.	11
12	459			12
13	460	070000 0000 INTADR, ...	/MAIN-MEMORY INTERRUPTS PC STORED HERE	13
14	461	070001* 5402 INT1, JMP I INT2	/WHAT'S PUT HERE IN MAIN MEMORY FIELD 0	14
15	462	070002* 0250 INT2, INTHND	/POINTER TO MAIN MEMORY INTERRUPT HANDLER	15
16	463	070003* 0000 TIMEOUT, ...	/TIME-OUT COUNTER FOR LOOPBACK TEST	16
17	464	070004* 0000 CLKTICK, ...	/CLOCK TICK COUNTER FOR LOOPBACK TEST	17
18	465	070005* 0000 CSTATUS, ...	/COUNT PATTERN PROGRESS STATUS	18
19	466	070006* 0000 PATTERN, ...	/COUNT PATTERN TEMPORARY	19
20	467	070007* 0000 TSTATUS, ...	/LOOPBACK CP TEST STATUS	20
21	468			21
22	469	*0010	*10 /GET TO AUTO-INDEX AREA	22
23	470			23
24	471	070010* 0000 XR0, ...	/AUTO-INDEX REGISTER 0	24
25	472	070011* 0000 XR1, ...	/AUTO-INDEX REGISTER 1	25
26	473	070012* 0000 XR2, ...	/AUTO-INDEX REGISTER 2	26
27	474	070013* 0000 XR3, ...	/AUTO-INDEX REGISTER 3	27
28	475	070014* 0000 XR4, ...	/AUTO-INDEX REGISTER 4	28
29	476	070015* 0000 XR5, ...	/AUTO-INDEX REGISTER 5	29
30	477	070016* 0000 XR6, ...	/AUTO-INDEX REGISTER 6	30
31	478	070017* 0000 XR7, ...	/AUTO-INDEX REGISTER 7	31
32	479			32
33	480	*0020	*20 /GET PAST AUTO-INDEX AREA	33
34	481			34
35	482	070020* 0000 CTMP1, ...	/OUTPUT COUNTING TEMPORARY FOR LOOPBACK TEST	35
36	483	070021* 0000 CTMP2, ...	/INPUT COUNTING TEMPORARY FOR LOOPBACK TEST	36
37	484	070022* 0000 CNTLO, ...	/LOW-ORDER COUNTER FOR LOOPBACK TESTING	37
38	485	070023* 0000 CNTHI, ...	/HIGH-ORDER COUNTER FOR LOOPBACK TESTING	38
39	486	070024* 0000 TEMP1, ...	/TEMPORARY	39
40	487	070025* 0000 TEMP2, ...	/TEMPORARY	40
41	488	070026* 0000 TEMP3, ...	/TEMPORARY	41
42	489			42
43	490	070027* 0000 ZBLOCK 1	/EMPTY SPACE	43
44	491			44
45	492	/	THESE SIX ARE PRINTED AS A GROUP.	45
46	493			46
47	494	070030* 0000 TMO, ...	/TEMPORARY	47
48	495	070031* 0000 COMTM1, ...	/COMMUNICATIONS PORT TESTING TEMPORARY	48
49	496	070032* 0000 MTM1, ...	/TEMPORARY	49
50	497	070033* 0000 MTM2, ...	/TEMPORARY	50
51	498	070034* 0000 SP1SAVE, ...	/FIRST STACK POINTER SAVED HERE	51
52	499	070035* 0000 SP2SAVE, ...	/SECOND STACK POINTER SAVED HERE	52
53				53
54				54
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56				56
57				57
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1	524	/	START OF ROM-BASED CODE.	1
2	525			2
3	526	*0000	*0	3
4	527		/START AT THE BEGINNING	4
5	528	0000	ROMADR= .	5
6	529		/WHERE IT IS LOADED WHILE IN ROM	6
7	530	*0100	RELOC 100	7
8	531		/FOOL THE ASSEMBLER	8
9	532	0100	CODEXC= .	9
10	533		/WHERE IT RUNS IN FIELD 7	10
11	534	070100* 3732	3732	11
12	535		/THIS LOCATION IS PROBABLY A VERSION NUMBER	12
13	536	070101* 4204	PCHRPRT, CHRPRT	13
14	537	070102* 1400	PPPTABL, PPTABLE	14
15	538	070103* 1451	PROWLOA, ROWLOAD	15
16	539	070104* 0007	Z7, 7	16
17	540	070105* 0010	Z10, 10	17
18	541	070106* 0012	Z12, 12	18
19	542	070107* 0017	Z17, 17	19
20	543	070110* 0020	Z20, 20	20
21	544	070111* 0102	Z102, 102	21
22	545	070112* 0200	Z200, 200	22
23	546	070113* 0237	Z237, 237	23
24	547	070114* 0301	Z0301, 0301	24
25	548	070115* 0377	Z377, 377	25
26	549	070116* 0400	Z400, 400	26
27	550	070117* 2525	Z2525, 2525	27
28	551	070120* 3777	Z3777, 3777	28
29	552	070121* 5776	JMPIM1, ROMSP&177+JMPIC	29
30	553	070122* 6201	ZCDF, CDF 00	30
31	554	070123* 6400	PWRIBUF, WRIBUFFER	31
32	555	070124* 6500	PAPTR0U, APTROUTINE	32
33	556		/POINTER TO APT ROUTINE	33
34	557	070125* 6777	6777	34
35	558		/CONSTANT 6777 (UNUSED?)	35
36	559	070126* 7766	Z7766, 7766	36
37	560	070127* 7760	Z7760, 7760	37
38	561	070130* 7741	Z7741, 7741	38
39	562	070131* 7400	Z7400, 7400	39
40	563	070132* 7400	DTIMOUT, -DTIME	40
41	564	070133* 7775	Z7775, 7775	41
42	565		/DISK WAIT TIMEOUT FACTOR	42
43	566	070134* 7776	Z7776, 7776	43
44	567	070135* 7777	PROMST, ROMST	44
45	568	070136* 2577	LBLADR, LBLOAD-1	45
46	569	070137* 7631	LBMCNT, LBTEST-LBTEND	46
47	570	070140* 2777	CPLADR, CPLOAD-1	47
48	571	070141* 5177	CPMCNT, CPLOAD-CPEND2	48
49	572	070142* 5137	PCOMLOA, COMLOAD	49
50	573	070143* 5115	PCOMREA, COMREAD	50
51	574	070144* 1513	PCLR23, CLR23	51
52	575	070145* 5200	PRXCOMD, RXCOMD	52
53	576	070146* 2000	PSSTRIN, SSTRING	53
54			/POINTER TO SSTRING ROUTINE	54
55				55
56				56
57				57
58				58

1	578	/	THE FOLLOWING LOCATION IS UNREFERENCED; PERHAPS THERE IS A CONVENTION ABOUT			1
2	579	/	WHERE TO RESTART THE ROM PROGRAM?			2
3	580					3
4	581	070147* 0214	PRGDONE	/POINTER TO PRGDONE ROUTINE (UNUSED?)		4
5	582					5
6	583	070150* 1421	PMEMCOM, MEMCOMP	/POINTER TO MEMORY COMPARE ROUTINE		6
7	584	070151* 3672	PKBDIN, KBDIN	/POINTER TO KEYBOARD INPUT ROUTINE		7
8	585	070152* 5600	PCALLSU, CALLSUB	/POINTER TO TEST (SUBROUTINE) LOCATION		8
9	586	070153* 0205	Z205, 205	/CONSTANT 0205		9
10	587					10
11	588	070154* 0000	ZBLOCK 21	/EMPTY SPACE		11
12	589					12
13	590	/	*****			13
14	591					14
15	592	070175* 3412	SUBR	/POINTER TO SUBR		15
16	593	070176* 0000	SUB176, .-. .	/ENTRY HERE		16
17	594	070177* 5575	JMP I .-2/[SUBR]	/GO THERE		17
18						18
19						19
20						20
21						21
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[illegible]

1	648	/	FOR SOME PURPOSE, WE NEED TO STORE 0100 INTO CP20301.	1
2	649			2
3	650	070240* 7303	NL0100 CLL /GET VALUE	3
4	651	070241* 6221	CDF 20 /GOTO BUFFER FIELD	4
5	652	070242* 3514	DCA I Z0301/[0301] /STORE IN BUFFER	5
6	653	070243* 6271	CDF 70 /BACK TO OUR FIELD	6
7	654			7
8	655	/	IF THE PRINTER PORT HAS THE APTEEN LINE GROUNDED, WE GO IMMEDIATELY INTO THE	8
9	656	/	AUTOMATIC PRODUCT TEST DOWNLOAD MODE. THIS COULD BE A SECONDARY CALL TO THE	9
10	657	/	DOWNLOAD ROUTINE IF A PREVIOUSLY LOADED ROUTINE EXITED VIA A PANEL REQUEST	10
11	658	/	INSTEAD OF HLT.	11
12	659			12
13	660	070244* 6121	APTSKP /AUTOMATIC PRODUCT TEST REQUESTED?	13
14	661	070245* 5251	JMP MEMTEST /NO, JUST KEEP GOING	14
15	662	070246* 4647	JMS I .+1; DOWNLOAD /YES, GO DOWNLOAD SOME STUFF.	15
16	663	070247* 0640		16
17	664			17
18	665	070250* 0072 L72, 72	/CONSTANT 0072	18
19	666			19
20	667	/	MEMORY TESTING. ALTERNATING ONES AND ZEROES ARE STORED INTO ALL FIELDS IN	20
21	668	/	MAIN MEMORY AND FIELDS 2-6 IN CP MEMORY. CHECK IF THE PATTERN WORKED AND THEN	21
22	669	/	REVERSE ALL PARAMETERS EACH PASS FOR SEVERAL ITERATIONS.	22
23	670			23
24	671	070251* 6266 MEMTEST, CPD	/INDIRECT TO MAIN MEMORY FROM NOW ON	24
25	672	070252* 6271	CDF 70 /RESET TO OUR FIELD FIRST (WHY?)	25
26	673	070253* 3044	DCA T2 /CLEAR FIELD COUNTER	26
27	674			27
28	675	/	EXECUTE CDF TO LATEST TEST FIELD.	28
29	676			29
30	677	070254* 1044 MLODLP, TAD T2	/GET LATEST FIELD	30
31	678	070255* 0104	AND Z7/[7] /JUST FIELD BITS	31
32	679	070256* 7014	R3L /MOVE UP	32
33	680	070257* 1122	TAD ZCDF/[CDF] /MAKE IT A CDF INSTRUCTION	33
34	681	070260* 3261	DCA .+1 /STORE IN-LINE	34
35	682	070261* 7402	HLT+.-. /WILL BE CDF TEST FIELD INSTRUCTION	35
36	683	070262* 3043	DCA T1 /CLEAR ADDRESS POINTER	36
37	684			37
38	685	/	STORE TEST PATTERN IN LATEST TEST FIELD.	38
39	686			39
40	687	070263* 1117 FILLIT, TAD Z2525	/GET ALTERNATING TEST PATTERN	40
41	688	070264* 3443	DCA I T1 /STASH INTO TEST MEMORY	41
42	689	070265* 2043	ISZ T1 /BUMP TO NEXT	42
43	690	070266* 5263	JMP FILLIT /GO BACK FOR MORE	43
44	691			44
45	692	/	BUMP TO NEXT FIELD AND CHECK WHICH MEMORY BANK WE ARE TESTING.	45
46	693			46
47	694	070267* 2044	ISZ T2 /BUMP TO NEXT FIELD	47
48	695	070270* 1044	TAD T2 /GET LATEST FIELD	48
49	696	070271* 0105	AND Z10/[10] /JUST GROUP BIT	49
50	697	070272* 7650	SNA CLA /SKIP IF FIRST GROUP DONE	50
51	698	070273* 5254	JMP MLODLP /GO BACK AGAIN	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	700	/	CP MEMORY IS BEING TESTED; CHECK IF WE JUST LOADED FIELD 6.	1
2	701			2
3	702	070274* 1044	TAD T2 /GET LATEST FIELD	3
4	703	070275* 7001	IAC /LOOK AT NEXT TENTATIVE FIELD	4
5	704	070276* 0110	AND Z20/[20] /CHECK IF AT LAST MEMBER (WHICH WOULD BE US!)	5
6	705	070277* 7640	SZA CLA /SKIP IF NOT THERE YET	6
7	706	070300* 5311	JMP MEMLOADED /JUMP IF AT END	7
8	707			8
9	708	/	CHECK IF WE ARE JUST FINISHED LOADING MAIN MEMORY.	9
10	709			10
11	710	070301* 1044	TAD T2 /GET LATEST FIELD	11
12	711	070302* 0104	AND Z7/[7] /JUST FIELD BITS	12
13	712	070303* 7640	SZA CLA /SKIP IF GROUP DONE	13
14	713	070304* 5254	JMP MLODLP /GO BACK AGAIN	14
15	714			15
16	715	/	SWITCH INDIRECTS BACK TO CP MEMORY. AVOID TESTS IN FIELDS 0 AND 1.	16
17	716			17
18	717	070305* 6276	SPD /WILL BE CP INDIRECTS FROM NOW ON	18
19	718	070306* 2044	ISZ T2 /BUMP LATEST FIELD	19
20	719	070307* 2044	ISZ T2 /DO IT TWICE	20
21	720	070310* 5254	JMP MLODLP /KEEP GOING	21
22	721			22
23	722	/	COMES HERE WHEN ALL TESTABLE FIELDS ARE FILLED WITH THE TEST PATTERN.	23
24	723			24
25	724	070311* 3043	MEMLOAD, DCA T1 /CLEAR ADDRESS POINTER	25
26	725	070312* 3044	DCA T2 /CLEAR FIELD COUNTER	26
27	726	070313* 1117	TAD Z2525/[2525] /GET TEST VALUE	27
28	727	070314* 7040	CMA /INVERT FOR CHECKING LATER	28
29	728	070315* 3042	DCA T0 /STASH IT	29
30	729	070316* 6266	CPD /MAIN MEMORY INDIRECTS FROM NOW ON	30
31	730	070317* 7201	NL0001 /SETUP FOR SINGLE	31
32	731	070320* 3041	DCA TINC /FIELD INCREMENT	32
33	732	070321* 7346	NL7775 /SETUP FOR THREE	33
34	733	070322* 3011	DCA XR1 /COMPLETE TEST PASSES	34
35	734	070323* 4550	MTSTLP, JMS I PMEMCOMP/[MEMCOMP] /CHECK LATEST FIELD FOR PROPER TEST VALUE	35
36	735	070324* 5251	JMP MEMTEST /MEMORY FAILED TO MATCH, GO BACK FOR MORE	36
37	736	070325* 1041	TAD TINC /GET INCREMENT	37
38	737	070326* 1044	TAD T2 /UPDATE FIELD	38
39	738	070327* 3044	DCA T2 /STORE BACK	39
40	739	070330* 1041	TAD TINC /GET INCREMENTAL FACTOR	40
41	740	070331* 7710	SPA CLA /SKIP IF STILL ADDING	41
42	741	070332* 5344	JMP CDWNTST /JUMP IF NOT	42
43	742			43
44	743	/	FOUND OUT IF WE ARE FINISHED (JUST TESTED CP FIELD 6) WHILE COUNTING FORWARD.	44
45	744			45
46	745	070333* 1044	TAD T2 /GET CURRENT FIELD	46
47	746	070334* 7001	IAC /TENTATIVELY ADD ONE	47
48	747	070335* 0110	AND Z20/[20] /CHECK IF AT LAST FIELD	48
49	748	070336* 7650	SNA CLA /SKIP IF SO	49
50	749	070337* 5347	JMP CPMTST /JUMP IF NOT	50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
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1	751	/	BACKUP TO FIELD 6 TO START NEXT TEST PASS.	1
2	752			2
3	753	070340* 7240	NL7777 /BACKUP	3
4	754	070341* 1044	TAD T2 /ADD ON LATEST FIELD	4
5	755	070342* 3044	BACKEND, DCA T2 /STORE BACK	5
6	756	070343* 5377	JMP PASSEND /CONTINUE THERE	6
7	757			7
8	758	/	COUNTING FIELDS DOWN; CHECK IF WE JUST FINISHED A PASS.	8
9	759			9
10	760	070344* 1044	CDWNTST, TAD T2 /GET CURRENT FIELD	10
11	761	070345* 7710	SPA CLA /SKIP IF STILL VALID	11
12	762	070346* 5342	JMP BACKEND /JUMP IF NOT	12
13	763			13
14	764	/	CHECK IF WE ARE ABOUT TO START TESTING CP MEMORY.	14
15	765			15
16	766	070347* 1044	CPMTST, TAD T2 /GET CURRENT FIELD	16
17	767	070350* 0105	AND Z10/[10] /CHECK IF INTO CP GROUP	17
18	768	070351* 7650	SNA CLA /SKIP IF SO	18
19	769	070352* 5323	JMP MTSTLP /KEEP GOING	19
20	770	070353* 1041	TAD TINC /GET INCREMENTAL FACTOR	20
21	771	070354* 7710	SPA CLA /SKIP IF STILL POSITIVE	21
22	772	070355* 5366	JMP NEGST /JUMP IF NOT	22
23	773			23
24	774	/	CHECK IF FIRST TIME IN CP MEMORY; SWITCH OVER TO CP INDIRECTS AND BYPASS	24
25	775	/	TESTING OF FIELD 0 AND FIELD 1.	25
26	776			26
27	777	070356* 1044	TAD T2 /GET LATEST FIELD	27
28	778	070357* 0104	AND Z7/[7] /JUST FIELD BITS	28
29	779	070360* 7650	SNA CLA /SKIP IF NOT AT ZERO	29
30	780	070361* 7305	NL002A /ELSE START HERE	30
31	781	070362* 1044	TAD T2 /ADD ON CURRENT FIELD	31
32	782	070363* 3044	DCA T2 /STORE BACK	32
33	783	070364* 6276	SPD /PANEL INDIRECTS FROM NOW ON	33
34	784	070365* 5323	JMP MTSTLP /CONTINUE THERE	34
35	785			35
36	786	/	IF COUNTING FIELDS DOWN, WE MUST AVOID FIELD 0 AND FIELD 1; IF CP FIELD 2 WAS	36
37	787	/	JUST FINISHED, THEN SWITCH OVER TO MAIN FIELD 7.	37
38	788			38
39	789	070366* 7240	NEGST, NL7777 /-1	39
40	790	070367* 1044	TAD T2 /COMPARE TO LATEST FIELD	40
41	791	070370* 0104	AND Z7/[7] /JUST FIELD BITS	41
42	792	070371* 7640	SZA CLA /SKIP IF WE ARE AT FIELD 1	42
43	793	070372* 5323	JMP MTSTLP /JUST KEEP GOING	43
44	794	070373* 6266	CPD /MAIN MEMORY INDIRECTS FROM NOW ON	44
45	795	070374* 1104	TAD Z7/[7] /SETUP FOR FIELD 7	45
46	796	070375* 3044	DCA T2 /STASH IN FIELD COUNTER	46
47	797	070376* 5323	JMP MTSTLP /CONTINUE THERE	47
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1	799	/	COMES HERE AT THE END OF A TEST PASS OVER ALL TESTED FIELDS.			1
2	800					2
3	801	070377* 1041	PASSEND, TAD	TINC	/GET INCREMENT FACTOR	3
4	802	070400* 7041	CIA		/INVERT IT	4
5	803	070401* 3041	DCA	TINC	/STORE BACK	5
6	804	070402* 1042	TAD	T0	/GET TEST PATTERN COMPARE VALUE	6
7	805	070403* 7040	CMA		/INVERT THE BITS	7
8	806	070404* 3042	DCA	T0	/STORE BACK	8
9	807	070405* 1041	TAD	TINC	/GET INCREMENT AGAIN	9
10	808	070406* 7710	SPA CLA		/SKIP IF POSITIVE	10
11	809	070407* 7240	NL7777		/ELSE SET -1	11
12	810	070410* 3043	DCA	T1	/STORE EITHER WAY	12
13	811	070411* 2011	ISZ	XR1	/DONE ENOUGH PASSES?	13
14	812	070412* 5664	JMP I	PMTSTLP/(MTSTLP)	/NO, KEEP GOING	14
15	813	070413* 4544	JMS I	PCLR23/[CLR23]	/CLEAR FIELDS 2, 3 (AND BITS IN 13705, 13707)	15
16	814	070414* 6271	CDF	70	/BACK TO OUR FIELD	16
17	815	070415* 6121	APTSKP		/AUTOMATIC PRODUCT TESTING?	17
18	816	070416* 7410	SKP		/SKIP IF NOT	18
19	817	070417* 4524	JMS I	PAPTRoutine/[APTRoutine]	/CALL ROUTINE IF SO	19
20	818					20
21	819	/	MOVE CP 72600-72746 TO MAIN 00200-00346. THIS IS THE MAIN MEMORY ROUTINE AND			21
22	820	/	INTERRUPT HANDLER FOR THE PRINTER LOOPBACK TEST.			22
23	821					23
24	822	070420* 1136	TAD	LBLADR/[LBLOAD-1]	/SETUP THE	24
25	823	070421* 3010	DCA	XR0	/SOURCE POINTER	25
26	824	070422* 1137	TAD	LBMcnt/[LBTEST-LBTEND]	/SETUP THE	26
27	825	070423* 3011	DCA	XR1	/MOVE COUNTER	27
28	826	070424* 1263	TAD	LBADR/(LBTEST-1)	/SETUP THE	28
29	827	070425* 3012	DCA	XR2	/DESTINATION POINTER	29
30	828	070426* 1410	LBMVLP, TAD I	XR0	/GET A WORD	30
31	829	070427* 6201	CDF	00	/GOING TO FIELD 0	31
32	830	070430* 6266	CPD		/GOING TO MAIN MEMORY	32
33	831	070431* 3412	DCA I	XR2	/PUT A WORD	33
34	832	070432* 6276	SPD		/INDIRECTS BACK TO CP MEMORY AGAIN	34
35	833	070433* 6271	CDF	70	/BACK TO OUR FIELD	35
36	834	070434* 2011	ISZ	XR1	/DONE YET?	36
37	835	070435* 5226	JMP	LBMVLP	/NO, GO BACK	37
38						38
39						39
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1	837	/	MOVE CP 73000-75600 TO CP 04000-06600. THIS IS THE CP-INTERRUPT HANDLER FOR	1
2	838	/	THE PRINTER LOOPBACK TEST.	2
3	839			3
4	840	/	PROGRAMMING NOTE: THE LENGTH OF EXECUTABLE CODE REQUIRED TO BE RELOCATED IS	4
5	841	/	APPARENTLY MUCH SMALLER THAN THE ACTUAL AMOUNT OF CODE MOVED. THE LENGTH	5
6	842	/	SHOULD BE CPEND-CPINT INSTEAD OF CPEND2-CPLoad. THIS LENGTH MAY INTERFERE	6
7	843	/	WITH AUTOMATIC PRODUCT TESTING OR CAUSE SOME OTHER BUG. IT IS NOT KNOWN AT	7
8	844	/	THIS TIME IF THE ORIGINAL PROGRAMMER'S INTENTION WAS TO OVERLAY ANY OTHER CODE	8
9	845	/	BEYOND THE INTERRUPT HANDLER, BUT THIS SOURCE CODE FOLLOWS THE ROM CONTENTS,	9
10	846	/	BUGS AND ALL :-).	10
11	847			11
12	848	070436* 1140	TAD CPLADR/[CPLOAD-1] /SETUP THE	12
13	849	070437* 3010	DCA XR0 /SOURCE POINTER	13
14	850	070440* 1141	TAD CPMCNT/[CPLOAD-CPEND2] /SETUP THE	14
15	851	070441* 3011	DCA XR1 /MOVE COUNTER	15
16	852	070442* 7350	NL3777/TAD (CPINT-1) /SETUP THE	16
17	853	070443* 3012	DCA XR2 /DESTINATION POINTER	17
18	854	070444* 1410 CPMVLP, TAD I	XR0 /GET A WORD	18
19	855	070445* 6201	CDF 00 /GOING TO FIELD 0	19
20	856	070446* 3412	DCA I XR2 /PUT A WORD	20
21	857	070447* 6271	CDF 70 /BACK TO OUR FIELD	21
22	858	070450* 2011	ISZ XR1 /DONE YET?	22
23	859	070451* 5244	JMP CPMVLP /NO, KEEP GOING	23
24	860			24
25	861	/	NOW WE HAVE TO MOVE JMP I .-1 TO CP 07777 AND 4000 TO CP 07776 TO COMPLETE THE	25
26	862	/	INTERRUPT HANDLER.	26
27	863			27
28	864	070452* 7330	NL4000/TAD (CPINT) /GET INTERRUPT HANDLING ADDRESS	28
29	865	070453* 6201	CDF 00 /GOING TO FIELD 0	29
30	866	070454* 3534	DCA I CPIADDRESS/[ROMSP] /STORE IN POINTER LOCATION	30
31	867	070455* 1121	TAD JMPIM1/[ROMSP&177+JMPIC] /GET JMP I .-1 INSTRUCTION	31
32	868	070456* 3535	DCA I PROMST/[ROMST] /STORE IN RESTART ADDRESS	32
33	869	070457* 6203	CIF CDF 00 /GOING TO FIELD 0	33
34	870	070460* 6266	CPD /INDIRECTS TO MAIN MEMORY	34
35	871	070461* 6004	PEX /LEAVING CP MEMORY	35
36	872	070462* 5512	JMP I Z200/[200] /GO START IT UP	36
37	873			37
38	874	070463* 0177 LBADR, LBTEST-1	/LOOPBACK TEST ADDRESS (-1)	38
39	875	070464* 0323 PMTSTLP, MTSTLP	/POINTER TO LOOP BRANCH	39
40				40
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1	877	/	COMES HERE WHEN THE PRINTER LOOPBACK TEST IS DONE TO TEST THE KEYBOARD IN	1
2	878	/	LOOPBACK MODE.	2
3	879			3
4	880	070465* 3054	LBDONE, DCA ESTATUS /SAVE (POSSIBLE ERROR) RESULTS	4
5	881	070466* 1320	TAD L16/(16) /GET 9600 BAUD VALUE	5
6	882	070467* 6333	PRSB /SET BAUD RATE	6
7	883	070470* 7200	CLA /CLEAN UP	7
8	884	070471* 3020	DCA CTMP1 /CLEAR OUTPUT TEST VALUE	8
9	885	070472* 3021	DCA CTMP2 /CLEAR INPUT TEST VALUE	9
10	886	070473* 3022	DCA CNTLO /CLEAR LOW-ORDER COUNTER	10
11	887	070474* 1346	TAD M52/(-52) /SETUP THE	11
12	888	070475* 3023	DCA CNTHI /HIGH-ORDER COUNTER	12
13	889	070476* 6121	APTSKP /AUTOMATIC PRODUCT TESTING?	13
14	890	070477* 1110	TAD Z20/[20] /NO, GET LOOPBACK MODE VALUE	14
15	891	070500* 6126	VIDCON /LOAD NEW MODE EITHER WAY	15
16	892	070501* 6116	KBIRB /READ KEYBOARD DATA TO SET UART	16
17	893	070502* 7200	CLA /THROW IT AWAY	17
18	894	070503* 6054	KBOPC /OUTPUT <NUL> TO UART NOW	18
19	895	070504* 6111	LUPLUP, KBISF /KEYBOARD INPUT FLAG UP?	19
20	896	070505* 7410	SKP /SKIP IF NOT	20
21	897	070506* 5317	JMP LUPIN /JUMP IF SO	21
22	898	070507* 6051	KBOSF /KEYBOARD OUTPUT FLAG UP?	22
23	899	070510* 7410	SKP /SKIP IF NOT	23
24	900	070511* 5334	JMP LUPOUT /JUMP IF SO	24
25	901	070512* 2022	ISZ CNTLO /DONE WAITING?	25
26	902	070513* 5304	JMP LUPLUP /NO, KEEP GOING	26
27	903	070514* 2023	ISZ CNTHI /ENOUGH TIMES?	27
28	904	070515* 5304	JMP LUPLUP /NO, KEEP GOING	28
29	905	070516* 5353	JMP LUPTIMOUT /YES, CONTINUE THERE	29
30	906			30
31	907	070517* 6111	LUPIN, KBISF /SKIP ON, CLEARING THE INPUT FLAG	31
32	908	070520* 0016	L16, 16 /9600 BAUD RATE VALUE; HERE IN CASE IT SKIPS	32
33	909	070521* 6116	KBIRB /GET THE INPUT VALUE	33
34	910	070522* 0115	AND Z377/[377] /JUST 8-BIT CHARACTER	34
35	911	070523* 7041	CIA /INVERT FOR TEST	35
36	912	070524* 1021	TAD CTMP2 /COMPARE TO TEST VALUE	36
37	913	070525* 7640	SZA CLA /SKIP IF IT MATCHES	37
38	914	070526* 6126	VIDCON /CLEAR LOOPBACK IF NOT	38
39	915	070527* 1021	TAD CTMP2 /GET INPUT PATTERN VALUE	39
40	916	070530* 4747	JMS I PCOUNTUP/(COUNTUP) /BUMP TO NEXT VALUE	40
41	917	070531* 5354	JMP LUPDONE /ALL VALUES USED	41
42	918	070532* 3021	DCA CTMP2 /SAVE UPDATED VALUE	42
43	919	070533* 5304	JMP LUPLUP /KEEP GOING	43
44	920			44
45	921	070534* 6051	LUPOUT, KBOSF /SKIP ON, THUS CLEARING OUTPUT FLAG	45
46	922	070535* 7000	NOP /JUST IN CASE IT SKIPS	46
47	923	070536* 1020	TAD CTMP1 /GET COUNT PATTERN VALUE	47
48	924	070537* 4747	JMS I PCOUNTUP/(COUNTUP) /BUMP TO NEXT VALUE	48
49	925	070540* 5304	JMP LUPLUP /ALL VALUES USED; KEEP GOING FOR INPUT'S SAKE	49
50	926	070541* 3020	DCA CTMP1 /SAVE UPDATED VALUE	50
51	927	070542* 1020	TAD CTMP1 /GET IT BACK	51
52	928	070543* 6056	KBOLS /OUTPUT IT	52
53	929	070544* 7200	CLA /CLEAN UP	53
54	930	070545* 5304	JMP LUPLUP /KEEP GOING	54
55				55
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1	932	070546*	7726	M52,	-52	/CONSTANT 7726	1
2	933	070547*	2734	PCOUNTU,	COUNTUP-LBTEST+LBLOAD	/POINTER TO COUNTUP ROUTINE	2
3	934	070550*	0375	L375,	375	/CONSTANT 0375	3
4	935	070551*	5600	PTSTBUF,	TSTBUFFER-1	/POINTER TO TEST BUFFER	4
5	936	070552*	0630	PKBTEND,	KBTEND	/POINTER TO END OF KEYBOARD TEST	5
6	937						6
7	938			/	COMES HERE WHEN THE LOOPBACK TEST TIMES OUT.		7
8	939						8
9	940	070553*	7315	LUPTIMO,	NL0010 CLL	/INDICATE TIME-OUT ENDED THE TEST	9
10	941						10
11	942			/	COMES HERE WHEN THE COUNT PATTERN IS FINISHED TO (POSSIBLY) TEST THE KEYBOARD.		11
12	943						12
13	944	070554*	1054	LUPDONE,	TAD	ESTATUS	13
14	945	070555*	3054	DCA		ESTATUS	14
15	946	070556*	6126		VIDCON		15
16	947	070557*	6121		APTSKP		16
17	948	070560*	7410		SKP		17
18	949	070561*	5752	JMP I	PKBTEND/(KBTEND)		18
19	950	070562*	3012	DCA	XR2		19
20	951	070563*	6116		KBIRB		20
21	952	070564*	7200		CLA		21
22	953	070565*	1350	TAD	L375/[375]		22
23	954	070566*	6056		KBOLS		23
24	955	070567*	6051		KBOSF		24
25	956	070570*	5367	JMP	.-1		25
26	957	070571*	6051		KBOSF		26
27	958	070572*	0040		40/NOP		27
28	959	070573*	7200		CLA		28
29	960	070574*	1126	TAD	Z7766/[-12]		29
30	961	070575*	3023	DCA	CNTHI		30
31	962	070576*	3022	DCA	CNTLO		31
32	963	070577*	1351	TAD	PTSTBUFF/(TSTBUFFER-1)		32
33	964	070600*	3011	DCA	XR1		33
34	965	070601*	6111	KEYLUP,	KBISF		34
35	966	070602*	7410		SKP		35
36	967	070603*	5211		JMP	GOTKEY	36
37	968	070604*	2022		ISZ	CNTLO	37
38	969	070605*	5201		JMP	KEYLUP	38
39	970	070606*	2023		ISZ	CNTHI	39
40	971	070607*	5201		JMP	KEYLUP	40
41	972						41
42	973			/	NO (FURTHER) KEYBOARD INPUT.		42
43	974						43
44	975	070610*	5220	JMP	KBTIMOUT	/CONTINUE THERE	44
45							45
46							46
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1	977	/	COMES HERE WHEN A KEYBOARD CHARACTER IS RECEIVED.	1
2	978			2
3	979	070611* 6111	GOTKEY, KBISF /SKIP ON, THUS CLEARING THE FLAG	3
4	980	070612* 7000	NOP /IN CASE IT SKIPS	4
5	981	070613* 6116	KBIRB /GET THE CHARACTER	5
6	982	070614* 0115	AND Z377/[377] /JUST 8 BITS	6
7	983	070615* 3411	DCA I XR1 /STORE INTO THE BUFFER	7
8	984	070616* 2012	ISZ XR2 /BUMP CHARACTER COUNT	8
9	985	070617* 5201	JMP KEYLUP /KEEP GOING	9
10	986			10
11	987	/	COMES HERE WHEN KEYBOARD INPUT TIMES OUT.	11
12	988			12
13	989	070620* 1233	KBTIMOU, TAD L7774/(-4) /GET EXPECTED COUNT	13
14	990	070621* 1012	TAD XR2 /COMPARE TO ACTUAL	14
15	991	070622* 7450	SNA /SKIP IF OTHER THAN EXPECTED	15
16	992	070623* 1632	TAD I PTSTBEND/(TSTBEND) /ELSE LOOK AT LAST CHARACTER RECEIVED	16
17	993	070624* 7640	SZA CLA /SKIP IF LAST CHARACTER WAS A <NUL>	17
18	994	070625* 1110	TAD Z20/[20] /ELSE ADD ON KEYBOARD ERROR VALUE	18
19	995	070626* 1054	TAD ESTATUS /GET CUMULATIVE STATUS	19
20	996	070627* 3054	DCA ESTATUS /STORE BACK	20
21	997	070630* 5631	KBTEND, JMP I .+1/(COMTEST) /CONTINUE THERE	21
22	998	070631* 4652	COMTEST /THROUGH HERE	22
23	999			23
24	1000	070632* 5604	PTSTBEN, TSTBEND-1 /POINTER TO END OF TEST KEYBOARD BUFFER	24
25	1001	070633* 7774	L7774, 7774 /CONSTANT 7774	25
26	1002	070634* 0342	L342, 342 /CONSTANT 0342	26
27	1003	070635* 0320	L320, 320 /CONSTANT 0320	27
28	1004	070636* 7740	A7740, 7740 /CONSTANT 7740	28
29	1005	070637* 6000	PAPTSTR, APTSTR /STARTING ADDRESS FOR DOWNLOAD	29
30				30
31				31
32				32
33				33
34				34
35				35
36				36
37				37
38				38
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1	1007	/			DIRECT DOWN-LOAD ROUTINE. CALLED VIA JMS I .+1;DOWNLOAD IF APTEN IS SET. THE	1
2	1008	/			FORMAT IS STANDARD PAPER-TAPE RIM FORMAT WITHOUT A LEADER. LOADING IS TO MAIN	2
3	1009	/			MEMORY FIELD 7. THE ROUTINE IS ALSO CALLABLE AFTER A PREVIOUSLY LOADED	3
4	1010	/			PROGRAM EXITS VIA A PANEL REQUEST INSTEAD OF A HLT INSTRUCTION. THIS CAUSES A	4
5	1011	/			TOTAL RESTART OF ALL CODE INITIALLY MOVED FROM THE ROM TO CP FIELD 7. IN THIS	5
6	1012	/			SITUATION, CODE LOADED INTO MAIN MEMORY 76000-77777 WILL BE MOVED TO CP	6
7	1013	/			76000-7777 TO BECOME THE NEXT CP-INTERRUPT HANDLER, ETC. EXIT VIA HLT	7
8	1014	/			INSTRUCTION RETURNS TO A DESIGNATED ENTRY POINT TO LOAD MORE CODE.	8
9	1015					9
10	1016	070640*	0000	DOWNLOA, .-. .	/DOWN-LOAD ROUTINE	10
11	1017	070641*	2240	ISZ DOWNLOAD	/BUMP PAST CALL POINTER	11
12	1018	070642*	6271	CDF 70	/ENSURE OUR FIELD	12
13	1019	070643*	1116	TAD Z400/[400]	/SET POWER-ON MASK	13
14	1020	070644*	0061	AND PSTATUS	/GET POWER-ON BIT	14
15	1021	070645*	7650	SNA CLA	/SKIP IF SET	15
16	1022	070646*	5337	JMP DWNAGN	/JUMP IF NOT	16
17	1023	070647*	3355	DCA CMSW	/INDICATE MOVABLE CODE NOT LOADED YET	17
18	1024					18
19	1025	/			THE CODE IS REJOINED HERE WHEN DOWN-LOADED CODE EXITS WITH A HLT INSTRUCTION.	19
20	1026					20
21	1027	070650*	6271	DWNENTR, CDF 70	/GOTO MAIN MEMORY LOADING FIELD	21
22	1028	070651*	1255	TAD PBAUD	/GET BAUD RATE	22
23	1029	070652*	6333	PRSB	/SETUP 9600 BAUD RATE	23
24	1030	070653*	7200	CLA	/CLEAN UP	24
25	1031	070654*	6321	PRISKP	/SKIP ON, THUS CLEARING INPUT FLAG	25
26	1032	070655*	0016	PBAUD, 16	/RATE VALUE FOR 9600 BAUD; ALSO SKIP PROTECT	26
27	1033	070656*	1234	TAD L342/(342)	/GET CHARACTER CODE	27
28	1034	070657*	6336	PROLS	/OUTPUT IT	28
29	1035	070660*	6326	PRIRB	/READ KEYBOARD BUFFER NOW FOR CHARACTERS LATER	29
30	1036	070661*	6266	CPD	/MAIN MEMORY INDIRECTS FROM NOW ON	30
31	1037	070662*	6321	RIMLUP, PRISKP	/FLAG UP	31
32	1038	070663*	5262	JMP RIMLUP	/NO, WAIT FOR IT	32
33	1039	070664*	6321	PRISKP	/YES, MAKE SURE IT CLEARS	33
34	1040	070665*	7000	NOP	/IN CASE IT SKIPS AGAIN	34
35	1041	070666*	6326	PRIRB	/READ IN THE CHARACTER	35
36	1042	070667*	7106	CLL RTL;RTL	/MOVE UP	36
37	1043	070670*	7006			37
38	1044	070671*	7510	SPA	/SKIP IF NOT AT TRAILER	38
39	1045	070672*	5305	JMP RIMEND	/JUMP IF DONE	39
40	1046	070673*	7006	RTL	/MOVE UP TO HIGH-ORDER	40
41	1047	070674*	6321	PRISKP	/WAIT FOR	41
42	1048	070675*	5274	JMP .-1	/NEXT CHARACTER	42
43	1049	070676*	6321	PRISKP	/MAKE SURE IT CLEARS	43
44	1050	070677*	7000	NOP	/IN CASE IT SKIPS AGAIN	44
45	1051	070700*	6324	PRIRS	/OR IN LATEST CHARACTER	45
46	1052	070701*	7420	SNL	/SKIP IF ORIGIN SETTING	46
47	1053	070702*	3400	DCA I RIMADR	/STORE THE DATA	47
48	1054	070703*	3000	DCA RIMADR	/STORE AS NEW ORIGIN	48
49	1055	070704*	5262	JMP RIMLUP	/KEEP GOING	49
50						50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	1057	/	COMES HERE AT THE END OF THE RIM LOADING.	1
2	1058			2
3	1059	070705* 7200	RIMEND, CLA /CLEAN UP	3
4	1060	070706* 1235	TAD L320/(320) /GET THIS VALUE	4
5	1061	070707* 3636	DCA I A7740/(7740) /SOMEONE NEEDS THIS THERE	5
6	1062	070710* 6276	SPD /CP MEMORY INDIRECTS	6
7	1063	070711* 3010	DCA XR0 /CLEAR INDEX (WHY?)	7
8	1064	070712* 6201	CDF 00 /GOING TO FIELD 0	8
9	1065	070713* 1357	TAD PCPIHND/(CPIHND) /GET INTERRUPT HANDLER ADDRESS	9
10	1066	070714* 3534	DCA I CPIADDR/[ROMSP] /STORE IN INTERRUPT POINTER	10
11	1067	070715* 1356	TAD JMPIPM/(ROMSP&177+JMPIC) /GET JMP I .-1	11
12	1068	070716* 3535	DCA I PROMST/[ROMST] /STORE IN RESTART ADDRESS	12
13	1069	070717* 7240	NL7777 /BACKUP	13
14	1070	070720* 1357	TAD PCPIHND/(CPIHND) /NOW HAVE AUTO-INCREMENT LOADING ADDRESS	14
15	1071	070721* 3010	DCA XR0 /STASH THE POINTER	15
16	1072	070722* 1010	TAD XR0 /GET IT AGAIN	16
17	1073	070723* 3011	DCA XR1 /STASH FOR MOVING ALSO	17
18	1074	070724* 6271	CDF 70 /GOTO OUR FIELD	18
19	1075	070725* 1410	RCPLUP, TAD I XR0 /GET A WORD	19
20	1076	070726* 7450	SNA /END OF LIST?	20
21	1077	070727* 5334	JMP RIMST /JUMP IF SO	21
22	1078	070730* 6201	CDF 00 /GOTO FIELD 0	22
23	1079	070731* 3411	DCA I XR1 /PUT A WORD	23
24	1080	070732* 6271	CDF 70 /BACK TO OUR FIELD	24
25	1081	070733* 5325	JMP RCPLUP /KEEP GOING	25
26	1082			26
27	1083	070734* 6273	RIMST, CIF CDF 70 /GOING TO FIELD 7	27
28	1084	070735* 6004	PEX /LEAVING PANEL MEMORY	28
29	1085	070736* 5637	JMP I PAPTSTRT/(APTSTRT) /GO THERE	29
30	1086			30
31	1087	/	COMES HERE IF A PREVIOUSLY LOADED ROUTINE EXITED VIA A PANEL REQUEST WHICH	31
32	1088	/	CAUSES A TOTAL RESTART OF THE FORMERLY ROM-BASED CODE MOVED TO FIELD 7. CODE	32
33	1089	/	LOADED INTO MAIN MEMORY 76000-77777 SHOULD BE MOVED TO CP 76000-77777.	33
34	1090			34
35	1091	070737* 1355	DWNAGN, TAD CMSW /GET LOAD SWITCH	35
36	1092	070740* 7640	SZA CLA /SKIP IF NOT LOADED YET	36
37	1093	070741* 5640	JMP I DOWNLOAD /RETURN TO CALLER	37
38	1094	070742* 7240	NL7777 /INDICATE CODE MOVED	38
39	1095	070743* 3355	DCA CMSW /FOR NEXT TIME	39
40	1096	070744* 1237	TAD PAPTSTRT/(APTSTRT) /GET STARTING ADDRESS	40
41	1097	070745* 3000	DCA RIMADR /STASH THE POINTER	41
42	1098	070746* 6266	MCLUP, CPD /MAIN MEMORY INDIRECTS	42
43	1099	070747* 1400	TAD I RIMADR /GET A WORD	43
44	1100	070750* 6276	SPD /CP MEMORY INDIRECTS	44
45	1101	070751* 3400	DCA I RIMADR /PUT A WORD	45
46	1102	070752* 2000	ISZ RIMADR /BUMP TO NEXT	46
47	1103	070753* 5346	JMP MCLUP /KEEP GOING	47
48	1104	070754* 5640	JMP I DOWNLOAD /RETURN TO CALLER	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1106	070755*	0000	CMSW, .-. /CODE MOVED SWITCH	1
2	1107	070756*	5776	JMPIPM, ROMSP&177+JMPIC /JMP I .-1 CONSTANT FOR RESTART ADDRESS	2
3	1108	070757*	2536	PCPIHND, CPIHND /CP-INTERRUPT HANDLING ADDRESS	3
4	1109				4
5	1110			/ VALIDATION PATTERN FOR TRACK ZERO, SECTOR 4 DATA.	5
6	1111				6
7	1112	070760*	7777	T0S4DAT, -1	7
8	1113	070761*	7670	- "H!200	8
9	1114	070762*	7664	- "L!200	9
10	1115	070763*	7654	- "T!200	10
11	1116	070764*	7672	- "F!200	11
12	1117	070765*	7664	- "L!200	12
13	1118	070766*	7671	- "G!200	13
14	1119	070767*	7776	-2	14
15	1120	070770*	0000	0 /THIS ENDS THE LIST	15
16	1121				16
17	1122			/ DATA PATTERN USED TO TEST RX50 SILO BUFFER.	17
18	1123				18
19	1124	070771*	0000	TPATTRN, 000	19
20	1125	070772*	0252	252	20
21	1126	070773*	0125	125	21
22	1127	070774*	0063	063	22
23	1128	070775*	0017	017	23
24	1129	070776*	0377	377	24
25	1130				25
26	1131	070777*	0000	ZBLOCK 1 /EMPTY SPACE	26
27	1132				27
28	1133	071000*	0000	DINDEX, .-. /TEST DATA INDEX	28
29	1134	071001*	0000	DCNT, .-. /TEST DATA COUNTER	29
30	1135	071002*	0000	DPTR, .-. /TEST DATA POINTER	30
31	1136	071003*	7773	M5, -5 /CONSTANT 7773	31
32	1137	071004*	7000	M1000, -1000 /CONSTANT 7000	32
33	1138	071005*	0771	PTPATTR, TPATTRN /POINTER TO SILO TEST DATA	33
34	1139	071006*	5356	PRDTWAI, RDTWAIT /POINTER TO RDTWAIT ROUTINE	34
35					35
36					36
37					37
38					38
39					39
40					40
41					41
42					42
43					43
44					44
45					45
46					46
47					47
48					48
49					49
50					50
51					51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	1141		/	COMES HERE AFTER DEALING WITH THE COMMUNICATIONS PORT.	1
2	1142				2
3	1143	071007*	4606	COMDONE, JMS I PRDTPWAIT/(RDTWAIT)	3
4	1144	071010*	6751	LCD	4
5	1145	071011*	6007	CAF	5
6	1146	071012*	7240	NL7777	6
7	1147	071013*	3010	DCA XR0	7
8	1148	071014*	1132	TAD DTIMOUT/[-DTIME]	8
9	1149	071015*	3050	DCA T6	9
10	1150	071016*	3051	DCA T7	10
11	1151	071017*	6755	DONWAIT, SDN	11
12	1152	071020*	5224	JMP NOTDONE	12
13	1153	071021*	6754	SER	13
14	1154	071022*	5231	JMP DONEOK	14
15	1155	071023*	5376	JMP RXERROR	15
16	1156				16
17	1157	071024*	2051	NOTDONE, ISZ T7	17
18	1158	071025*	5217	JMP DONWAIT	18
19	1159	071026*	2050	ISZ T6	19
20	1160	071027*	5217	JMP DONWAIT	20
21	1161	071030*	5376	JMP RXERROR	21
22	1162				22
23	1163	071031*	6121	DONEOK, APTSKP	23
24	1164	071032*	7410	SKP	24
25	1165	071033*	4524	JMS I PAPTRROUTINE/[APTRROUTINE]	25
26	1166	071034*	7300	CLA CLL	26
27	1167	071035*	3200	DCA DINDEX	27
28	1168	071036*	3201	DCA DCNT	28
29	1169	071037*	7303	NL0100 CLL	29
30	1170	071040*	6751	LCD	30
31	1171	071041*	4340	JMS RXWAIT	31
32	1172	071042*	5245	JMP CMNDOK	32
33	1173	071043*	5376	JMP RXERROR	33
34	1174	071044*	5376	JMP RXERROR	34
35	1175				35
36	1176	071045*	6752	CMNDOK, XDR	36
37	1177	071046*	4340	FILLUP, JMS RXWAIT	37
38	1178	071047*	5252	JMP TRNSUP	38
39	1179	071050*	5267	JMP DONEUP	39
40	1180	071051*	5376	JMP RXERROR	40
41	1181				41
42	1182	071052*	1200	TRNSUP, TAD DINDEX	42
43	1183	071053*	1205	TAD PTPATTRN/(TPATTRN)	43
44	1184	071054*	3202	DCA DPTR	44
45	1185	071055*	1602	TAD I DPTR	45
46	1186	071056*	6752	XDR	46
47	1187	071057*	7200	CLA	47
48	1188	071060*	2201	ISZ DCNT	48
49	1189	071061*	2200	ISZ DINDEX	49
50	1190	071062*	1200	TAD DINDEX	50
51	1191	071063*	1203	TAD M5/(-5)	51
52	1192	071064*	7740	SMA SZA CLA	52
53	1193	071065*	3200	DCA DINDEX	53
54	1194	071066*	5246	JMP FILLUP	54
55					55
56					56
57					57
58					58

1	1196		/	DONE FLAG IS UP; CHECK IF PROPER BYTE COUNT.		1
2	1197					2
3	1198	071067*	1201	DONEUP, TAD DCNT	/GET DATA COUNTER	3
4	1199	071070*	1204	TAD M1000/(-1000)	/COMPARE TO EXPECTED VALUE	4
5	1200	071071*	7640	SZA CLA	/SKIP IF IT MATCHES	5
6	1201	071072*	5376	JMP RXERROR	/ELSE COMPLAIN	6
7	1202	071073*	7300	CLA CLL	/CLEAN UP	7
8	1203	071074*	3200	DCA DINDEX	/CLEAR DATA INDEX	8
9	1204	071075*	3201	DCA DCNT	/CLEAR DATA COUNTER	9
10	1205	071076*	1111	TAD Z102/[102]	/GET EMPTY BUFFER COMMAND (8-BIT) VALUE	10
11	1206	071077*	6751	LCD	/LOAD THE COMMAND	11
12	1207	071100*	4340	JMS RXWAIT	/CALL WAIT ROUTINE	12
13	1208	071101*	5304	JMP CMDOK	/TRANSFER FLAG CAME UP	13
14	1209	071102*	5376	JMP RXERROR	/DONE FLAG CAME UP, COMPLAIN	14
15	1210	071103*	5376	JMP RXERROR	/ERROR OR TIME-OUT	15
16	1211					16
17	1212	071104*	7200	CMDOK, CLA	/CLEAN UP	17
18	1213	071105*	6752	XDR	/FINISH THE COMMAND	18
19	1214	071106*	4340	EMPTLUP, JMS RXWAIT	/CALL WAIT ROUTINE	19
20	1215	071107*	7410	SKP	/TRANSFER FLAG CAME UP	20
21	1216	071110*	5332	JMP DONUP	/DONE FLAG CAME UP	21
22	1217					22
23	1218		/	OOPS! FORGOT THE "JMP RXERROR" IF IT GETS AN ERROR OR TIMES OUT.		23
24	1219					24
25	1220	071111*	1200	TAD DINDEX	/GET DATA INDEX	25
26	1221	071112*	1205	TAD PTPATTRN/(TPATTRN)	/POINT TO PROPER ELEMENT IN LIST	26
27	1222	071113*	3202	DCA DPTR	/STASH THE POINTER	27
28	1223	071114*	1602	TAD I DPTR	/GET THE DATA	28
29	1224	071115*	3202	DCA DPTR	/SAVE IT	29
30	1225	071116*	6752	XDR	/GET THE BYTE	30
31	1226	071117*	7041	CIA	/INVERT FOR TESTING	31
32	1227	071120*	1202	TAD DPTR	/COMPARE TO TEST VALUE	32
33	1228	071121*	7640	SZA CLA	/SKIP IF IT MATCHES	33
34	1229	071122*	5376	JMP RXERROR	/ELSE COMPLAIN	34
35	1230	071123*	2201	ISZ DCNT	/BUMP DATA COUNTER	35
36	1231	071124*	2200	ISZ DINDEX	/BUMP DATA INDEX	36
37	1232	071125*	1200	TAD DINDEX	/GET DATA INDEX	37
38	1233	071126*	1203	TAD M5/(-5)	/COMPARE TO LARGEST VALUE	38
39	1234	071127*	7740	SMA SZA CLA	/SKIP IF NOT TOO FAR	39
40	1235	071130*	3200	DCA DINDEX	/ELSE RESET DATA INDEX	40
41	1236	071131*	5306	JMP EMPTLUP	/KEEP GOING	41
42	1237					42
43	1238		/	DONE FLAG IS UP; CHECK IF PROPER BYTE COUNT.		43
44	1239					44
45	1240	071132*	1201	DONUP, TAD DCNT	/GET DATA COUNTER	45
46	1241	071133*	1204	TAD M1000/(-1000)	/COMPARE TO EXPECTED VALUE	46
47	1242	071134*	7640	SZA CLA	/SKIP IF IT MATCHES	47
48	1243	071135*	5376	JMP RXERROR	/ELSE COMPLAIN	48
49	1244	071136*	5737	JMP I .+1/(RXOK)	/CONTINUE THERE	49
50	1245	071137*	1201	RXOK	/THROUGH HERE	50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	1247	/				RX50 WAIT ROUTINE. TAKES IMMEDIATE RETURN IF TRANSFER FLAG RAISES. TAKES	1
2	1248	/				SKIP RETURN IF DONE FLAG RAISES. TAKES DOUBLE-SKIP RETURN IF AN ERROR OCCURS	2
3	1249	/				OR THE HARDWARE TIMES OUT.	3
4	1250						4
5	1251	071140*	0000	RXWAIT,	...		5
6	1252	071141*	1132	TAD	DTIMOUT/[-DIME]	/SETUP THE	6
7	1253	071142*	3050	DCA	T6	/TIME-OUT COUNTER	7
8	1254	071143*	3051	DCA	T7	/CLEAR INNER TIME-OUT COUNTER	8
9	1255	071144*	6753	RXWATLP,	STR	/TRANSFER FLAG UP?	9
10	1256	071145*	7410	SKP		/SKIP IF NOT	10
11	1257	071146*	5740	JMP I	RXWAIT	/YES, TAKE IMMEDIATE RETURN	11
12	1258	071147*	6755	SDN		/DONE FLAG UP?	12
13	1259	071150*	5354	JMP	RXWATING	/NO, KEEP TRYING	13
14	1260	071151*	6754	SER		/YES, WERE THERE ERRORS?	14
15	1261	071152*	5361	JMP	RXSKIP	/NO, TAKE SKIP RETURN	15
16	1262	071153*	5360	JMP	RXSKP2	/YES, TAKE DOUBLE SKIP RETURN	16
17	1263						17
18	1264	071154*	2051	RXWATIN,	ISZ T7	/WAITING TOO LONG?	18
19	1265	071155*	5344	JMP	RXWATLP	/NO, KEEP TRYING	19
20	1266	071156*	2050	ISZ	T6	/WAITED TOO LONG?	20
21	1267	071157*	5344	JMP	RXWATLP	/NO, KEEP TRYING	21
22	1268	071160*	2340	RXSKP2,	ISZ RXWAIT	/YES, TAKE DOUBLE-SKIP RETURN	22
23	1269	071161*	2340	RXSKIP,	ISZ RXWAIT	/SET SKIP RETURN FOR DONE FLAG UP CONDITION	23
24	1270	071162*	5740	JMP I	RXWAIT	/RETURN AS NECESSARY	24
25	1271						25
26	1272	071163*	7700	M100,	-100	/CONSTANT 7700	26
27	1273						27
28	1274	071164*	0000	TIMOUT,	...	/GENERAL TIME-OUT ROUTINE	28
29	1275	071165*	3050	DCA	T6	/CLEAR INNER COUNTER	29
30	1276	071166*	1363	TAD	M100/(-100)	/SETUP THE	30
31	1277	071167*	3051	DCA	T7	/OUTER COUNTER	31
32	1278	071170*	2050	TIMLUP,	ISZ T6	/WAITING LONG ENOUGH?	32
33	1279	071171*	5370	JMP	TIMLUP	/NO, KEEP GOING	33
34	1280	071172*	2051	ISZ	T7	/WAITED TOO LONG?	34
35	1281	071173*	5370	JMP	TIMLUP	/NO, KEEP GOING	35
36	1282	071174*	5764	JMP I	TIMOUT	/YES, RETURN TO CALLER	36
37	1283						37
38	1284	071175*	0040	L40,	40	/CONSTANT 0040	38
39	1285						39
40	1286	071176*	1375	RXERROR,	TAD L40/(40)	/GET RX50 ERROR VALUE	40
41	1287	071177*	1054	TAD	ESTATUS	/UPDATE CUMULATIVE ERROR STATUS	41
42	1288	071200*	3054	DCA	ESTATUS	/STORE BACK	42
43							43
44							44
45							45
46							46
47							47
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56							56
57							57
58							58

1	1290	071201*	7315	RXOK,	NL0010	CLL	/SET CRT VALUE	1
2	1291	071202*	6126		VIDCON		/LOAD ECHRMOD	2
3	1292	071203*	7200		CLA		/CLEAN UP	3
4	1293	071204*	4643		JMS I	PENDUP/(ENDUP)	/FINISH UP SCREEN DISPLAY (POSSIBLY ERRORS)	4
5	1294	071205*	4635		JMS I	PRDSETUP/(RDSETUP)	/GET RD STARTUP CODE READ IN (IF PRESENT)	5
6	1295	071206*	4634		JMS I	PRXRDY/(RXRDY)	/WAIT UNTIL DRIVE ZERO IS READY	6
7	1296	071207*	5206		JMP	.-1	/WAIT FOR IT TO BE READY INDEFINITELY	7
8	1297	071210*	4544		JMS I	PCLR23/[CLR23]	/CLEAR FIELDS 2, 3 (AND BITS IN 13705, 13707)	8
9	1298	071211*	7240	RTRK1,	NL7777		/SETUP THE	9
10	1299	071212*	3010		DCA	XR0	/STORAGE POINTER	10
11	1300	071213*	6201		CDF	00	/STORING IN FIELD 0	11
12	1301	071214*	1111		TAD	Z102/[102]	/GET EMPTY SILO (8-BIT) COMMAND	12
13	1302	071215*	3053		DCA	RXCMD	/STASH IT	13
14	1303			/	CLA		/CALL WITH AC CLEAR (1-1)	14
15	1304	071216*	4311		JMS	FREAD	/CALL FLOPPY READ ROUTINE	15
16	1305	071217*	7776		-1-1		/ONE SECTOR	16
17	1306	071220*	0001		1		/ON TRACK 1	17
18	1307	071221*	5211		JMP	RTRK1	/ERROR, TRY AGAIN	18
19	1308	071222*	4632		JMS I	PVALID8/(VALID8)	/VALIDATE THE PREFERENCE SETTINGS	19
20	1309	071223*	5236		JMP	FLPBAD	/FLOPPY IS BAD, GO COMPLAIN	20
21	1310	071224*	4244	SLSHAGN,	JMS	RXLOAD	/LOAD IN VARIOUS SLUSHWARE ITEMS	21
22	1311	071225*	5240		JMP	SLSHBAD	/COULDN'T DO IT	22
23	1312	071226*	4544		JMS I	PCLR23/[CLR23]	/CLEAR FIELDS 2, 3 (AND BITS IN 13705, 13707)	23
24	1313	071227*	7200		CLA		/CLEAN UP	24
25	1314	071230*	6203		CIF CDF	00	/GOING TO FIELD 0	25
26	1315	071231*	5512		JMP I	Z200/[200]	/GO START IT UP	26
27	1316							27
28	1317	071232*	1600	PVALID8,	VALID8		/POINTER TO VALIDATION ROUTINE	28
29	1318	071233*	1554	PFBLINK,	FBLINK		/POINTER TO FBLINK ROUTINE	29
30	1319	071234*	3701	PRXRDY,	RXRDY		/POINTER TO RX50 READY CHECK ROUTINE	30
31	1320	071235*	5000	PRDSETU,	RDSETUP		/POINTER TO RD51 SETUP ROUTINE	31
32	1321							32
33	1322			/		COMES HERE TO DISPLAY A BLINKING BIG FLOPPY PATTERN IF THE CURRENT DISKETTE		33
34	1323			/		HAS AN INVALID BOOT BLOCK IN TRACK 1, SECTOR 1.		34
35	1324							35
36	1325	071236*	4633	FLPBAD,	JMS I	PFBLINK/(FBLINK)	/GO DISPLAY THE PATTERN ONCE	36
37	1326	071237*	5211		JMP	RTRK1	/TRY AGAIN	37
38	1327							38
39	1328			/		COMES HERE TO DISPLAY THE BIG FLOPPY PATTERN IF THE SLUSHWARE STUFF CAN'T BE		39
40	1329			/		COMPLETELY READ IN.		40
41	1330							41
42	1331	071240*	4633	SLSHBAD,	JMS I	PFBLINK/(FBLINK)	/GO DISPLAY THE PATTERN ONCE	42
43	1332	071241*	5224		JMP	SLSHAGN	/TRY AGAIN	43
44	1333							44
45	1334	071242*	1140	PRXWAIT,	RXWAIT		/POINTER TO RX50 WAIT ROUTINE	45
46	1335	071243*	2444	PENDUP,	ENDUP		/POINTER TO ENDUP ROUTINE	46
47								47
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1	1337	071244*	0000	RXLOAD,	...	/MULTIPLE RX50 LOADING ROUTINE	1
2	1338	071245*	7240		NL7777	/SETUP THE	2
3	1339	071246*	3010	DCA	XR0	/STORAGE POINTER	3
4	1340	071247*	1111	TAD	Z102/[102]	/SETUP THE	4
5	1341	071250*	3053	DCA	RXCMD	/EMPTY BUFFER COMMAND (8-BIT)	5
6	1342	071251*	4707	JMS I	PT0S3RD/(T0S3RD)	/READ AND VALIDATE TRACK 0, SECTOR 3	6
7	1343	071252*	5644	JMP I	RXLOAD	/COULDN'T DO IT, TAKE FAILURE RETURN	7
8	1344	071253*	4710	JMS I	PDOCG/(DOCG)	/READ AND LOAD CHARACTER GENERATOR RAM DATA	8
9	1345	071254*	7305		NL002A	/SETUP 12-BIT EMPTY SILO COMMAND VALUE	9
10	1346	071255*	3053	DCA	RXCMD	/STASH THE COMMAND	10
11	1347	071256*	7240		NL7777	/SETUP THE	11
12	1348	071257*	3010	DCA	XR0	/DATA POINTER	12
13	1349	071260*	6201	CDF	00	/SETUP THE DATA FIELD	13
14	1350			/	CLA	/CALL WITH AC CLEAR (1-1)	14
15	1351	071261*	4311	JMS	FREAD	/CALL FLOPPY READ ROUTINE	15
16	1352	071262*	7765		-12-1	/READ 10 SECTORS (1-10)	16
17	1353	071263*	0116		116	/TRACK 78	17
18	1354	071264*	5644	JMP I	RXLOAD	/COULDN'T DO IT, TAKE FAILURE RETURN	18
19	1355	071265*	6201	CDF	00	/SETUP THE DATA FIELD	19
20	1356			/	CLA	/CALL WITH AC CLEAR (1-1)	20
21	1357	071266*	4311	JMS	FREAD	/CALL FLOPPY READ ROUTINE	21
22	1358	071267*	7771		-6-1	/READ 6 SECTORS (1-6)	22
23	1359	071270*	0117		117	/TRACK 79	23
24	1360	071271*	5644	JMP I	RXLOAD	/COULDN'T DO IT, TAKE FAILURE RETURN	24
25	1361	071272*	6211	CDF	10	/SETUP THE DATA FIELD	25
26	1362	071273*	7327		NL0006	/SETUP FOR SECTOR 7 FIRST	26
27	1363	071274*	4311	JMS	FREAD	/CALL FLOPPY READ ROUTINE	27
28	1364	071275*	7765		-12-1	/READ 4 SECTORS (7-10)	28
29	1365	071276*	0117		117	/TRACK 79	29
30	1366	071277*	5644	JMP I	RXLOAD	/COULDN'T DO IT, TAKE FAILURE RETURN	30
31	1367	071300*	6201	CDF	00	/BACK TO PRIMARY CODE FIELD	31
32	1368	071301*	3010	DCA	XR0	/CLEAR THE POINTER	32
33	1369	071302*	4706	JMS I	PSLCHECK/(SLCHECK)	/GO VALIDATE THE DATA	33
34	1370	071303*	5644	JMP I	RXLOAD	/BAD DATA, TAKE FAILURE RETURN	34
35	1371	071304*	2244	ISZ	RXLOAD	/GOOD DATA, BUMP RETURN ADDRESS	35
36	1372	071305*	5644	JMP I	RXLOAD	/TAKE SKIP RETURN TO CALLER	36
37	1373						37
38	1374	071306*	4153	PSLCHEC,	SLCHECK	/POINTER TO SLUSHWARE VALIDATION ROUTINE	38
39	1375	071307*	3111	PT0S3RD,	T0S3RD	/POINTER TO TRACK 0, SECTOR 3 READ ROUTINE	39
40	1376	071310*	3152	PDOCG,	DOCG	/POINTER TO DOCG ROUTINE	40
41							41
42							42
43							43
44							44
45							45
46							46
47							47
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1	1378	/				RX50 READ ROUTINE. CALL WITH AC CONTAINING RELATIVE OFFSET ON THE TRACK OF	1
2	1379	/				FIRST SECTOR TO TRANSFER. THIS IS ONE LESS THAN THE ACTUAL SECTOR NUMBER.	2
3	1380	/				THE FIRST IN-LINE ARGUMENT IS THE LIMIT ON THE SECTOR COUNT, I.E., THE TWO'S	3
4	1381	/				COMPLEMENT OF THE SECTOR VALUE ONE HIGHER THAN DESIRED AS THE LAST SECTOR	4
5	1382	/				TRANSFERRED. THE DATA IS TRANSFERRED INTO THE SAME FIELD AS THE CURRENT DATA	5
6	1383	/				FIELD. XR0 SHOULD BE SET TO THE DATA STORAGE ADDRESS -1. RXCMD SHOULD BE SET	6
7	1384	/				TO A SILO EMPTY COMMAND, EITHER 8-BIT OR 12-BIT AS REQUIRED. THE COMMAND WILL	7
8	1385	/				BE ISSUED CORRECTLY EITHER WAY. CONTROL IS RETURNED TO JUST PAST THE SECOND	8
9	1386	/				ARGUMENT IF THERE IS AN ERROR. AN EXTRA SKIP RETURN IS PERFORMED IF THE READ	9
10	1387	/				WAS OK.	10
11	1388						11
12	1389	071311*	0000	FREAD,	...	/FLOPPY READ ROUTINE	12
13	1390	071312*	3041	DCA	TINC	/SAVE PASSED VALUE (SECTOR OFFSET ON TRACK)	13
14	1391	071313*	6214	RDF		/GET CALLING FIELD	14
15	1392	071314*	1122	TAD	ZCDF/[CDF]	/FORM CDF CALLING FIELD	15
16	1393	071315*	3361	DCA	TRNCDF	/STORE IN-LINE	16
17	1394	071316*	6271	CDF	70	/BACK TO OUR FIELD	17
18	1395	071317*	1711	TAD I	FREAD	/GET FIRST ARGUMENT (COUNT LIMIT)	18
19	1396	071320*	3042	DCA	T0	/STASH IT	19
20	1397	071321*	2311	ISZ	FREAD	/BUMP TO NEXT	20
21	1398	071322*	1711	TAD I	FREAD	/GET SECOND ARGUMENT (TRACK)	21
22	1399	071323*	3043	DCA	T1	/STASH IT	22
23	1400	071324*	2311	ISZ	FREAD	/BUMP PAST ARGUMENT	23
24	1401	071325*	6271	FNXTSCT, CDF	70	/BACK TO OUR FIELD	24
25	1402	071326*	2041	ISZ	TINC	/BUMP SECTOR OFFSET TO LATEST SECTOR	25
26	1403	071327*	1041	TAD	TINC	/GET THE LATEST OFFSET	26
27	1404	071330*	1042	TAD	T0	/COMPARE TO COUNT LIMIT	27
28	1405	071331*	7710	SPA CLA		/SKIP IF FINISHED	28
29	1406	071332*	5335	JMP	FNEXT	/ELSE KEEP GOING	29
30	1407	071333*	2311	ISZ	FREAD	/BUMP RETURN ADDRESS	30
31	1408	071334*	5711	JMP I	FREAD	/TAKE SUCCESSFUL RETURN	31
32	1409						32
33	1410	071335*	7327	FNEXT,	NL0006	/GET READ FUNCTION VALUE	33
34	1411	071336*	4545	JMS I	PRXCOMD/[RXCOMD]	/CALL READ COMMAND ROUTINE	34
35	1412	071337*	7410	SKP		/SKIP IF COMMAND FINISHED	35
36	1413	071340*	5711	JMP I	FREAD	/ELSE TAKE FAILURE RETURN	36
37	1414	071341*	6754	SER		/ANY ERRORS?	37
38	1415	071342*	5346	JMP	FRDOK	/JUMP IF NOT	38
39	1416	071343*	4546	FERR,	JMS I PSSTRING/[SSTRING]	/STORE PATTERN FOR BIG FLOPPY	39
40	1417	071344*	2047	BIGFLPY		/SCREEN DATA POINTER	40
41	1418	071345*	5711	JMP I	FREAD	/TAKE FAILURE RETURN	41
42	1419						42
43	1420	071346*	1053	FRDOK,	TAD RXCMD	/GET SILO EMPTY COMMAND	43
44	1421	071347*	6751	LCD		/LOAD THE COMMAND	44
45	1422	071350*	7303	NL0100	CLL	/SET THE 8-BIT MASK	45
46	1423	071351*	0053	AND	RXCMD	/ISOLATE THE BIT	46
47	1424	071352*	7650	SNA CLA		/SKIP IF SET	47
48	1425	071353*	5361	JMP	TRNCDF	/JUMP IF NOT	48
49	1426	071354*	4642	JMS I	PRXWAIT/(RXWAIT)	/CALL FLAG WAIT ROUTINE	49
50	1427	071355*	5360	JMP	COMFINISH	/TRANSFER FLAG UP AS EXPECTED	50
51	1428	071356*	5343	JMP	FERR	/DONE FLAG CAME UP, ERROR	51
52	1429	071357*	5343	JMP	FERR	/TIME-OUT, ERROR	52
53							53
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1	1431	071360*	6752	COMFINI, XDR	/FINISH THE COMMAND	1
2	1432	071361*	7402	TRNCDF, HLT+.-.	/WILL BE CDF XX INSTRUCTION TO BUFFER FIELD	2
3	1433	071362*	7200	CLA	/CLEAN UP	3
4	1434	071363*	6753	FRDLUP, STR	/TRANSFER FLAG UP?	4
5	1435	071364*	5367	JMP TRYDONE	/NO, TRY DONE FLAG	5
6	1436	071365*	6752	XDR	/YES, GET THE LATEST	6
7	1437	071366*	3410	DCA I XR0	/STORE IT	7
8	1438	071367*	6755	TRYDONE, SDN	/DONE FLAG UP?	8
9	1439	071370*	5363	JMP FRDLUP	/NO, TRY TRANSFER	9
10	1440	071371*	6754	SER	/YES, ANY ERRORS?	10
11	1441	071372*	5325	JMP FNXTSCT	/NO, GO DO NEXT SECTOR	11
12	1442	071373*	5343	JMP FERR	/YES, GO COMPLAIN	12
13	1443					13
14	1444	071374*	0000	ZBLOCK 4	/EMPTY SPACE	14
15						15
16						16
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1	1446	*1400	PAGE		1
2	1447				2
3	1448	/	AUTO-INDEXED POINTER TO PIE CHIP PROGRAMMING TABLE (TABLE FOLLOWS).		3
4	1449				4
5	1450	071400* 1400	PPTABLE,PTABLE-1	/POINTER TO PROGRAMMING TABLE.	5
6	1451				6
7	1452	/	PROGRAMMING NOTE: BY PLACING THIS POINTER HERE, AN EXTRANEIOUS PAGE ZERO		7
8	1453	/	REFERENCE TO THE POINTER IS REQUIRED, INSTEAD OF MERELY A POINTER TO THE TABLE		8
9	1454	/	ON PAGE ZERO ITSELF.		9
10	1455				10
11	1456	/	PIE CHIP PROGRAMMING TABLE.		11
12	1457				12
13	1458	1401	PTABLE= .	/PROGRAMMING TABLE HERE	13
14	1459				14
15	1460	/	FIRST CHIP.		15
16	1461				16
17	1462	071401* 1034	KEY^10+1004	/DEVICE 03	17
18	1463	071402* 1044	TTY^10+1004	/DEVICE 04	18
19	1464	071403* 3130	CLKCON^10+3000	/DEVICE 13	19
20	1465	071404* 3322	PRICON^10+3002	/DEVICE 32	20
21	1466	071405* 3337	PROCON^10+3007	/DEVICE 33	21
22	1467				22
23	1468	/	SECOND CHIP.		23
24	1469				24
25	1470	071406* 1302	PORTIN^10+1002	/DEVICE 30	25
26	1471	071407* 1313	PORTOUT^10+1003	/DEVICE 31	26
27	1472	071410* 3364	PORTCON^10+3004	/DEVICE 36	27
28	1473	071411* 0754	RXCON^10+0004	/DEVICE 75	28
29	1474	071412* 1144	APUCON^10+1004	/DEVICE 14	29
30	1475				30
31	1476	/	THIRD CHIP (CP INTERRUPTS ONLY).		31
32	1477				32
33	1478	071413* 1070	TTYCON^10+1000	/DEVICE 07	33
34	1479	071414* 3060	VIDINT^10+3000	/DEVICE 06	34
35	1480	071415* 7053	KBOCON^10+7003	/DEVICE 05	35
36	1481	071416* 7112	KBICON^10+7002	/DEVICE 11	36
37	1482	071417* 4124	VIDDEV^10+4004	/DEVICE 12	37
38	1483				38
39	1484	071420* 0000	0	/THIS ENDS THE LIST	39
40					40
41					41
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1	1486		/		MEMORY COMPARE ROUTINE.	1
2	1487					2
3	1488		/		THIS ROUTINE CHECKS IF THE CURRENT TEST FIELD MATCHES THE TEST PATTERN IN	3
4	1489		/		EVERY WORD. ALL MATCHING WORDS ARE INVERTED FOR THE NEXT TEST. THE IMMEDIATE	4
5	1490		/		RETURN IS TAKEN ON A MATCH FAILURE; SKIP RETURN IS TAKEN IF ALL WORDS MATCH.	5
6	1491					6
7	1492	071421*	0000	MEMCOMP, .-. .	/MEMORY COMPARE ROUTINE	7
8	1493	071422*	1044	TAD T2	/GET LATEST FIELD	8
9	1494	071423*	0104	AND Z7/[7]	/JUST FIELD BITS	9
10	1495	071424*	7014	R3L	/MOVE UP	10
11	1496	071425*	1122	TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	11
12	1497	071426*	3227	DCA .+1	/STORE IN-LINE	12
13	1498	071427*	7402	HLT+-. .	/WILL BE CDF XX INSTRUCTION	13
14	1499	071430*	1443	MCHKLP, TAD I T1	/GET A WORD	14
15	1500	071431*	1042	TAD T0	/COMPARE TO TEST VALUE	15
16	1501	071432*	7001	IAC	/SHOULD BE ZERO	16
17	1502	071433*	7640	SZA CLA	/SKIP IF IT MATCHES	17
18	1503	071434*	5621	JMP I MEMCOMPARE	/TAKE IMMEDIATE RETURN IF IT FAILS	18
19	1504	071435*	1042	TAD T0	/GET INVERTED TEST VALUE	19
20	1505	071436*	3443	DCA I T1	/STORE BACK	20
21	1506	071437*	1043	TAD T1	/GET ADDRESS	21
22	1507	071440*	1041	TAD TINC	/BUMP UP	22
23	1508	071441*	3043	DCA T1	/STORE BACK	23
24	1509	071442*	2010	ISZ XR0	/DONE ALL IN THIS FIELD	24
25	1510	071443*	5230	JMP MCHKLP	/NO, GO BACK FOR MORE	25
26	1511	071444*	2221	ISZ MEMCOMPARE	/YES, BUMP RETURN ADDRESS	26
27	1512	071445*	5621	JMP I MEMCOMPARE	/TAKE SKIP RETURN TO CALLER	27
28	1513					28
29	1514	071446*	1746	PROWTAB, ROWTABLE-1	/POINTER TO HIGH-ORDER ROW TABLE	29
30	1515	071447*	1663	PREGLIS, REGLIST-1	/POINTER TO REGISTER AND CONTENTS LIST	30
31	1516	071450*	3657	PROWADR, ROWADR-1	/POINTER TO ROWTABLE ADDRESS	31
32	1517					32
33	1518	071451*	0000	ROWLOAD, .-. .	/ROUTINE TO LOAD ROWTABLE AND VIDEO REGISTERS	33
34	1519	071452*	7300	CLA CLL	/CLEAN UP	34
35	1520	071453*	6126	VIDCON	/CLEAR SCREEN CONTROL REGISTER	35
36	1521	071454*	6271	CDF 70	/ENSURE OUR FIELD	36
37	1522	071455*	1250	TAD PROWADR/(ROWADR-1)	/GET ROWTABLE TABLE POINTER	37
38	1523	071456*	3010	DCA XR0	/STASH THE POINTER	38
39	1524	071457*	1246	TAD PROWTABLE/(ROWTABLE-1)	/GET HIGH-ORDER ROW ADDRESS TABLE VALUE	39
40	1525	071460*	3011	DCA XR1	/STASH THE POINTER	40
41	1526	071461*	1411	ROWLUP, TAD I XR1	/GET A LIST ELEMENT	41
42	1527	071462*	7510	SPA	/SKIP IF NOT AT END OF LIST	42
43	1528	071463*	5274	JMP FHERE	/JUMP IF AT END OF LIST	43
44	1529	071464*	7421	MQL	/STASH IT FOR NOW	44
45	1530	071465*	6211	CDF 10	/GOTO TABLE FIELD	45
46	1531	071466*	1153	TAD Z205/[205]	/GET LOW-ORDER BITS VALUE	46
47	1532	071467*	3410	DCA I XR0	/STASH IN TABLE	47
48	1533	071470*	7701	CLA MQA	/GET HIGH-ORDER AND ATTRIBUTES	48
49	1534	071471*	3410	DCA I XR0	/STASH IN TABLE	49
50	1535	071472*	6271	CDF 70	/BACK TO OUR FIELD	50
51	1536	071473*	5261	JMP ROWLUP	/KEEP GOING	51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	1538	071474*	7300	FHERE,	CLA CLL		/CLEAN UP	1
2	1539	071475*	1247		TAD	PREGLIST/(REGLIST-1)	/POINT TO	2
3	1540	071476*	3010		DCA	XR0	/REGISTER LIST	3
4	1541	071477*	6271	REGLUP,	CDF	70	/BACK TO OUR FIELD	4
5	1542	071500*	1410		TAD I	XR0	/GET A LIST ITEM	5
6	1543	071501*	7510		SPA		/SKIP IF NOT END OF LIST	6
7	1544	071502*	5311		JMP	REGDONE	/JUMP IF DONE	7
8	1545	071503*	6122		LSCREG		/LOAD REGISTER SELECT	8
9	1546	071504*	7200		CLA		/CLEAN UP	9
10	1547	071505*	1410		TAD I	XR0	/GET THE VALUE	10
11	1548	071506*	6124		VLOAD		/LOAD THE REGISTER WITH DESIRED VALUE	11
12	1549	071507*	7300		CLA CLL		/CLEAN UP	12
13	1550	071510*	5277		JMP	REGLUP	/KEEP GOING	13
14	1551							14
15	1552	071511*	7300	REGDONE,	CLA CLL		/CLEAN UP	15
16	1553	071512*	5651		JMP I	ROWLOAD	/RETURN	16
17	1554							17
18	1555			/		ROUTINE TO CLEAR CP FIELD 2 AND 3; ALSO CLEARS HIGH-ORDER HALF OF CP 13705		18
19	1556			/		AND 13707 TO MAKE THOSE LINES HAVE NORMAL ATTRIBUTES.		19
20	1557							20
21	1558	071513*	0000	CLR23,	...		/CLEAR FIELD 2, FIELD 3 ROUTINE	21
22	1559	071514*	7200		CLA		/CLEAN UP	22
23	1560	071515*	3044		DCA	T2	/CLEAR EVEN/ODD FIELD FLAG	23
24	1561	071516*	7240	ZERNXT,	NL7777		/SETUP FOR STORING	24
25	1562	071517*	3010		DCA	XR0	/INTO LOCATION 0000 OF THE LATEST FIELD	25
26	1563	071520*	4340		JMS	CDFSETUP	/SETUP PROPER FIELD	26
27	1564	071521*	1127		TAD	Z7760/[-20]	/SETUP THE	27
28	1565	071522*	3045		DCA	T3	/BUFFER COUNT	28
29	1566	071523*	1131	BUFLUP,	TAD	Z7400/[-400]	/SETUP FOR TWO PAGES WORTH	29
30	1567	071524*	3046		DCA	T4	/IN EACH BUFFER	30
31	1568	071525*	3410	ZERLUP,	DCA I	XR0	/CLEAR A WORD	31
32	1569	071526*	2046		ISZ	T4	/DONE YET?	32
33	1570	071527*	5325		JMP	ZERLUP	/NO, GO BACK	33
34	1571	071530*	2045		ISZ	T3	/DONE ENOUGH BUFFERS?	34
35	1572	071531*	5323		JMP	BUFLUP	/NO, GO BACK	35
36	1573	071532*	1044		TAD	T2	/GET FLAG	36
37	1574	071533*	7640		SZA CLA		/SKIP IF ZERO	37
38	1575	071534*	5350		JMP	CLRFINISH	/ELSE FINISH THERE	38
39	1576	071535*	7240		NL7777		/SET THE FLAG	39
40	1577	071536*	3044		DCA	T2	/FOR NEXT FIELD	40
41	1578	071537*	5316		JMP	ZERNXT	/GO DO IT AGAIN	41
42	1579							42
43	1580	071540*	0000	CDFSETU,	...		/CDF SETTING ROUTINE	43
44	1581	071541*	1044		TAD	T2	/GET THE FLAG	44
45	1582	071542*	7640		SZA CLA		/SKIP IF EVEN FIELD	45
46	1583	071543*	5346		JMP	SETODD	/JUMP IF ODD	46
47	1584	071544*	6221		CDF	20	/SETUP FOR FIELD 2	47
48	1585	071545*	5740		JMP I	CDFSETUP	/RETURN	48
49	1586							49
50	1587	071546*	6231	SETODD,	CDF	30	/SETUP FOR FIELD 3	50
51	1588	071547*	5740		JMP I	CDFSETUP	/RETURN	51
52	1589							52
53	1590	071550*	4752	CLRFINI,	JMS I	PCLR2/(CLR2)	/CLEAR ATTRIBUTE BITS OF 13705 AND 13707	53
54	1591	071551*	5713		JMP I	CLR23	/RETURN	54
55								55
56								56
57								57
58								58

1	1593	071552*	2765	PCLR2, CLR2	/POINTER TO CLR2 ROUTINE	1
2	1594	071553*	1164	PTIMOUT, TIMEOUT	/POINTER TO TIME-OUT ROUTINE	2
3	1595					3
4	1596	071554*	0000	FBLINK, .-. .	/BIG FLOPPY DISPLAY ROUTINE	4
5	1597	071555*	7315	NL0010 CLL	/SET EXTENDED CHARACTER MODE	5
6	1598	071556*	6126	VIDCON	/LOAD NEW CONTROL VALUE	6
7	1599	071557*	7200	CLA	/CLEAN UP	7
8	1600	071560*	1370	TAD BL40/(" &177)	/GET SPACE CODE	8
9	1601	071561*	4546	JMS I PSSTRING/[SSTRING]	/CLEAR ALL ITEMS OF	9
10	1602	071562*	2047	BIGFLPY	/THE BIG FLOPPY PATTERN	10
11	1603	071563*	4753	JMS I PTIMOUT/(TIMEOUT)	/GO WAIT FOR AWHILE	11
12	1604	071564*	4546	JMS I PSSTRING/[SSTRING]	/GO DISPLAY ALL ITEMS OF	12
13	1605	071565*	2047	BIGFLPY	/THE BIG FLOPPY PATTERN	13
14	1606	071566*	4753	JMS I PTIMOUT/(TIMEOUT)	/GO WAIT FOR AWHILE	14
15	1607	071567*	5754	JMP I FBLINK	/RETURN	15
16	1608					16
17	1609	071570*	0040	BL40, " &177	/CONSTANT 0040	17
18	1610					18
19	1611	071571*	0000	ZBLOCK 7	/EMPTY SPACE	19
20						20
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1	1613	*1600	PAGE			1
2	1614					2
3	1615	/	ROUTINE TO VALIDATE THE CONTENTS OF TRACK ONE, SECTOR ONE AS A BOOTABLE			3
4	1616	/	FLOPPY. THE FIRST FEW BYTES OF THIS SECTOR HAVE TO BE A VALID PREFERENCE			4
5	1617	/	SETTING BLOCK IN ORDER TO ALLOW FURTHER BOOTING. NON-SKIP RETURN IS TAKEN IF			5
6	1618	/	THE BYTES ARE BAD, SKIP RETURN IF THEY ARE GOOD.			6
7	1619					7
8	1620	071600* 0000	VALID8, .-. /TRACK ONE VALIDATION ROUTINE			8
9	1621	071601* 6271	CDF 70 /RESET TO OUR FIELD (WHY?)			9
10	1622	071602* 7240	NL7777 /SETUP POINTER TO 0000			10
11	1623	071603* 3010	DCA XR0 /STASH THE POINTER			11
12	1624	071604* 6201	CDF 00 /GOTO THE DATA FIELD			12
13	1625	071605* 1410	TAD I XR0 /GET THE FIRST BYTE			13
14	1626	071606* 1410	TAD I XR0 /ADD TO THE SECOND BYTE			14
15	1627	071607* 6271	CDF 70 /BACK TO OUR FIELD			15
16	1628					16
17	1629	/	THE SUM SHOULD BE ZERO MEANING THE TWO WORDS ARE THE TWO'S COMPLEMENT OF EACH			17
18	1630	/	OTHER (OR ARE BOTH ZERO).			18
19	1631					19
20	1632	071610* 7640	SZA CLA /SKIP IF OK			20
21	1633	071611* 5600	JMP I VALID8 /TAKE BAD RETURN IF NOT			21
22	1634	071612* 6201	CDF 00 /GOTO THE DATA FIELD			22
23	1635					23
24	1636	/	THE NEXT BYTE IS TREATED AS A POINTER FOR LATER.			24
25	1637					25
26	1638	071613* 1410	TAD I XR0 /GET THE NEXT BYTE			26
27	1639	071614* 7104	CLL RAL /*2			27
28	1640	071615* 1010	TAD XR0 /ADD 2			28
29	1641	071616* 3011	DCA XR1 /STASH THE POINTER FOR LATER			29
30	1642					30
31	1643	/	THE NEXT BYTE HAS TO BE 001.			31
32	1644					32
33	1645	071617* 7240	NL7777 /GET -1			33
34	1646	071620* 1410	TAD I XR0 /ADD ON NEXT BYTE			34
35	1647	071621* 6271	CDF 70 /BACK TO OUR FIELD			35
36	1648	071622* 7640	SZA CLA /SKIP IF IT MATCHES			36
37	1649	071623* 5600	JMP I VALID8 /ELSE TAKE BAD RETURN			37
38						38
39						39
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1	1651	/	THE REST OF THE TESTED BYTES ARE POINTED TO THE THIRD BYTE. THE FOUR BYTES				1
2	1652	/	SHOULD SUM TO 377. THE POINTER WAS INCREMENTED BY TWO INITIALLY, SO WE MAY				2
3	1653	/	HAVE TO SUBTRACT TWO TO LOCATE THE PROPER FOUR BYTE SEQUENCE.				3
4	1654						4
5	1655	071624* 3012	VALAGN, DCA	XR2	/SET TO 0000 (OR 7776)		5
6	1656	071625* 7240	NL7777		/BACKUP		6
7	1657	071626* 1011	TAD	XR1	/ADD ON POINTER		7
8	1658	071627* 1012	TAD	XR2	/ADD 0000 OR 7776		8
9	1659	071630* 3010	DCA	XR0	/STASH THE POINTER		9
10	1660	071631* 6201	CDF	00	/GOTO THE DATA FIELD		10
11	1661	071632* 1410	TAD I	XR0	/\		11
12	1662	071633* 1410	TAD I	XR0	/ \GET THE NEXT		12
13	1663	071634* 1410	TAD I	XR0	/ /FOUR BYTES		13
14	1664	071635* 1410	TAD I	XR0	//		14
15	1665	071636* 6271	CDF	70	/BACK TO OUR FIELD		15
16	1666	071637* 7001	IAC		/ADD ONE		16
17	1667	071640* 0115	AND	Z377/[377]	/JUST LOW-ORDER BITS		17
18	1668	071641* 7640	SZA CLA		/SKIP IF ALL TOGETHER IT'S ZERO		18
19	1669	071642* 5257	JMP	VALALT	/JUMP IF NOT		19
20	1670						20
21	1671	/	HAVING LOCATED THE CORRECT FOUR BYTE SEQUENCE, CONFIRM THAT THE FIRST TWO HAVE				21
22	1672	/	BIT[8] SET.				22
23	1673						23
24	1674	071643* 7240	NL7777		/BACKUP		24
25	1675	071644* 1011	TAD	XR1	/ADD ON POINTER		25
26	1676	071645* 1012	TAD	XR2	/ADD ON 0000 OR 7776		26
27	1677	071646* 3010	DCA	XR0	/STASH THE POINTER		27
28	1678	071647* 6201	CDF	00	/GOTO THE DATA FIELD		28
29	1679	071650* 7315	NL0010	CLL	/SET THE BIT MASK		29
30	1680	071651* 0410	AND I	XR0	/GET THE FIRST ONE'S BIT		30
31	1681	071652* 0410	AND I	XR0	/AND THE SECOND ONE'S BIT		31
32	1682	071653* 6271	CDF	70	/BACK TO OUR FIELD		32
33	1683	071654* 7640	SZA CLA		/SKIP IF OFF		33
34	1684	071655* 2200	ISZ	VALID8	/BUMP TO GOOD RETURN		34
35	1685	071656* 5600	JMP I	VALID8	/RETURN EITHER WAY		35
36	1686						36
37	1687	071657* 1012	VALALT, TAD	XR2	/GET THE OFFSET		37
38	1688	071660* 7640	SZA CLA		/SKIP IF ZERO		38
39	1689	071661* 5600	JMP I	VALID8	/RETURN IF 7776 WAS ALSO UNSUCCESSFUL		39
40	1690	071662* 7344	NL7776		/SET -2 VALUE		40
41	1691	071663* 5224	JMP	VALAGN	/TRY IT AGAIN		41
42							42
43							43
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1	1693	/	THIS IS THE LIST OF VALUES USED WITH LSCREG AND VLOAD IN PAIRS.	1
2	1694			2
3	1695	071664* 0026	REGLIST, 26	3
4	1696	071665* 0000	000	4
5	1697	071666* 0000	00	5
6	1698	071667* 0143	144-1	6
7	1699	071670* 0001	01	7
8	1700	071671* 0117	120-1	8
9	1701	071672* 0002	02	9
10	1702	071673* 0016	016	10
11	1703	071674* 0003	03	11
12	1704	071675* 0007	007	12
13	1705	071676* 0004	04	13
14	1706	071677* 0003	003	14
15	1707	071700* 0005	05	15
16	1708	071701* 0025	025	16
17	1709	071702* 0006	06	17
18	1710	071703* 0022	022	18
19	1711	071704* 0007	07	19
20	1712	071705* 0027	30-1	20
21	1713	071706* 0010	10	21
22	1714	071707* 0051	051	22
23	1715	071710* 0011	11	23
24	1716	071711* 0011	011	24
25	1717	071712* 0012	12	25
26	1718	071713* 0160	160	26
27	1719	071714* 0013	13	27
28	1720	071715* 0010	010	28
29	1721	071716* 0014	0014	29
30	1722	071717* 0260	ROWADR&377	30
31	1723	071720* 0015	0015	31
32	1724	071721* 0227	ROWADR%400+0220	32
33	1725	071722* 0016	16	33
34	1726	071723* 0000	000	34
35	1727	071724* 0017	17	35
36	1728	071725* 0000	000	36
37	1729	071726* 0020	20	37
38	1730	071727* 0077	077	38
39	1731	071730* 0021	21	39
40	1732	071731* 0000	000	40
41	1733	071732* 0022	22	41
42	1734	071733* 0030	030	42
43	1735	071734* 0027	27	43
44	1736	071735* 0000	000	44
45	1737	071736* 0030	30	45
46	1738	071737* 0000	000	46
47	1739	071740* 0031	31	47
48	1740	071741* 0000	000	48
49	1741	071742* 0032	32	49
50	1742	071743* 0100	100	50
51	1743	071744* 0025	25	51
52	1744	071745* 0000	000	52
53	1745	071746* 4000	4000	53
54				54
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1	1747	/	THIS TABLE IS USED TO LOAD THE HIGH-ORDER BYTES OF THE CONTIGUOUS ROW ADDRESS	1
2	1748	/	TABLE. THE FORMAT IS TWO ATTRIBUTE BITS AND SIX ADDRESS BITS. THE LOW-ORDER	2
3	1749	/	ADDRESS BYTE IS ALWAYS 205. THE ATTRIBUTE BITS ARE: 00=NORMAL DATA ROW,	3
4	1750	/	01=SINGLE HEIGHT, DOUBLE WIDTH; 10=TOP OF DOUBLE HEIGHT, DOUBLE WIDTH;	4
5	1751	/	11=BOTTOM OF DOUBLE HEIGHT, DOUBLE WIDTH.	5
6	1752			6
7	1753	071747* 0141	ROWTABL, 0141 /LINE 00 DATA STARTS AT 20605 (DOUBLE WIDTH)	7
8	1754	071750* 0042	0042 /LINE 01 DATA STARTS AT 21205	8
9	1755	071751* 0043	0043 /LINE 02 DATA STARTS AT 21605	9
10	1756	071752* 0044	0044 /LINE 03 DATA STARTS AT 22205	10
11	1757	071753* 0045	0045 /LINE 04 DATA STARTS AT 22605	11
12	1758	071754* 0046	0046 /LINE 05 DATA STARTS AT 23205	12
13	1759	071755* 0047	0047 /LINE 06 DATA STARTS AT 23605	13
14	1760	071756* 0050	0050 /LINE 07 DATA STARTS AT 24205	14
15	1761	071757* 0051	0051 /LINE 08 DATA STARTS AT 24605	15
16	1762	071760* 0052	0052 /LINE 09 DATA STARTS AT 25205	16
17	1763	071761* 0053	0053 /LINE 10 DATA STARTS AT 25605	17
18	1764	071762* 0054	0054 /LINE 11 DATA STARTS AT 26205	18
19	1765	071763* 0055	0055 /LINE 12 DATA STARTS AT 26605	19
20	1766	071764* 0056	0056 /LINE 13 DATA STARTS AT 27205	20
21	1767	071765* 0057	0057 /LINE 14 DATA STARTS AT 27605	21
22	1768	071766* 0072	0072 /LINE 15 DATA STARTS AT 35205	22
23	1769	071767* 0061	0061 /LINE 16 DATA STARTS AT 30605	23
24	1770	071770* 0062	0062 /LINE 17 DATA STARTS AT 31205	24
25	1771	071771* 0063	0063 /LINE 18 DATA STARTS AT 31605	25
26	1772	071772* 0064	0064 /LINE 19 DATA STARTS AT 32205	26
27	1773	071773* 0065	0065 /LINE 20 DATA STARTS AT 32605	27
28	1774	071774* 0066	0066 /LINE 21 DATA STARTS AT 33205	28
29	1775	071775* 0067	0067 /LINE 22 DATA STARTS AT 33605	29
30	1776	071776* 0070	0070 /LINE 23 DATA STARTS AT 34205	30
31	1777	071777* 7777	7777 /THIS ENDS THE LIST	31
32				32
33				33
34				34
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1	1779		*2000	PAGE			1
2	1780						2
3	1781	/		STORE DISPLAY STRING ROUTINE.	CALL WITH CLEAR AC TO DISPLAY VALUES IN		3
4	1782	/		ARGUMENT LIST OR WITH 0040 IN AC TO BLANK THE FIELD.	IN-LINE ARGUMENT IS A		4
5	1783	/		POINTER TO THE ARGUMENT LIST. LIST FORMAT IS 0000 IF ITEMS ARE FOR FIELD 2,			5
6	1784	/		OR 4000 IF ITEMS ARE FOR FIELD 3. RELATIVE STORAGE ADDRESS FOR SUBSEQUENT			6
7	1785	/		ITEMS FOLLOWS THE 0000 OR 4000. ALL OTHER ITEMS ARE SEQUENTIAL VALUES TO			7
8	1786	/		STORE INTO THE BUFFER STARTING AT THE GIVEN ADDRESS. ALL ADDRESSES ARE			8
9	1787	/		RELATIVE TO THE ABSOLUTE VALUE 0205, NOT THE ACTUAL ROW ADDRESS WHICH IS			9
10	1788	/		ALWAYS SOME NUMBER OF WHOLE PAGES DIFFERENT. ADDRESSES NEAR THE END OF A			10
11	1789	/		FIELD WILL NOT BE CORRECTED TO THE NEXT FIELD. THE ARGUMENT LIST ENDS WITH			11
12	1790	/		7777. CONTROL RETURNS PAST THE IN-LINE POINTER ARGUMENT.			12
13	1791						13
14	1792	072000*	0000	SSTRING, .-. .	/STORE STRING FOR DISPLAY ROUTINE		14
15	1793	072001*	6271	CDF 70	/ENSURE OUR FIELD		15
16	1794	072002*	3246	DCA SPACSW	/SAVE PASSED VALUE		16
17	1795	072003*	7240	NL7777	/BACKUP FOR AUTO-INDEX		17
18	1796	072004*	1600	TAD I SSTRING	/GET ARGUMENT ADDRESS		18
19	1797	072005*	2200	ISZ SSTRING	/BUMP PAST ARGUMENT		19
20	1798	072006*	3017	DCA XR7	/STASH THE POINTER		20
21	1799	072007*	1417	SSLUP, TAD I XR7	/GET A WORD		21
22	1800	072010*	7421	SQL	/STASH IT FOR NOW		22
23	1801	072011*	7701	CLA MQA	/GET IT BACK		23
24	1802	072012*	7001	IAC	/ADD ONE FOR TEST		24
25	1803	072013*	7650	SNA CLA	/SKIP IF NOT AT END OF LIST (7777)		25
26	1804	072014*	5600	JMP I SSTRING	/RETURN IF AT END		26
27	1805	072015*	7701	CLA MQA	/GET THE VALUE AGAIN		27
28	1806	072016*	0120	AND Z3777/[3777]	/IGNORE HIGH-ORDER BIT FOR NOW		28
29	1807	072017*	7650	SNA CLA	/SKIP IF SOMETHING SET		29
30	1808	072020*	5232	JMP SSADR	/JUMP IF NOT		30
31	1809	072021*	1246	TAD SPACSW	/GET PASSED ARGUMENT		31
32	1810	072022*	7450	SNA	/SKIP IF IT WAS SET (A SPACE CHARACTER)		32
33	1811	072023*	7701	CLA MQA	/USE PASSED VALUE		33
34	1812	072024*	7402	CDFINST, HLT+. -. .	/WILL BE CDF XX INSTRUCTION		34
35	1813	072025*	3447	DCA I T5	/STORE IN BUFFER		35
36	1814	072026*	2047	ISZ T5	/BUMP TO NEXT		36
37	1815	072027*	7000	NOP	/JUST IN CASE IT SKIPS		37
38	1816	072030*	6271	CDF 70	/BACK TO OUR FIELD		38
39	1817	072031*	5207	JMP SSLUP	/KEEP GOING		39
40	1818						40
41	1819	072032*	1417	SSADR, TAD I XR7	/GET NEXT WORD AFTER FLAG WORD		41
42	1820	072033*	3047	DCA T5	/STASH AS ADDRESS		42
43	1821	072034*	7701	CLA MQA	/GET THE WORD		43
44	1822	072035*	7710	SPA CLA	/SKIP IF IT FLAG WORD IS ZERO		44
45	1823	072036*	1105	TAD Z10/[10]	/MAKE IT CDF 30		45
46	1824	072037*	1245	TAD CDF20/(CDF 20)	/GET CDF 20		46
47	1825	072040*	3224	DCA CDFINST	/STORE CDF 20 OR CDF 30 IN-LINE		47
48	1826	072041*	1047	TAD T5	/GET ADDRESS WORD		48
49	1827	072042*	1153	TAD Z205/[205]	/ADD OFFSET		49
50	1828	072043*	3047	DCA T5	/STORE BACK		50
51	1829	072044*	5207	JMP SSLUP	/KEEP GOING		51
52	1830						52
53	1831	072045*	6221	CDF20, CDF 20	/CONSTANT 6221		53
54	1832	072046*	0000	SPACSW, .-. .	/PASSED SPACE PARAMETER		54
55							55
56							56
57							57
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1	1834	/	DATA PATTERN THAT LOOKS LIKE A BIG FLOPPY (USED FOR ERROR DISPLAY).	1
2	1835			2
3	1836	072047* 0000	BIGFLPY, 0000 /FIELD 2	3
4	1837	072050* 1445	1445 /ADDRESS	4
5	1838	072051* 0015	0015	5
6	1839	072052* 0022	0022	6
7	1840	072053* 0022	0022	7
8	1841	072054* 0022	0022	8
9	1842	072055* 0022	0022	9
10	1843	072056* 0022	0022	10
11	1844	072057* 0022	0022	11
12	1845	072060* 0022	0022	12
13	1846	072061* 0022	0022	13
14	1847	072062* 0022	0022	14
15	1848	072063* 0022	0022	15
16	1849	072064* 0022	0022	16
17	1850	072065* 0022	0022	17
18	1851	072066* 0022	0022	18
19	1852	072067* 0022	0022	19
20	1853	072070* 0022	0022	20
21	1854	072071* 0022	0022	21
22	1855	072072* 0022	0022	22
23	1856	072073* 0022	0022	23
24	1857	072074* 0022	0022	24
25	1858	072075* 0022	0022	25
26	1859	072076* 0022	0022	26
27	1860	072077* 0022	0022	27
28	1861	072100* 0022	0022	28
29	1862	072101* 0022	0022	29
30	1863	072102* 0022	0022	30
31	1864	072103* 0014	0014	31
32	1865	072104* 0000	0000 /FIELD 2	32
33	1866	072105* 2045	2045 /ADDRESS	33
34	1867	072106* 0031	0031	34
35	1868	072107* 0000	0000 /FIELD 2	35
36	1869	072110* 2070	2070 /ADDRESS	36
37	1870	072111* 0015	0015	37
38	1871	072112* 0022	0022	38
39	1872	072113* 0022	0022	39
40	1873	072114* 0022	0022	40
41	1874	072115* 0022	0022	41
42	1875	072116* 0022	0022	42
43	1876	072117* 0014	0014	43
44	1877	072120* 0031	0031	44
45	1878	072121* 0000	0000 /FIELD 2	45
46	1879	072122* 2445	2445 /ADDRESS	46
47	1880	072123* 0031	0031	47
48	1881	072124* 0000	0000 /FIELD 2	48
49	1882	072125* 2470	2470 /ADDRESS	49
50	1883	072126* 0031	0031	50
51	1884	072127* 0000	0000 /FIELD 2	51
52	1885	072130* 2476	2476 /ADDRESS	52
53	1886	072131* 0031	0031	53
54	1887	072132* 0031	0031	54
55				55
56				56
57				57
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1	1889	072133*	0000	0000	/FIELD 2	1
2	1890	072134*	3045	3045	/ADDRESS	2
3	1891	072135*	0031	0031		3
4	1892	072136*	0000	0000	/FIELD 2	4
5	1893	072137*	3070	3070	/ADDRESS	5
6	1894	072140*	0031	0031		6
7	1895	072141*	0000	0000	/FIELD 2	7
8	1896	072142*	3076	3076	/ADDRESS	8
9	1897	072143*	0031	0031		9
10	1898	072144*	0031	0031		10
11	1899	072145*	0000	0000	/FIELD 2	11
12	1900	072146*	3445	3445	/ADDRESS	12
13	1901	072147*	0031	0031		13
14	1902	072150*	0000	0000	/FIELD 2	14
15	1903	072151*	3462	3462	/ADDRESS	15
16	1904	072152*	4025	4025		16
17	1905	072153*	0000	0000	/FIELD 2	17
18	1906	072154*	3470	3470	/ADDRESS	18
19	1907	072155*	0016	0016		19
20	1908	072156*	0022	0022		20
21	1909	072157*	0022	0022		21
22	1910	072160*	0022	0022		22
23	1911	072161*	0022	0022		23
24	1912	072162*	0022	0022		24
25	1913	072163*	0013	0013		25
26	1914	072164*	0031	0031		26
27	1915	072165*	0000	0000	/FIELD 2	27
28	1916	072166*	4045	4045	/ADDRESS	28
29	1917	072167*	0031	0031		29
30	1918	072170*	0000	0000	/FIELD 2	30
31	1919	072171*	4060	4060	/ADDRESS	31
32	1920	072172*	4026	4026		32
33	1921	072173*	4031	4031		33
34	1922	072174*	0000	0000	/FIELD 2	34
35	1923	072175*	4063	4063	/ADDRESS	35
36	1924	072176*	4031	4031		36
37	1925	072177*	4026	4026		37
38	1926	072200*	0000	0000	/FIELD 2	38
39	1927	072201*	4072	4072	/ADDRESS	39
40	1928	072202*	0015	0015		40
41	1929	072203*	0022	0022		41
42	1930	072204*	0022	0022		42
43	1931	072205*	0022	0022		43
44	1932	072206*	0014	0014		44
45	1933	072207*	0031	0031		45
46	1934	072210*	0000	0000	/FIELD 2	46
47	1935	072211*	4445	4445	/ADDRESS	47
48	1936	072212*	0031	0031		48
49	1937	072213*	0050	0050		49
50	1938	072214*	0440	" +200		50
51	1939	072215*	0440	" +200		51
52	1940	072216*	0440	" +200		52
53	1941	072217*	0440	" +200		53
54	1942	072220*	0440	" +200		54
55	1943	072221*	0051	0051		55
56						56
57						57
58						58

1	1945	072222*	0000	0000	/FIELD 2	1
2	1946	072223*	4460	4460	/ADDRESS	2
3	1947	072224*	4027	4027		3
4	1948	072225*	0000	0000	/FIELD 2	4
5	1949	072226*	4464	4464	/ADDRESS	5
6	1950	072227*	4027	4027		6
7	1951	072230*	0000	0000	/FIELD 2	7
8	1952	072231*	4472	4472	/ADDRESS	8
9	1953	072232*	0031	0031		9
10	1954	072233*	0000	0000	/FIELD 2	10
11	1955	072234*	4476	4476	/ADDRESS	11
12	1956	072235*	0031	0031		12
13	1957	072236*	0031	0031		13
14	1958	072237*	0000	0000	/FIELD 2	14
15	1959	072240*	5045	5045	/ADDRESS	15
16	1960	072241*	0031	0031		16
17	1961	072242*	0000	0000	/FIELD 2	17
18	1962	072243*	5061	5061	/ADDRESS	18
19	1963	072244*	4030	4030		19
20	1964	072245*	4027	4027		20
21	1965	072246*	4030	4030		21
22	1966	072247*	0000	0000	/FIELD 2	22
23	1967	072250*	5072	5072	/ADDRESS	23
24	1968	072251*	0031	0031		24
25	1969	072252*	0000	0000	/FIELD 2	25
26	1970	072253*	5076	5076	/ADDRESS	26
27	1971	072254*	0031	0031		27
28	1972	072255*	0031	0031		28
29	1973	072256*	0000	0000	/FIELD 2	29
30	1974	072257*	5445	5445	/ADDRESS	30
31	1975	072260*	0031	0031		31
32	1976	072261*	0000	0000	/FIELD 2	32
33	1977	072262*	5460	5460	/ADDRESS	33
34	1978	072263*	0007	0007		34
35	1979	072264*	0000	0000	/FIELD 2	35
36	1980	072265*	5472	5472	/ADDRESS	36
37	1981	072266*	0031	0031		37
38	1982	072267*	0000	0000	/FIELD 2	38
39	1983	072270*	5476	5476	/ADDRESS	39
40	1984	072271*	0031	0031		40
41	1985	072272*	0031	0031		41
42	1986	072273*	0000	0000	/FIELD 2	42
43	1987	072274*	6045	6045	/ADDRESS	43
44	1988	072275*	0031	0031		44
45	1989	072276*	0000	0000	/FIELD 2	45
46	1990	072277*	6072	6072	/ADDRESS	46
47	1991	072300*	0031	0031		47
48	1992	072301*	0000	0000	/FIELD 2	48
49	1993	072302*	6076	6076	/ADDRESS	49
50	1994	072303*	0031	0031		50
51	1995	072304*	0031	0031		51
52	1996	072305*	0000	0000	/FIELD 2	52
53	1997	072306*	6445	6445	/ADDRESS	53
54	1998	072307*	0031	0031		54
55						55
56						56
57						57
58						58

1	2000	072310*	0000	0000	/FIELD 2	1
2	2001	072311*	6472	6472	/ADDRESS	2
3	2002	072312*	0031	0031		3
4	2003	072313*	0000	0000	/FIELD 2	4
5	2004	072314*	6476	6476	/ADDRESS	5
6	2005	072315*	0031	0031		6
7	2006	072316*	0031	0031		7
8	2007	072317*	0000	0000	/FIELD 2	8
9	2008	072320*	7045	7045	/ADDRESS	9
10	2009	072321*	0031	0031		10
11	2010	072322*	0000	0000	/FIELD 2	11
12	2011	072323*	7070	7070	/ADDRESS	12
13	2012	072324*	0015	0015		13
14	2013	072325*	0014	0014		14
15	2014	072326*	0016	0016		15
16	2015	072327*	0022	0022		16
17	2016	072330*	0022	0022		17
18	2017	072331*	0022	0022		18
19	2018	072332*	0013	0013		19
20	2019	072333*	0031	0031		20
21	2020	072334*	0000	0000	/FIELD 2	21
22	2021	072335*	7445	7445	/ADDRESS	22
23	2022	072336*	0016	0016		23
24	2023	072337*	0022	0022		24
25	2024	072340*	0022	0022		25
26	2025	072341*	0022	0022		26
27	2026	072342*	0022	0022		27
28	2027	072343*	0022	0022		28
29	2028	072344*	0022	0022		29
30	2029	072345*	0022	0022		30
31	2030	072346*	0022	0022		31
32	2031	072347*	0022	0022		32
33	2032	072350*	0022	0022		33
34	2033	072351*	0022	0022		34
35	2034	072352*	0022	0022		35
36	2035	072353*	0022	0022		36
37	2036	072354*	0022	0022		37
38	2037	072355*	0022	0022		38
39	2038	072356*	0022	0022		39
40	2039	072357*	0022	0022		40
41	2040	072360*	0022	0022		41
42	2041	072361*	0013	0013		42
43	2042	072362*	0016	0016		43
44	2043	072363*	0022	0022		44
45	2044	072364*	0022	0022		45
46	2045	072365*	0022	0022		46
47	2046	072366*	0022	0022		47
48	2047	072367*	0022	0022		48
49	2048	072370*	0013	0013		49
50	2049	072371*	7777	7777	/END OF LIST	50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	2051		/	ERROR NUMBER DISPLAY LIST.		1
2	2052					2
3	2053	072372*	0000	EDISPLI, 0000	/FIELD 2	3
4	2054	072373*	5422	5422	/ADDRESS	4
5	2055	072374*	0000	..	/HIGH-ORDER ERROR DIGIT	5
6	2056	072375*	0000	..	/LOW-ORDER ERROR DIGIT	6
7	2057	072376*	0000	0000	/FIELD 2	7
8	2058	072377*	6022	6022	/ADDRESS	8
9	2059	072400*	0040	J00, 0040+..	/HIGH-ORDER ERROR DIGIT	9
10	2060	072401*	0040	J01, 0040+..	/LOW-ORDER ERROR DIGIT	10
11	2061	072402*	7777	7777	/END OF LIST	11
12	2062					12
13	2063		/	NON-ERROR (DECMATE II LOGO) DISPLAY LIST.		13
14	2064					14
15	2065	072403*	0000	DMLOGO, 0000	/FIELD 2	15
16	2066	072404*	5415	5415	/ADDRESS	16
17	2067	072405*	0440	" +200		17
18	2068	072406*	0504	"D+200		18
19	2069	072407*	0505	"E+200		19
20	2070	072410*	0503	"C+200		20
21	2071	072411*	0555	"M+240		21
22	2072	072412*	0541	"A+240		22
23	2073	072413*	0564	"T+240		23
24	2074	072414*	0545	"E+240		24
25	2075	072415*	0440	" +200		25
26	2076	072416*	0511	"I+200		26
27	2077	072417*	0511	"I+200		27
28	2078	072420*	0440	" +200		28
29	2079	072421*	0000	0000	/FIELD 2	29
30	2080	072422*	6015	6015	/ADDRESS	30
31	2081	072423*	0440	" +200		31
32	2082	072424*	0504	"D+200		32
33	2083	072425*	0505	"E+200		33
34	2084	072426*	0503	"C+200		34
35	2085	072427*	0555	"M+240		35
36	2086	072430*	0541	"A+240		36
37	2087	072431*	0564	"T+240		37
38	2088	072432*	0545	"E+240		38
39	2089	072433*	0440	" +200		39
40	2090	072434*	0511	"I+200		40
41	2091	072435*	0511	"I+200		41
42	2092	072436*	0440	" +200		42
43	2093	072437*	7777	7777	/END OF LIST	43
44	2094					44
45	2095	072440*	0023	L23, 23	/CONSTANT 0023	45
46	2096	072441*	0247	L247, 247	/CONSTANT 0247	46
47	2097	072442*	0202	L202, 202	/CONSTANT 0202	47
48	2098	072443*	0460	L460, "0+200	/CONSTANT 0460	48
49						49
50						50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	2100	072444*	0000	ENDUP,	...	/FINISH UP ROUTINE	1
2	2101	072445*	7300		CLA CLL	/CLEAN UP	2
3	2102	072446*	4730		JMS I PSET2/(SET2)	/SET ATTRIBUTE BITS IN 13705, 13707	3
4	2103	072447*	1054		TAD	ESTATUS	4
5	2104	072450*	7650		SNA CLA	/SKIP IF ANY PROBLEMS	5
6	2105	072451*	5315		JMP	NOERRORS	6
7	2106	072452*	3010		DCA	XR0	7
8	2107	072453*	1054		TAD	ESTATUS	8
9	2108	072454*	6121		APTSKP	/AUTOMATIC PRODUCT TEST?	9
10	2109	072455*	7410		SKP	/SKIP IF NOT	10
11	2110	072456*	5735		JMP I	PAPT2/(APTR02)	11
12	2111	072457*	1126	SUBLUP,	TAD	Z7766/[7766]	12
13	2112	072460*	7510		SPA	/GONE TOO FAR?	13
14	2113	072461*	5264		JMP	ADDFIX	14
15	2114	072462*	2010		ISZ	XR0	15
16	2115	072463*	5257		JMP	SUBLUP	16
17	2116					/KEEP GOING	17
18	2117	072464*	1106	ADDFIX,	TAD	Z12/[12]	18
19	2118	072465*	1243		TAD	L460/("0+200)	19
20	2119	072466*	3731		DCA I	LOW1/(EDISPLIST+3)	20
21	2120	072467*	1731		TAD I	LOW1/(EDISPLIST+3)	21
22	2121	072470*	3732		DCA I	LOW2/(EDISPLIST+7)	22
23	2122	072471*	1010		TAD	XR0	23
24	2123	072472*	1243		TAD	L460/("0+200)	24
25	2124	072473*	3733		DCA I	HIGH1/(EDISPLIST+2)	25
26	2125	072474*	1733		TAD I	HIGH1/(EDISPLIST+2)	26
27	2126	072475*	3734		DCA I	HIGH2/(EDISPLIST+6)	27
28	2127	072476*	1240		TAD	L23/(23)	28
29	2128	072477*	6056		KBOLS	/GET KEYBOARD CODE TO LIGHT LEDS	29
30	2129	072500*	3012		DCA	XR2	30
31	2130	072501*	2012		ISZ	XR2	31
32	2131	072502*	5301		JMP	.-1	32
33	2132	072503*	1242		TAD	L202/(202)	33
34	2133	072504*	6056		KBOLS	/GET KEYBOARD CODE FOR LED PATTERN	34
35	2134	072505*	7200		CLA	/SEND IT TO KEYBOARD	35
36	2135	072506*	4546		JMS I	PSSTRING/[SSTRING]	36
37	2136	072507*	2372		EDISPLIST	/CALL STRING STORE FOR DISPLAY ROUTINE	37
38	2137	072510*	6111		KBISF	/ERROR DIGITS DISPLAY LIST POINTER	38
39	2138	072511*	5310		JMP	.-1	39
40	2139	072512*	6116		KBIRB	/KEYBOARD FLAG UP?	40
41	2140	072513*	7200		CLA	/NO, WAIT FOR IT	41
42	2141	072514*	5644		JMP I	ENDUP	42
43	2142					/YES, READ IN THE CHARACTER	43
44	2143			/		/THROW IT AWAY	44
45	2144					/RETURN	45
46	2145	072515*	6121	NOERROR,	APTSKP	/AUTOMATIC PRODUCT TESTING?	46
47	2146	072516*	5322		JMP	NOTAPT	47
48	2147	072517*	4524		JMS I	PAPTRoutine/[APTRoutine]	48
49	2148	072520*	5721		JMP I	+.1/(PRGDONE)	49
50	2149	072521*	0214		PRGDONE	/GO THERE	50
51						/THROUGH HERE	51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

1	2151	072522*	1241	NOTAPT,	TAD	L247/(247)	/GET KEYBOARD CODE FOR THE BEEPER	1
2	2152	072523*	6056		KBOLS		/SEND IT	2
3	2153	072524*	7200		CLA		/CLEAN UP	3
4	2154	072525*	4546		JMS I	PSSTRING/[SSTRING]	/DISPLAY THE	4
5	2155	072526*	2403		DMLOGO		/DECMATE II LOGO	5
6	2156	072527*	5644		JMP I	ENDUP	/RETURN	6
7	2157							7
8	2158	072530*	2747	PSET2,	SET2		/POINTER TO 2-WORD ATTRIBUTE BITS SET ROUTINE	8
9	2159	072531*	2375	LOW1,	EDISPLIST+3		/POINTER TO LOW-ORDER DIGIT STORAGE (1)	9
10	2160	072532*	2401	LOW2,	EDISPLIST+7		/POINTER TO LOW-ORDER DIGIT STORAGE (2)	10
11	2161	072533*	2374	HIGH1,	EDISPLIST+2		/POINTER TO HIGH-ORDER DIGIT STORAGE (1)	11
12	2162	072534*	2400	HIGH2,	EDISPLIST+6		/POINTER TO HIGH-ORDER DIGIT STORAGE (2)	12
13	2163	072535*	6520	PAPT2,	APTR02		/POINTER TO APTRoutine ERROR HANDLING ENTRY	13
14	2164							14
15	2165			/		CP-INTERRUPT HANDLER FOR DOWN-LOADED CODE. THE DOWNLOADED CODE EXITS VIA A		15
16	2166			/		HLT INSTRUCTION OR PANEL REQUEST. HLT ENDS THE EXECUTION AND RETURNS CONTROL		16
17	2167			/		TO OUR CODE; PANEL REQUEST CAUSES A TOTAL RESTART OF ALL MOVED ROM CODE.		17
18	2168							18
19	2169	072536*	3017	CPIHND,	DCA	XR7	/SAVE AC	19
20	2170	072537*	6256		GCF		/GET CURRENT FIELDS, ETC. STATUS	20
21	2171	072540*	3016		DCA	XR6	/STASH IT	21
22	2172	072541*	6000		PRS		/GET PANEL STATUS	22
23	2173	072542*	3010		DCA	XR0	/STASH IT	23
24	2174	072543*	6276		SPD		/CP INDIRECTS FROM NOW ON	24
25	2175	072544*	1010		TAD	XR0	/GET STATUS BACK	25
26	2176	072545*	7500		SMA		/SKIP IF BOOTSTRAP (CPREQ) SET	26
27	2177	072546*	5365		JMP	HALTEST	/JUMP IF NOT	27
28	2178	072547*	7200		CLA		/CLEAN UP	28
29	2179	072550*	6061		VSF		/SKIP ON, CLEAR VIDEO INTERRUPT FLAG	29
30	2180	072551*	7000		NOP		/IN CASE IT SKIPS	30
31	2181	072552*	6071		TTYSF		/SKIP ON, CLEAR TTY: TRAP FLAG	31
32	2182	072553*	7000		NOP		/IN CASE IT SKIPS	32
33	2183	072554*	6111		KBISF		/SKIP ON, CLEAR KEYBOARD INPUT FLAG	33
34	2184	072555*	7000		NOP		/IN CASE IT SKIPS	34
35	2185	072556*	6051		KBOSF		/SKIP ON, CLEAR KEYBOARD OUTPUT FLAG	35
36	2186	072557*	7000		NOP		/IN CASE IT SKIPS	36
37	2187	072560*	1016		TAD	XR6	/GET FIELDS BACK	37
38	2188	072561*	6005		RTF		/RESTORE THEM	38
39	2189	072562*	1017		TAD	XR7	/RESTORE AC	39
40	2190	072563*	6004		PEX		/LEAVING PANEL MEMORY	40
41	2191	072564*	5400		JMP I	INTADR	/RETURN TO INTERRUPTED MAIN MEMORY PROGRAM	41
42	2192							42
43	2193	072565*	6273	HALTEST,	CIF	70	/STAYING IN OUR FIELD	43
44	2194	072566*	0374		AND	L200/(200)	/JUST HLTFLG BIT	44
45	2195	072567*	7650		SNA	CLA	/SKIP IF SET	45
46	2196	072570*	5774		JMP I	L200/(CODESTART)	/JUMP IF NOT (PANEL REQUEST)	46
47	2197	072571*	6003		PG0		/RESET RUN FLIP-FLOP	47
48	2198	072572*	5773		JMP I	.+1/(DWNENTRY)	/GO THERE	48
49	2199	072573*	0650		DWNENTRY		/THROUGH HERE	49
50	2200							50
51	2201	072574*	0200	L200,	200/CODESTART		/CONSTANT 0200	51
52	2202							52
53	2203	072575*	0000		0000		/THIS ENDS THE LIST OF INSTRUCTIONS TO MOVE	53
54	2204							54
55	2205	072576*	0000		ZBLOCK	2	/EMPTY SPACE	55
56								56
57								57
58								58

1	2207	*2600	PAGE			1
2	2208					2
3	2209	/	INTERRUPT-DRIVEN LOOPBACK TEST ROUTINE.	MOVED TO MAIN MEMORY FIELD 0.		3
4	2210					4
5	2211	2600	LBLOAD=	/WHERE WE ARE NOW		5
6	2212					6
7	2213	*0200	RELOC 0200	/WHERE THIS EXECUTES IN MAIN MEMORY FIELD 0		7
8	2214					8
9	2215	070200* 6007	LBTEST, CAF	/CLEAN UP		9
10	2216	070201* 6326	PRIRB	/READ PRINTER INPUT NOW		10
11	2217	070202* 7402	HLT	/STOP NOW		11
12	2218	070203* 7307	HLTPC, NL0004	/THIS SHOULD BE SKIPPED BECAUSE OF CP ROUTINES		12
13	2219	070204* 6060	VFL	/SET VIDEO INTERRUPT FLAG		13
14	2220	070205* 7307	VIDPC, NL0004	/THIS SHOULD BE SKIPPED BECAUSE OF CP ROUTINES		14
15	2221	070206* 6034	KRS	/DO A TRAPPED KEYBOARD INSTRUCTION		15
16	2222	070207* 7307	TTYPC, NL0004	/THIS SHOULD BE SKIPPED BECAUSE OF CP ROUTINES		16
17	2223	070210* 6050	KBOFL	/SET KEYBOARD OUTPUT FLAG		17
18	2224	070211* 7307	KBOIPC, NL0004	/THIS SHOULD BE SKIPPED BECAUSE OF CP ROUTINES		18
19	2225	070212* 6110	KBIFL	/SET KEYBOARD INPUT FLAG		19
20	2226	070213* 7307	KBIPC, NL0004	/THIS SHOULD BE SKIPPED BECAUSE OF CP ROUTINES		20
21	2227	070214* 3007	DCA TSTATUS	/SAVE AC TEST STATUS		21
22	2228	070215* 1247	TAD LB20/(20)	/GET LOOPBACK ENABLE VALUE		22
23	2229	070216* 6126	VIDCON	/MAKE THE INTERFACES LOOP BACK		23
24	2230	070217* 7200	CLA	/CLEAN UP		24
25	2231	070220* 7201	NL0001	/SET INTERRUPT ENABLE BIT		25
26	2232	070221* 6135	CLIE	/ENABLE CLOCK INTERRUPTS		26
27	2233	070222* 3005	DCA CSTATUS	/SET INITIAL COUNT STATUS		27
28	2234	070223* 1302	TAD L17/(17)	/GET BAUD RATE VALUE FOR 19200 BAUD		28
29	2235	070224* 6333	PRSB	/SET BAUD RATE		29
30	2236	070225* 7300	CLA CLL	/CLEAN UP		30
31	2237	070226* 3020	DCA CTMP1	/CLEAR INITIAL OUTPUT CHARACTER		31
32	2238	070227* 1245	TAD JMP12/(JMP I INT2)	/GET INTERRUPT INSTRUCTION		32
33	2239	070230* 3001	DCA INT1	/STORE IN INTERRUPT STARTUP		33
34	2240	070231* 1244	TAD PINTHND/(INTHND)	/GET INTERRUPT ADDRESS		34
35	2241	070232* 3002	DCA INT2	/STORE IN INTERRUPT POINTER		35
36	2242	070233* 3003	DCA TIMEOUT	/CLEAR TIME-OUT COUNTER		36
37	2243	070234* 3004	DCA CLKTICK	/CLEAR CLOCK-TICK COUNTER		37
38	2244	070235* 1020	TAD CTMP1	/GET OUTPUT CHARACTER		38
39	2245	070236* 6334	PROPC	/OUTPUT TO PRINTER PORT		39
40	2246	070237* 3021	DCA CTMP2	/SAVE AS INITIAL INPUT TEST VALUE ALSO		40
41	2247	070240* 6001	ION	/TURN ON INTERRUPTS		41
42	2248	070241* 2003	ISZ TIMEOUT	/DONE WAITING YET?		42
43	2249	070242* 5241	JMP .-1	/NO, KEEP WAITING		43
44	2250	070243* 5321	JMP TIMCHK	/WE'RE DONE, FINISH IT THERE		44
45	2251					45
46	2252	070244* 0250	PINTHND, INT2	/POINTER TO INTERRUPT HANDLER		46
47	2253	070245* 5402	JMP12, JMP I	/INTERRUPT STARTUP INSTRUCTION		47
48	2254	070246* 0377	L377, 377	/CONSTANT 0377		48
49	2255	070247* 0020	LB20, 20	/LOOPBACK ENABLE VALUE		49
50						50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
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1	2257		/		INTERRUPTS COME HERE.		1
2	2258						2
3	2259	070250*	6131	INTHND,	CLSK	/CLOCK FLAG UP?	3
4	2260	070251*	7410		SKP	/SKIP IF NOT	4
5	2261	070252*	5277		JMP DOCLK	/JUMP IF SO	5
6	2262	070253*	6331		PROSKP	/PRINTER OUTPUT FLAG UP?	6
7	2263	070254*	7410		SKP	/SKIP IF NOT	7
8	2264	070255*	5304		JMP PROINT	/JUMP IF SO	8
9	2265	070256*	6321		PRISKP	/PRINTER INPUT FLAG UP?	9
10	2266	070257*	5275		JMP INTXIT	/JUMP IF NOT	10
11	2267	070260*	6321		PRISKP	/SKIP ON, CLEARING THE FLAG	11
12	2268	070261*	7000		NOP	/HERE IN CASE IT SKIPS	12
13	2269	070262*	6326		PRIRB	/GET THE CHARACTER	13
14	2270	070263*	7041		CIA	/INVERT FOR TESTING	14
15	2271	070264*	1021		TAD CTMP2	/COMPARE TO TEST VALUE	15
16	2272	070265*	0246		AND L377/(377)	/JUST DATA BITS	16
17	2273	070266*	7640		SZA CLA	/SKIP IF IT MATCHES	17
18	2274	070267*	6325		PRIIE	/PREVENT FURTHER PRINTER INPUT INTERRUPTS	18
19	2275	070270*	1021		TAD CTMP2	/GET LATEST COUNT PATTERN VALUE	19
20	2276	070271*	4334		JMS COUNTUP	/GET NEXT IN THE PATTERN	20
21	2277	070272*	5320		JMP COUNTDONE	/AT END OF COUNT PATTERN	21
22	2278	070273*	3021		DCA CTMP2	/STORE UPDATED VALUE	22
23	2279	070274*	3003		DCA TIMEOUT	/RESET TIME-OUT COUNTER	23
24	2280	070275*	6001	INTXIT,	ION	/RE-ENABLE INTERRUPTS	24
25	2281	070276*	5400		JMP I INTADR	/BACK TO INTERRUPTED CODE	25
26	2282						26
27	2283	070277*	6131	DOCLK,	CLSK	/MAKE SURE CLOCK FLAG CLEARS	27
28	2284	070300*	7000		NOP	/IN CASE IT SKIPS	28
29	2285	070301*	2004		ISZ CLKTICK	/COUNT THIS TICK	29
30	2286	070302*	0017	L17,	17	/19200 BAUD RATE VALUE; HERE IN CASE IT SKIPS	30
31	2287	070303*	5275		JMP INTXIT	/CONTINUE THERE	31
32	2288						32
33	2289	070304*	6331	PROINT,	PROSKP	/SKIP ON, CLEARING PRINTER OUTPUT FLAG	33
34	2290	070305*	7000		NOP	/JUST IN CASE IT SKIPS	34
35	2291	070306*	1020		TAD CTMP1	/GET COUNT PATTERN VALUE	35
36	2292	070307*	4334		JMS COUNTUP	/GET NEXT IN THE PATTERN	36
37	2293	070310*	7410		SKP	/AT END OF COUNT PATTERN	37
38	2294	070311*	5314		JMP NEXTOUT	/JUMP IF NOT AT END	38
39	2295	070312*	6335		PROIE	/PREVENT FURTHER PRINTER OUTPUT INTERRUPTS	39
40	2296	070313*	5275		JMP INTXIT	/CONTINUE THERE	40
41	2297						41
42	2298	070314*	3020	NEXTOUT,	DCA CTMP1	/SAVE NEW VALUE	42
43	2299	070315*	1020		TAD CTMP1	/GET IT BACK	43
44	2300	070316*	6336		PROLS	/OUTPUT IT	44
45	2301	070317*	5275		JMP INTXIT	/CONTINUE THERE	45
46	2302						46
47	2303			/		COMES HERE WHEN ALL CHARACTER VALUES USED.	47
48	2304						48
49	2305	070320*	3005	COUNTDO,	DCA CSTATUS	/CLEAR COUNT STATUS	49
50							50
51							51
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1	2307	/	COMES HERE WHEN NULL JOB TIMES OUT.			1
2	2308					2
3	2309	070321* 7344	TIMCHK,	NL7776	/SETUP VALUE	3
4	2310	070322* 1004	TAD	CLKTICK	/COMPARE TO CLOCK TICK COUNT	4
5	2311	070323* 7710	SPA	CLA	/SKIP IF >=2	5
6	2312	070324* 7305	NL002A		/SET STATUS BIT IF NOT	6
7	2313	070325* 1005	TAD	CSTATUS	/GET COUNT STATUS EITHER WAY	7
8	2314	070326* 3005	DCA	CSTATUS	/STORE BACK	8
9	2315	070327* 6007	CAF		/CLEAR ALL FLAGS NOW	9
10	2316	070330* 1005	TAD	CSTATUS	/GET COUNT STATUS	10
11	2317	070331* 1007	TAD	TSTATUS	/GET CP TEST STATUS	11
12	2318	070332* 6236	PRQ3		/DO A PANEL REQUEST TO EXIT	12
13	2319	070333* 5333	JMP	.	/WILL HANG HERE IF IT FAILS!	13
14	2320					14
15	2321	070334* 0000	COUNTUP,	...	/PATTERN COUNT ROUTINE	15
16	2322	070335* 3006	DCA	PATTERN	/SAVE PASSED VALUE	16
17	2323	070336* 1006	TAD	PATTERN	/GET IT BACK	17
18	2324	070337* 7001	IAC		/ADD ONE	18
19	2325	070340* 0246	AND	L377/(377)	/JUST GOOD BITS	19
20	2326	070341* 7450	SNA		/SKIP IF NOT AT END	20
21	2327	070342* 5734	JMP I	COUNTUP	/TAKE IMMEDIATE RETURN IF AT END	21
22	2328	070343* 3006	DCA	PATTERN	/SAVE UPDATED VALUE	22
23	2329	070344* 2334	ISZ	COUNTUP	/BUMP RETURN ADDRESS	23
24	2330	070345* 1006	TAD	PATTERN	/GET NEW PATTERN VALUE	24
25	2331	070346* 5734	JMP I	COUNTUP	/TAKE SKIP RETURN TO CALLER	25
26	2332					26
27	2333	0347	LBTEST=	.	/END OF CODE TO BE MOVED	27
28	2334					28
29	2335	*2747	RELOC	LBTEST-LBTEST+LBLOAD	/RE-ESTABLISH ORIGIN	29
30	2336					30
31	2337	/	SUBROUTINE TO SET HIGH-ORDER BITS IN CP 13705 AND 13707. BIT[4] IS SET IN			31
32	2338	/	13705; BITS[4-5] ARE SET IN 13707. THIS MAKES THE LINES INTO DOUBLE-HEIGHT,			32
33	2339	/	DOUBLE WIDTH UPPER AND LOWER HALVES RESPECTIVELY. THIS ROUTINE DEPENDS ON THE			33
34	2340	/	PREVIOUS CONTENTS OF THE BITS BEING ZERO.			34
35	2341					35
36	2342	072747* 0000	SET2,	...	/SET HIGH BITS IN 2 WORDS ROUTINE	36
37	2343	072750* 6211	CDF	10	/GOTO BUFFER FIELD	37
38	2344	072751* 1761	TAD I	L3705/(3705)	/GET CURRENT CONTENTS OF 13705	38
39	2345	072752* 1112	TAD	Z200/[200]	/ADD ON A HIGH-ORDER BIT	39
40	2346	072753* 3761	DCA I	L3705/(3705)	/STORE BACK	40
41	2347	072754* 1762	TAD I	L3707/(3707)	/GET CURRENT CONTENTS OF 13707	41
42	2348	072755* 1363	TAD	L300/(300)	/ADD ON HIGH-ORDER BITS	42
43	2349	072756* 3762	DCA I	L3707/(3707)	/STORE BACK	43
44	2350	072757* 6271	CDF	70	/BACK TO OUR FIELD	44
45	2351	072760* 5747	JMP I	SET2	/RETURN	45
46	2352					46
47	2353	072761* 3705	L3705,	3705	/CONSTANT 3705	47
48	2354	072762* 3707	L3707,	3707	/CONSTANT 3707	48
49	2355	072763* 0300	L300,	300	/CONSTANT 0300	49
50	2356	072764* 0077	L77,	77	/CONSTANT 0077	50
51						51
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1	2358	/	SUBROUTINE TO CLEAR THE ATTRIBUTE BITS OF CP 13705 AND 13707. THIS MAKES THE			1
2	2359	/	CORRESPONDING LINES NORMAL WIDTH AND HEIGHT.			2
3	2360					3
4	2361	072765* 0000	CLR2,	...	/CLEAR BITS IN TWO WORDS ROUTINE.	4
5	2362	072766* 6211	CDF	10	/GOTO BUFFER FIELD	5
6	2363	072767* 1761	TAD I	L3705/(3705)	/GET CURRENT CONTENTS OF 13705	6
7	2364	072770* 0364	AND	L77/(77)	/JUST KEEP LOW-ORDER HALF	7
8	2365	072771* 3761	DCA I	L3705/(3705)	/STORE BACK	8
9	2366	072772* 1762	TAD I	L3707/(3707)	/GET CURRENT CONTENTS OF 13707	9
10	2367	072773* 0364	AND	L77/(77)	/JUST KEEP LOW-ORDER HALF	10
11	2368	072774* 3762	DCA I	L3707/(3707)	/STORE BACK	11
12	2369	072775* 6271	CDF	70	/BACK TO OUR FIELD	12
13	2370	072776* 5765	JMP I	CLR2	/RETURN	13
14	2371					14
15	2372	072777* 0000	ZBLOCK	1	/EMPTY SPACE	15
16						16
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[illegible]

1	2421	074034*	6071	TSTTRAP, TTYSF		/DEVICE 03, 04 TRAP FLAG SET?	1
2	2422	074035*	5245	JMP	TSTKYIN	/JUMP IF NOT	2
3	2423	074036*	6075	TTYIE		/PREVENT FURTHER TTY: TRAP INTERRUPTS	3
4	2424	074037*	1305	TAD	PTTYPC/(TTYPC)	/GET EXPECTED INTERRUPTED ADDRESS	4
5	2425	074040*	7041	CIA		/INVERT FOR COMPARISON	5
6	2426	074041*	1000	TAD	INTADR	/COMPARE TO ACTUAL INTERRUPT PC	6
7	2427	074042*	7650	SNA CLA		/SKIP IF OTHER THAN EXPECTED	7
8	2428	074043*	5265	JMP	BUMPIT	/JUMP IF OK	8
9	2429						9
10	2430			/	INTERRUPT PC WRONG.		10
11	2431						11
12	2432	074044*	5266	JMP	JUSTEXIT	/CONTINUE THERE	12
13	2433						13
14	2434	074045*	6111	TSTKYIN, KBISF		/KEYBOARD INPUT FLAG SET?	14
15	2435	074046*	5256	JMP	TSTKYOUT	/NO, TRY OUTPUT	15
16	2436	074047*	6115	KBIIE		/PREVENT FURTHER KEYBOARD INPUT INTERRUPTS	16
17	2437	074050*	1306	TAD	PKBIPC/(KBIPC)	/GET EXPECTED INTERRUPTED ADDRESS	17
18	2438	074051*	7041	CIA		/INVERT FOR COMPARISON	18
19	2439	074052*	1000	TAD	INTADR	/COMPARE TO ACTUAL INTERRUPT PC	19
20	2440	074053*	7650	SNA CLA		/SKIP IF OTHER THAN EXPECTED	20
21	2441	074054*	5265	JMP	BUMPIT	/JUMP IF OK	21
22	2442						22
23	2443			/	INTERRUPT PC WRONG.		23
24	2444						24
25	2445	074055*	5266	JMP	JUSTEXIT	/CONTINUE THERE	25
26	2446						26
27	2447	074056*	6051	TSTKYOU, KBOSF		/KEYBOARD OUTPUT FLAG SET?	27
28	2448	074057*	5266	JMP	JUSTEXIT	/NO, FORGET IT	28
29	2449	074060*	6055	KBOIE		/PREVENT FURTHER KEYBOARD OUTPUT INTERRUPTS	29
30	2450	074061*	1307	TAD	PKBOIPC/(KBOIPC)	/GET EXPECTED INTERRUPTED ADDRESS	30
31	2451	074062*	7041	CIA		/INVERT FOR COMPARISON	31
32	2452	074063*	1000	TAD	INTADR	/COMPARE TO ACTUAL INTERRUPT PC	32
33	2453	074064*	7650	SNA CLA		/SKIP IF OTHER THAN EXPECTED	33
34	2454						34
35	2455			/	COMES HERE TO BUMP THE PC PAST THE NL0004 INSTRUCTIONS IF THE PROPER INTERRUPT		35
36	2456			/	ADDRESS WAS PRESENT AT THE POINT OF INTERRUPT.		36
37	2457						37
38	2458	074065*	2000	BUMPIT, ISZ	INTADR	/BUMP INTERRUPTED ADDRESS	38
39	2459						39
40	2460			/	SKIPS PREVIOUS TO EXIT NORMALLY; THIS COULD POSSIBLY BE CONSIDERED AN ERROR.		40
41	2461						41
42	2462	074066*	6004	JUSTEXI, PEX		/EXITING FROM PANEL MEMORY	42
43	2463	074067*	5400	JMP I	INTADR	/BACK TO INTERRUPTED MAIN MEMORY PROGRAM	43
44							44
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1	2465		/	COMES HERE IF THE HALT FLAG WAS SET		1
2	2466					2
3	2467	074070*	7200	TESTHLT,CLA	/CLEAN UP	3
4	2468	074071*	6003	PG0	/RESET THE HALT FLAG	4
5	2469	074072*	1310	TAD PHLTPC/(HLTPC)	/GET EXPECTED INTERRUPTED ADDRESS	5
6	2470	074073*	7041	CIA	/INVERT FOR COMPARISON	6
7	2471	074074*	1000	TAD INTADR	/COMPARE TO ACTUAL INTERRUPT PC	7
8	2472	074075*	7650	SNA CLA	/SKIP IF OTHER THAN EXPECTED	8
9	2473	074076*	5265	JMP BUMPIT	/JUMP IF OK	9
10	2474					10
11	2475		/	INTERRUPT PC WRONG.		11
12	2476					12
13	2477	074077*	5266	JMP JUSTEXIT	/CONTINUE THERE	13
14	2478					14
15	2479		/	COMES HERE IF MAIN MEMORY PROGRAM EXITED VIA PANEL REQUEST.		15
16	2480					16
17	2481	074100*	7200	PRQEXIT,CLA	/CLEAN UP	17
18	2482	074101*	1017	TAD XR7	/RESTORE EXITING AC	18
19	2483	074102*	6273	CIF CDF 70	/BACK TO OUR FIELD	19
20	2484	074103*	5620	JMP I PLBDONE/(LBDONE)	/RETURN TO CALLER	20
21	2485					21
22	2486	074104*	0205	PVIDPC, VIDPC	/EXPECTED PC FOR VIDEO INTERRUPT	22
23	2487	074105*	0207	PTTYPC, TTYPC	/EXPECTED PC FOR TTY: TRAP INTERRUPT	23
24	2488	074106*	0213	PKBIPC, KBIPC	/EXPECTED PC FOR KEYBOARD INPUT INTERRUPT	24
25	2489	074107*	0211	PKBOIPC, KBOIPC	/EXPECTED PC FOR KEYBOARD OUTPUT INTERRUPT	25
26	2490	074110*	0203	PHLTPC, HLTPC	/EXPECTED PC FOR HALT INTERRUPT	26
27	2491					27
28	2492		4111	CPEND= .	/END OF CP-INTERRUPT HANDLER CODE THAT OUGHT	28
29	2493				/TO BE MOVED	29
30	2494					30
31	2495		*3111	RELOC CPEND-CPINT+CPLOAD	/RE-ESTABLISH ORIGIN	31
32						32
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1	2497	/	THIS ROUTINE READS IN TRACK 0, SECTOR 3 INTO CP 15350-16347. THE FIRST EIGHT	1
2	2498	/	BYTES ARE VALIDATED AGAINST A KNOWN PATTERN. ALL ELEMENTS READ IN AS 377 ARE	2
3	2499	/	REPLACED WITH 4000 IN THE RUNNING COPY.	3
4	2500			4
5	2501	073111* 0000	T0S3RD, .-. /TRACK 0, SECTOR 3 READ ROUTINE	5
6	2502	073112* 1344	TAD PT0S3BUFF/(T0S3BUFFER-1)/SETUP THE	6
7	2503	073113* 3010	DCA XR0 /DATA STORAGE POINTER	7
8	2504	073114* 6211	CDF 10 /SET THE DATA FIELD FOR THE TRANSFER	8
9	2505	073115* 7305	NL002A /SET SECTOR OFFSET (3-1)	9
10	2506	073116* 4742	JMS I PFREAD/(FREAD) /CALL FLOPPY READ ROUTINE	10
11	2507	073117* 7774	-3-1 /SECTOR THREE IS THE LAST ONE	11
12	2508	073120* 0000	0 /TRACK 0000	12
13	2509	073121* 5711	JMP I T0S3RD /ERROR, TAKE NON-SKIP RETURN	13
14	2510	073122* 1343	TAD PT0S3DATA/(T0S3DATA-1) /SETUP THE VALIDATION LIST VALUE	14
15	2511	073123* 3010	DCA XR0 /STASH THE POINTER	15
16	2512	073124* 1344	TAD PT0S3BUFF/(T0S3BUFFER-1)/POINT TO THE	16
17	2513	073125* 3011	DCA XR1 /READ-IN DATA	17
18	2514	073126* 1410	LISTLUP, TAD I XR0 /GET A TEST VALUE	18
19	2515	073127* 7450	SNA /END OF LIST?	19
20	2516	073130* 5337	JMP LISTOK /JUMP IF SO	20
21	2517	073131* 6211	CDF 10 /GOTO DATA FIELD	21
22	2518	073132* 1411	TAD I XR1 /COMPARE TO READ-IN VALUE	22
23	2519	073133* 6271	CDF 70 /BACK TO OUR FIELD	23
24	2520	073134* 7640	SZA CLA /SKIP IF IT MATCHES	24
25	2521	073135* 5711	JMP I T0S3RD /TAKE FAILURE RETURN IF NOT	25
26	2522	073136* 5326	JMP LISTLUP /KEEP GOING	26
27	2523			27
28	2524	073137* 4745	LISTOK, JMS I PRUBCHK/(RUBCHK) /REPLACE ALL 377 CODES WITH 4000	28
29	2525	073140* 2311	ISZ T0S3RD /BUMP TO SUCCESSFUL RETURN	29
30	2526	073141* 5711	JMP I T0S3RD /TAKE SKIP-RETURN TO CALLER	30
31	2527			31
32	2528	073142* 1311	PFREAD, FREAD /POINTER TO FLOPPY READ ROUTINE	32
33	2529	073143* 3774	PT0S3DA, T0S3DATA-1 /POINTER TO TEST DATA FOR TRACK 0, SECTOR 3	33
34	2530	073144* 5347	PT0S3BU, T0S3BUFFER-1 /POINTER TO TRACK 0, SECTOR 3 LOADING ADDRESS	34
35	2531	073145* 5400	PRUBCHK, RUBCHK /POINTER TO RUBCHK ROUTINE	35
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1	2533	/	COMES BACK HERE AFTER LOADING THE CHARACTER GENERATOR RAM TO EXIT.	1
2	2534			2
3	2535	073146* 5752	CGDONE, JMP I DOCG /RETURN TO ORIGINAL CALLER	3
4	2536			4
5	2537	073147* 3763	PT0S7DA,T0S7DATA-1 /POINTER TO TEST DATA FOR TRACK 0, SECTOR 7	5
6	2538	073150* 3205	PUSEDAT,USEDATA /POINTER TO USE DATA ROUTINE	6
7	2539	073151* 3200	PNODATA,NODATA /POINTER TO NO DATA ROUTINE	7
8	2540			8
9	2541	/	THE CHARACTER GENERATOR RAM LOAD ROUTINE IS ENTERED HERE.	9
10	2542			10
11	2543	073152* 0000	DOCG, .-. /CHARACTER GENERATOR RAM LOAD ROUTINE	11
12	2544	073153* 7240	NL7777 /SETUP THE	12
13	2545	073154* 3010	DCA XR0 /STORE POINTER	13
14	2546	073155* 6201	CDF 00 /GOTO DATA FIELD	14
15	2547	073156* 7327	NL0006 /SET OFFSET FOR SECTOR 7 (7-10)	15
16	2548	073157* 4742	JMS I PFREAD/(FREAD) /CALL FLOPPY READ ROUTINE	16
17	2549	073160* 7765	-12-1 /WANT SECTORS 7-10	17
18	2550	073161* 0000	0 /ON TRACK 0	18
19	2551	073162* 5751	JMP I PNODATA/(NODATA) /ERROR, TRY TO KEEP GOING	19
20	2552			20
21	2553	/	THE DATA READ IN OK, SO VALIDATE IT.	21
22	2554			22
23	2555	073163* 1347	TAD PT0S7DATA/(T0S7DATA-1) /SETUP THE	23
24	2556	073164* 3010	DCA XR0 /TEST DATA POINTER	24
25	2557	073165* 7240	NL7777 /SETUP THE	25
26	2558	073166* 3011	DCA XR1 /DATA POINTER	26
27	2559	073167* 1410	TSTDLP, TAD I XR0 /GET A TEST WORD	27
28	2560	073170* 7450	SNA /SKIP IF NOT AT END OF LIST	28
29	2561	073171* 5750	JMP I PUSEDATA/(USEDATA) /END OF LIST, GO USE GOOD DATA	29
30	2562	073172* 6201	CDF 00 /GOTO DATA FIELD	30
31	2563	073173* 1411	TAD I XR1 /COMPARE DATA TO TEST WORD	31
32	2564	073174* 6271	CDF 70 /BACK TO OUR FIELD	32
33	2565	073175* 7650	SNA CLA /SKIP IF NO MATCH	33
34	2566	073176* 5367	JMP TSTDLP /KEEP GOING IF IT MATCHES	34
35	2567			35
36	2568	/	THE DATA ISN'T VALID, SO IGNORE IT AS IF IT HAD BEEN AN I/O ERROR.	36
37	2569			37
38	2570	073177* 5751	JMP I PNODATA/(NODATA) /CONTINUE THERE	38
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1	2572	*3200	PAGE				1
2	2573						2
3	2574	/	COMES HERE TO CLEAR THE DATA WHICH WAS EITHER INVALID OR COULDN'T GET READ IN.				3
4	2575	/	BETTER TO CLEAR THE TABLE THAN LEAVE IT AT RANDOM.				4
5	2576						5
6	2577	073200* 3277	NODATA, DCA	CGPTR	/CLEAR THE DATA POINTER		6
7	2578	073201* 6201	CLRLUP, CDF	00	/GOTO DATA FIELD		7
8	2579	073202* 3677	DCA I	CGPTR	/CLEAR A WORD		8
9	2580	073203* 2277	ISZ	CGPTR	/DONE ALL YET?		9
10	2581	073204* 5201	JMP	CLRLUP	/NO, GO BACK		10
11	2582						11
12	2583	/	COMES HERE TO LOAD THE CHARACTER GENERATOR RAM FROM (HOPEFULLY) THE READ-IN				12
13	2584	/	DATA (OR JUST ZEROES IF IT WASN'T READABLE).				13
14	2585						14
15	2586	073205* 6271	USEDATA, CDF	70	/GOTO OUR FIELD		15
16	2587						16
17	2588	/	WE HAVE TO CLEAR THE FIRST ROW OF DATA BECAUSE IT ISN'T A PART OF THE USEFUL				17
18	2589	/	DATA (VERIFICATION HEADER AND OTHER JUNK).				18
19	2590						19
20	2591	073206* 7240	NL7777		/SETUP THE		20
21	2592	073207* 3010	DCA	XR0	/DATA POINTER		21
22	2593	073210* 1127	TAD	Z7760/[-20]	/SETUP THE		22
23	2594	073211* 3011	DCA	XR1	/ROW COUNTER		23
24	2595	073212* 6201	CDF	00	/GOTO THE DATA FIELD		24
25	2596	073213* 3410	CLRIT, DCA I	XR0	/CLEAR A WORD		25
26	2597	073214* 2011	ISZ	XR1	/DONE ENOUGH?		26
27	2598	073215* 5213	JMP	CLRIT	/NO, GO BACK		27
28	2599	073216* 1255	TAD	L110/(110)	/DISABLE SCREEN AND ENABLE CG RAM		28
29	2600	073217* 6126	VIDCON		/LOAD VIDEO CONTROL		29
30	2601	073220* 7240	NL7777		/SETUP THE		30
31	2602	073221* 3010	DCA	XR0	/DATA POINTER		31
32	2603	073222* 3277	DCA	CGPTR	/CLEAR CG RAM ADDRESS POINTER		32
33	2604						33
34	2605	073223* 1256	CGLDLP, TAD	PCGINDX/(CGINDX-1)	/SETUP THE		34
35	2606	073224* 3012	DCA	XR2	/INDEX POINTER		35
36	2607	073225* 1127	TAD	Z7760/[-20]	/SETUP THE		36
37	2608	073226* 3011	DCA	XR1	/LINE COUNT		37
38	2609	073227* 6271	NXTLINE, CDF	70	/BACK TO OUR (ODD) FIELD		38
39	2610	073230* 1412	TAD I	XR2	/GET AN INDEX		39
40	2611	073231* 1277	TAD	CGPTR	/ADD ON CURRENT CHARACTER BASE		40
41	2612	073232* 6123	CGLOAD		/LOAD CHARACTER GENERATOR ADDRESS		41
42	2613	073233* 7200	CLA		/CLEAN UP		42
43	2614	073234* 6201	CDF	00	/GOTO (EVEN) FIELD 0		43
44	2615	073235* 1410	TAD I	XR0	/GET LATEST CELL DATA LINE		44
45	2616	073236* 6123	CGLOAD		/LOAD CHARACTER GENERATOR VALUE		45
46	2617	073237* 7200	CLA		/CLEAN UP		46
47	2618	073240* 2011	ISZ	XR1	/DONE ALL LINES YET?		47
48	2619	073241* 5227	JMP	NXTLINE	/NO, GO BACK		48
49	2620	073242* 1110	TAD	Z20/[20]	/SETUP CHARACTER UPDATE		49
50	2621	073243* 1277	TAD	CGPTR	/ADD TO CHARACTER ADDRESS		50
51	2622	073244* 3277	DCA	CGPTR	/STORE BACK		51
52	2623	073245* 1277	TAD	CGPTR	/GET THE POINTER		52
53	2624	073246* 7700	SMA CLA		/SKIP IF DONE ALL 2048 LINES		53
54	2625	073247* 5223	JMP	CGLDLP	/JUMP IF NOT DONE YET		54
55							55
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57							57
58							58

1	2627	073250*	7315	NL0010	CLL	/SETUP RAM ENABLE AND SCREEN ENABLE	1
2	2628	073251*	6126		VIDCON	/LOAD NEW CONTROL VALUE	2
3	2629	073252*	7200		CLA	/CLEAN UP	3
4	2630	073253*	5654	JMP I	.+1/(CGDONE)	/GO THERE	4
5	2631	073254*	3146		CGDONE	/THROUGH HERE	5
6	2632						6
7	2633	073255*	0110	L110,	110	/CONSTANT 0110	7
8	2634	073256*	3256	PCGINDX,	CGINDX-1	/POINTER TO CG INDEX TABLE	8
9	2635						9
10	2636			/	TABLE OF CG CELL LINE INDICES; (THE FOUR BITS ARE READ BACKWARDS).		10
11	2637						11
12	2638	073257*	0000	CGINDX,	00	/LINE 00	12
13	2639	073260*	0010		10	/LINE 01	13
14	2640	073261*	0004		04	/LINE 02	14
15	2641	073262*	0014		14	/LINE 03	15
16	2642	073263*	0002		02	/LINE 04	16
17	2643	073264*	0012		12	/LINE 05	17
18	2644	073265*	0006		06	/LINE 06	18
19	2645	073266*	0016		16	/LINE 07	19
20	2646	073267*	0001		01	/LINE 08	20
21	2647	073270*	0011		11	/LINE 09	21
22	2648	073271*	0005		05	/LINE 10	22
23	2649	073272*	0015		15	/LINE 11	23
24	2650	073273*	0003		03	/LINE 12	24
25	2651	073274*	0013		13	/LINE 13	25
26	2652	073275*	0007		07	/LINE 14	26
27	2653	073276*	0017		17	/LINE 15	27
28	2654						28
29	2655	073277*	0000	CGPTR,	0000	/CHARACTER GENERATOR DATA POINTER	29
30	2656						30
31	2657			/	NOTE: ALL ROUTINES AND DATA BEYOND THIS POINT MARKED WITH ***** ARE NOT		31
32	2658			/	ACCESSED BY THE NORMAL USAGE OF THE ROM EXCEPT POSSIBLY INDIRECTLY THROUGH A		32
33	2659			/	DOWN-LOADED AUTOMATIC PRODUCT TEST ROUTINE. ALL OF THE CODE IS SUBORDINATE TO		33
34	2660			/	THE SUBROUTINE ENTERED VIA JMS 0176, WHICH IS ITSELF NOT CALLED BY ANY		34
35	2661			/	EXISTING ROM ROUTINES.		35
36							36
37							37
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1	2663		/	*****		1
2	2664					2
3	2665	073300*	0000	GETPARM, .-. .	/ROUTINE TO GET SELECTED PARAMETERS	3
4	2666	073301*	7240	NL7777	/SETUP THE	4
5	2667	073302*	3010	DCA XR0	/DATA POINTER	5
6	2668	073303*	6201	CDF 00	/GOTO DATA FIELD	6
7	2669	073304*	1410	TAD I XR0	/GET A WORD	7
8	2670	073305*	3031	DCA COMTM1	/STASH IT	8
9	2671	073306*	7325	NL0003	/SETUP THE	9
10	2672	073307*	3010	DCA XR0	/DATA POINTER	10
11	2673	073310*	7240	NL7777	/BACKUP FOR AUTO-INDEX	11
12	2674	073311*	1410	TAD I XR0	/GET ADDRESS POINTER	12
13	2675	073312*	3011	DCA XR1	/STASH IT	13
14	2676	073313*	1410	TAD I XR0	/GET FIELD VALUE	14
15	2677	073314*	1122	TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	15
16	2678	073315*	3316	DCA .+1	/STORE IN-LINE	16
17	2679	073316*	7402	HLT+.-.	/WILL BE CDF XX INSTRUCTION	17
18	2680	073317*	1411	TAD I XR1	/GET ADDRESS	18
19	2681	073320*	3030	DCA TM0	/STASH IT	19
20	2682	073321*	1411	TAD I XR1	/GET FIELD	20
21	2683	073322*	3036	DCA MTM5	/STASH IT	21
22	2684	073323*	1411	TAD I XR1	/GET ADDRESS	22
23	2685	073324*	3032	DCA MTM1	/STASH IT	23
24	2686	073325*	1411	TAD I XR1	/GET FIELD	24
25	2687	073326*	3037	DCA MTM6	/STASH IT	25
26	2688	073327*	1411	TAD I XR1	/GET ADDRESS	26
27	2689	073330*	3033	DCA MTM2	/STASH IT	27
28	2690	073331*	1411	TAD I XR1	/GET FIELD	28
29	2691	073332*	3040	DCA MTM7	/STASH IT	29
30	2692	073333*	6207	RSP1	/GET FIRST STACK POINTER	30
31	2693	073334*	3034	DCA SP1SAVE	/SAVE IT	31
32	2694	073335*	6227	RSP2	/GET SECOND STACK POINTER	32
33	2695	073336*	3035	DCA SP2SAVE	/SAVE IT	33
34	2696	073337*	1036	TAD MTM5	/GET FIELD	34
35	2697	073340*	1122	TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	35
36	2698	073341*	3342	DCA .+1	/STORE IN-LINE	36
37	2699	073342*	7402	HLT+.-.	/WILL BE CDF XX INSTRUCTION	37
38	2700	073343*	1430	TAD I TM0	/GET VALUE	38
39	2701	073344*	6271	CDF 70	/BACK TO OUR FIELD	39
40	2702	073345*	3030	DCA TM0	/SAVE IT	40
41	2703	073346*	1037	TAD MTM6	/GET FIELD	41
42	2704	073347*	1122	TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	42
43	2705	073350*	3351	DCA .+1	/STORE IN-LINE	43
44	2706	073351*	7402	HLT+.-.	/WILL BE CDF XX INSTRUCTION	44
45	2707	073352*	1432	TAD I MTM1	/GET VALUE	45
46	2708	073353*	6271	CDF 70	/BACK TO OUR FIELD	46
47	2709	073354*	3032	DCA MTM1	/SAVE IT	47
48	2710	073355*	1040	TAD MTM7	/GET FIELD	48
49	2711	073356*	1122	TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	49
50	2712	073357*	3360	DCA .+1	/STORE IN-LINE	50
51	2713	073360*	7402	HLT+.-.	/WILL BE CDF XX INSTRUCTION	51
52	2714	073361*	1433	TAD I MTM2	/GET VALUE	52
53	2715	073362*	6271	CDF 70	/BACK TO OUR FIELD	53
54	2716	073363*	3033	DCA MTM2	/SAVE IT	54
55	2717	073364*	5700	JMP I GETPARM	/RETURN TO CALLER	55
56						56
57						57
58						58

1	2719	/	*****	1
2	2720			2
3	2721	/	COUNT FOR FIELDS PRIOR TO HIGHLIGHTED RATE FIELD.	3
4	2722			4
5	2723	073365* 7777	CLIST, -1 /50 BAUD	5
6	2724	073366* 7777	-1 /75 BAUD	6
7	2725	073367* 7777	-1 /110 BAUD	7
8	2726	073370* 7777	-1 /134.5 BAUD	8
9	2727	073371* 7777	-1 /150 BAUD	9
10	2728	073372* 7777	-1 /300 BAUD	10
11	2729	073373* 7776	-2 /600 BAUD	11
12	2730	073374* 7775	-3 /1200 BAUD	12
13	2731	073375* 7774	-4 /1800 BAUD	13
14	2732	073376* 7774	-4 /2000 BAUD	14
15	2733	073377* 7774	-4 /2400 BAUD	15
16	2734	073400* 7773	-5 /3600 BAUD	16
17	2735	073401* 7773	-5 /4800 BAUD	17
18	2736	073402* 7772	-6 /7200 BAUD	18
19	2737	073403* 7772	-6 /9600 BAUD	19
20	2738	073404* 7772	-6 /19200 BAUD	20
21	2739			21
22	2740	/	*****	22
23	2741			23
24	2742	073405* 6203	CIDF, CIF CDF 00 /CONSTANT 6203	24
25	2743	073406* 1311	AFREAD, FREAD /POINTER TO FLOPPY READ ROUTINE	25
26	2744	073407* 4007	PPRMPRT, PRMPRT /POINTER TO SETUP PARAMETER PRINT ROUTINE	26
27	2745	073410* 4321	PLOC36, LOC36 /POINTER TO 36 OR 37 LOCATOR ROUTINE	27
28	2746	073411* 3754	PT0S1DA, T0S1DAT-1 /TRACK 0 SECTORS 1-2 VALIDATION DATA POINTER	28
29				29
30				30
31				31
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35				35
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[illegible]

1	2804	073470*	4551	NEXTCHR, JMS I	PKBDIN/[KBDIN]	/GET A CHARACTER	1
2	2805	073471*	3314	METRENT, DCA	SUBAC	/SAVE LATEST CHARACTER	2
3	2806	073472*	1316		TAD PCOMTABLE/(COMTABLE-1)	/SETUP THE	3
4	2807	073473*	3010		DCA XR0	/TABLE POINTER	4
5	2808	073474*	1410	COMSRCH, TAD I	XR0	/GET A WORD	5
6	2809	073475*	7450		SNA	/SKIP IF NOT AT END OF LIST	6
7	2810	073476*	5270		JMP NEXTCHR	/END OF LIST, IGNORE AND GET ANOTHER	7
8	2811	073477*	1314		TAD SUBAC	/COMPARE TO CHARACTER	8
9	2812	073500*	7650		SNA CLA	/SKIP IF NOT A MATCH	9
10	2813	073501*	5304		JMP FINDIT	/JUMP IF IT MATCHES	10
11	2814	073502*	2010		ISZ XR0	/BUMP PAST DISPATCH ADDRESS	11
12	2815	073503*	5274		JMP COMSRCH	/TRY AGAIN	12
13	2816						13
14	2817	073504*	1410	FINDIT, TAD I	XR0	/GET DISPATCH ADDRESS	14
15	2818	073505*	3315		DCA DISPADR	/STASH IT	15
16	2819	073506*	1314		TAD SUBAC	/GET THE CHARACTER ITSELF AS AN ARGUMENT	16
17	2820	073507*	4715		JMS I DISPADR	/DISPATCH TO ROUTINE	17
18	2821	073510*	5270		JMP NEXTCHR	/NO-CHANGE RETURN, GO THERE	18
19	2822	073511*	5265		JMP NEXTPRM	/POSSIBLE-CHANGE RETURN, GO THERE	19
20	2823	073512*	7402	SUBCDF, HLT+.-.		/DOUBLE-SKIP RETURN; WILL BE CIF CDF RETURN	20
21	2824	073513*	5576		JMP I SUB176	/RETURN TO CALLER	21
22	2825						22
23	2826	073514*	0000	SUBAC, .-. .		/AC DATA SAVED HERE	23
24	2827	073515*	0000	DISPADR, .-. .		/DISPATCH ADDRESS TEMPORARY	24
25	2828	073516*	3735	PCOMTAB, COMTABLE-1		/POINTER TO COMMAND DISPATCH TABLE	25
26	2829	073517*	3715	PVIDCLR, VIDCLR		/POINTER TO VIDEO CLEAR ROUTINE	26
27							27
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1	2879	*3600	PAGE			1
2	2880					2
3	2881	/	*****			3
4	2882					4
5	2883	/	HANDLER FOR SELECT KEY IN SETUP ROUTINE.			5
6	2884					6
7	2885	073600*	0000	SELCT, .-. .	/SELECT LATEST ITEM	7
8	2886	073601*	7200	CLA	/CLEAN UP	8
9	2887	073602*	1060	TAD CCNT	/GET SEARCH POSITIION LIMIT	9
10	2888	073603*	1260	TAD M16/(-16)	/COMPARE TO LIMIT	10
11	2889	073604*	7700	SMA CLA	/SKIP IF LESS THAN THIS MAXIMUM	11
12	2890	073605*	5234	JMP TRYPRATE	/JUMP IF NOT	12
13	2891	073606*	1060	TAD CCNT	/GET POSITION	13
14	2892	073607*	7110	CLL RAR	/%2	14
15	2893	073610*	7040	CMA	/INVERT IT	15
16	2894	073611*	3257	DCA SELSHFT	/SAVE AS COUNTER	16
17	2895	073612*	7120	STL	/SET A STOP BIT JUST IN CASE	17
18	2896	073613*	7004	SHFTLUP, RAL	/MOVE UP	18
19	2897	073614*	2257	ISZ SELSHFT	/DONE ENOUGH?	19
20	2898	073615*	5213	JMP SHFTLUP	/NO, GO BACK	20
21	2899	073616*	7421	SQL	/SAVE FOR NOW	21
22	2900	073617*	7701	CLA MQA	/GET IT BACK	22
23	2901	073620*	7040	CMA	/INVERT IT	23
24	2902	073621*	0024	AND TEMP1	/SAVE THE OTHER BITS	24
25	2903	073622*	3024	DCA TEMP1	/STORE BACK FOR NOW	25
26	2904	073623*	1060	TAD CCNT	/GET THE POSITION COUNT	26
27	2905	073624*	7110	CLL RAR	/%2	27
28	2906	073625*	7200	CLA	/CLEAN UP	28
29	2907	073626*	1024	TAD TEMP1	/GET THE OTHER BITS	29
30	2908	073627*	7430	SZL	/SKIP IF BIT IS OFF	30
31	2909	073630*	7501	MQA	/ELSE OR IN SELECTED VALUE	31
32	2910	073631*	3024	DCA TEMP1	/SAVE COMPOSITE VALUE	32
33	2911	073632*	2200	SELXIT, ISZ SELCT	/TAKE SKIP RETURN	33
34	2912	073633*	5600	JMP I SELCT	/RETURN TO COMMAND ROUTINE	34
35	2913					35
36	2914	073634*	1060	TRYPRAT, TAD CCNT	/GET POSITION	36
37	2915	073635*	1260	TAD M16/(-16)	/ADD ON UPPER LIMIT	37
38	2916	073636*	1261	TAD L7773/(-5)	/ADD ON EXTRA LIMIT	38
39	2917	073637*	7540	SMA SZA	/SKIP IF IN THIS RANGE	39
40	2918	073640*	5247	JMP DOCOMRATE	/JUMP IF NOT	40
41	2919	073641*	1262	TAD L5/(5)	/SUBTRACT RELATIVE LIMIT	41
42	2920	073642*	1263	TAD PBTABLE/(BTABLE)	/ADD ON TABLE BASE	42
43	2921	073643*	3025	DCA TEMP2	/STASH THE POINTER	43
44	2922	073644*	1425	TAD I TEMP2	/GET THE PROPER BAUD RATE VALUE	44
45	2923	073645*	3025	DCA TEMP2	/STASH IT	45
46	2924	073646*	5232	JMP SELXIT	/FINISH THERE	46
47	2925					47
48	2926	073647*	3026	DOCOMRA, DCA TEMP3	/SAVE RELATIVE POSITION	48
49	2927	073650*	7240	NL7777	/-1	49
50	2928	073651*	1026	TAD TEMP3	/NOW HAVE PROPER RELATIVE VALUE	50
51	2929	073652*	1263	TAD PBTABLE/(BTABLE)	/ADD ON TABLE ADDRESS	51
52	2930	073653*	3026	DCA TEMP3	/STASH THE POINTER	52
53	2931	073654*	1426	TAD I TEMP3	/GET THE PROPER BAUD RATE VALUE	53
54	2932	073655*	3026	DCA TEMP3	/STASH IT	54
55	2933	073656*	5232	JMP SELXIT	/CONTINUE THERE	55
56						56
57						57
58						58

1	2935	/	*****			1
2	2936					2
3	2937	073657* 0000	SELSHFT, .-. .	/SELECT ROUTINE COUNTER		3
4	2938	073660* 7762	M16, -16	/CONSTANT 7762		4
5	2939	073661* 7773	L7773, -5	/CONSTANT 7773		5
6	2940	073662* 0005	L5, 5	/CONSTANT 0005		6
7	2941	073663* 3664	PBTABLE, BTABLE	/POINTER TO BAUD RATE TABLE		7
8	2942					8
9	2943	/	BAUD RATE TABLE FOR SUPPORTED RATES.			9
10	2944					10
11	2945	073664* 0005	BTABLE, 05	/300 BAUD		11
12	2946	073665* 0006	06	/600 BAUD		12
13	2947	073666* 0007	07	/1200 BAUD		13
14	2948	073667* 0012	12	/2400 BAUD		14
15	2949	073670* 0014	14	/4800 BAUD		15
16	2950	073671* 0016	16	/9600 BAUD		16
17	2951					17
18	2952	/	*****			18
19	2953					19
20	2954	073672* 0000	KBDIN, .-. .	/COMMAND INPUT ROUTINE		20
21	2955	073673* 6111	KBISF	/FLAG UP?		21
22	2956	073674* 5273	JMP .-1	/NO, WAIT FOR IT		22
23	2957	073675* 6116	KBIRB	/YES, READ IT IN		23
24	2958	073676* 0115	AND Z377/[377]	/JUST EIGHT BITS		24
25	2959	073677* 5672	JMP I KBDIN	/RETURN		25
26	2960					26
27	2961	073700* 1140	ARXWAIT, RXWAIT	/POINTER TO RX50 WAIT ROUTINE		27
28	2962					28
29	2963	073701* 0000	RXRDY, .-. .	/DRIVE READY CHECK ROUTINE		29
30	2964	073702* 1106	TAD Z12/[12]	/GET READ STATUS VALUE		30
31	2965	073703* 6751	LCD	/LOAD THE COMMAND		31
32	2966	073704* 4700	JMS I ARXWAIT/(RXWAIT)	/WAIT FOR IT		32
33	2967	073705* 5701	JMP I RXRDY	/TRANSFER FLAG CAME UP, FORGET IT		33
34	2968	073706* 7410	SKP	/DONE FLAG CAME UP, THERE'S HOPE		34
35	2969	073707* 5701	JMP I RXRDY	/ERROR OR TIME-OUT		35
36	2970	073710* 6752	XDR	/GET STATUS		36
37	2971	073711* 0112	AND Z200/[200]	/JUST DRIVE READY BIT		37
38	2972	073712* 7640	SZA CLA	/SKIP IF NOT READY		38
39	2973	073713* 2301	ISZ RXRDY	/ELSE TAKE SKIP RETURN		39
40	2974	073714* 5701	JMP I RXRDY	/RETURN EITHER WAY		40
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1	2976	/	*****				1
2	2977						2
3	2978	073715*	0000	VIDCLR,	.-.	/VIDEO CLEAR (?) ROUTINE	3
4	2979	073716*	7333		NL6000	/SET BOTH CONTROL BITS	4
5	2980	073717*	6126		VIDCON	/LOAD CONTROL REGISTER	5
6	2981	073720*	7200		CLA	/CLEAN UP	6
7	2982	073721*	3010		DCA XR0	/CLEAR INNER COUNTER	7
8	2983	073722*	7346		NL7775	/SET THE	8
9	2984	073723*	3011		DCA XR1	/OUTER COUNTER	9
10	2985	073724*	2010	VIDWAIT,	ISZ XR0	/WAITING ENOUGH?	10
11	2986	073725*	5324		JMP VIDWAIT	/NO, KEEP GOING	11
12	2987	073726*	2011		ISZ XR1	/WAITED TOO LONG?	12
13	2988	073727*	5324		JMP VIDWAIT	/NO, KEEP GOING	13
14	2989	073730*	6126		VIDCON	/RESET TO NORMAL VIDEO MODE	14
15	2990	073731*	5715		JMP I VIDCLR	/RETURN	15
16	2991						16
17	2992	/		SLUSHWARE TEST VALIDATION DATA.			17
18	2993						18
19	2994	073732*	1577	SLTDAT,	-6201		19
20	2995	073733*	7745		-0033		20
21	2996	073734*	7701		-0077		21
22	2997	073735*	0000		0	/THIS ENDS THE LIST	22
23	2998						23
24	2999	/		*****			24
25	3000						25
26	3001	/		COMMAND DISPATCH TABLE FOR BUILT-IN COMMANDS.			26
27	3002						27
28	3003	073736*	7564	COMTABL,-214;	BOOTIT	/REMOVE	28
29	3004	073737*	4401				29
30	3005	073740*	7563		-215; SELCT	/SELECT	30
31	3006	073741*	3600				31
32	3007	073742*	7603		-175; DOKEY	/DO	32
33	3008	073743*	4446				33
34	3009	073744*	7531		-247; MOVLEFT	/LEFT ARROW	34
35	3010	073745*	4275				35
36	3011	073746*	7530		-250; MOVRITE	/RIGHT ARROW	36
37	3012	073747*	4311				37
38	3013	073750*	7503		-275; RETURN	/RETURN	38
39	3014	073751*	4606				39
40	3015	073752*	7514		-264; METRONOME	/METRONOME	40
41	3016	073753*	4600				41
42	3017	073754*	0000		0	/THIS ENDS THE LIST	42
43							43
44							44
45							45
46							46
47							47
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1	3019	/	*****	1
2	3020			2
3	3021	/	VALIDATION DATA FOR TRACK 0, SECTORS 1-2.	3
4	3022			4
5	3023	073755* 7777	T0S1DAT, -1	5
6	3024	073756* 7655	-"S!200	6
7	3025	073757* 7673	-"E!200	7
8	3026	073760* 7654	-"T!200	8
9	3027	073761* 7653	-"U!200	9
10	3028	073762* 7660	-"P!200	10
11	3029	073763* 0000	0 /THIS ENDS THE LIST	11
12	3030			12
13	3031	/	VALIDATION DATA FOR TRACK 0, SECTORS 7-10.	13
14	3032			14
15	3033	073764* 7777	T0S7DAT, -1	15
16	3034	073765* 7653	-"U!200	16
17	3035	073766* 7675	-"C!200	17
18	3036	073767* 7671	-"G!200	18
19	3037	073770* 7656	-"R!200	19
20	3038	073771* 7677	-"A!200	20
21	3039	073772* 7663	-"M!200	21
22	3040	073773* 7776	-2	22
23	3041	073774* 0000	0 /THIS ENDS THE LIST	23
24	3042			24
25	3043	/	VALIDATION DATA FOR TRACK 0, SECTOR 3.	25
26	3044			26
27	3045	073775* 7777	T0S3DAT, -1	27
28	3046	073776* 7665	-"K!200	28
29	3047	073777* 7676	-"B!200	29
30	3048	074000* 7674	-"D!200	30
31	3049	074001* 7676	-"B!200	31
32	3050	074002* 7656	-"R!200	32
33	3051	074003* 7674	-"D!200	33
34	3052	074004* 7776	-2	34
35	3053	074005* 0000	0 /THIS ENDS THE LIST	35
36				36
37				37
38				38
39				39
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41				41
42				42
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44				44
45				45
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[illegible]

1	3107	074064*	1025	PRTRATE,	TAD	TEMP2	/GET SECOND PARAMETER	1
2	3108	074065*	4303		JMS	NPRT	/OUTPUT IT	2
3	3109	074066*	1026		TAD	TEMP3	/GET THIRD PARAMETER	3
4	3110	074067*	4303		JMS	NPRT	/OUTPUT IT	4
5	3111	074070*	1410	OUTLUP,	TAD I	XR0	/GET A CHARACTER	5
6	3112	074071*	7421		SQL		/LOAD FOR PRINTING	6
7	3113	074072*	7701		CLA MQA		/GET IT BACK	7
8	3114	074073*	1133		TAD	Z7775/[7775]	/COMPARE TO END CHARACTER	8
9	3115	074074*	7650		SNA CLA		/SKIP IF NOT AT END	9
10	3116	074075*	5607		JMP I	PRMPRT	/RETURN IF AT END	10
11	3117	074076*	4501		JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	11
12	3118	074077*	5270		JMP	OUTLUP	/GO DO ANOTHER	12
13	3119							13
14	3120			/		UNREFERENCED LOCATION. PERHAPS ERRONEOUS NOTION THAT CHRPRT ROUTINE MIGHT		14
15	3121			/		SKIP OR OTHER OBSOLETE CONVENTION?		15
16	3122							16
17	3123	074100*	5607		JMP I	PRMPRT	/RETURN	17
18	3124							18
19	3125	074101*	0000	PRMTM1,	...		/TEMPORARY	19
20	3126	074102*	0000	PRMTM2,	...		/TEMPORARY	20
21	3127							21
22	3128			/		*****		22
23	3129							23
24	3130			/		PRINT NUMERICAL PARAMETER ROUTINE. THE SELECTED PARAMETER IS A BAUD RATE.		24
25	3131			/		THE (CLOSEST TO THE) SELECTED RATE IS HIGHLIGHTED.		25
26	3132							26
27	3133	074103*	0000	NPRT,	...			27
28	3134	074104*	3351		DCA	NTMP	/SAVE THE PASSED VALUE	28
29	3135	074105*	1351		TAD	NTMP	/GET IT BACK	29
30	3136	074106*	0107		AND	Z17/[17]	/JUST RATE BITS	30
31	3137	074107*	1352		TAD	PCLIST/(CLIST)	/ADD ON TABLE BASE	31
32	3138	074110*	3351		DCA	NTMP	/STASH THE POINTER	32
33	3139	074111*	1751		TAD I	NTMP	/GET THE PROPER COUNT	33
34	3140	074112*	3351		DCA	NTMP	/STASH IT	34
35	3141	074113*	1410	NPLUP,	TAD I	XR0	/GET A CHARACTER	35
36	3142	074114*	7421		SQL		/LOAD IT FOR PRINTING	36
37	3143	074115*	7701		CLA MQA		/GET IT BACK	37
38	3144	074116*	1130		TAD	Z7741/[-37]	/COMPARE TO FIELD DELIMITER	38
39	3145	074117*	7450		SNA		/SKIP IF NOT A MATCH	39
40	3146	074120*	5336		JMP	FNDDELIM	/JUMP IF IT IS	40
41	3147	074121*	7001		IAC		/COMPARE TO ALTERNATE DELIMITER VALUE	41
42	3148	074122*	7650		SNA CLA		/SKIP IF NOT A MATCH THERE EITHER	42
43	3149	074123*	5336		JMP	FNDDELIM	/JUMP IF IT IS A MATCH	43
44	3150	074124*	4501	NEXTREG,	JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	44
45	3151	074125*	5313		JMP	NPLUP	/GO DO NEXT	45
46	3152							46
47	3153	074126*	4501	MOREPRM,	JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	47
48	3154	074127*	7701		CLA MQA		/GET THE CHARACTER	48
49	3155	074130*	1126		TAD	Z7766/[7766]	/COMPARE TO MASTER DELIMITER	49
50	3156	074131*	7650		SNA CLA		/SKIP IF NOT AT END	50
51	3157	074132*	5703		JMP I	NPRT	/RETURN IF AT END	51
52	3158	074133*	1410		TAD I	XR0	/GET NEXT CHARACTER	52
53	3159	074134*	7421		SQL		/LOAD IT FOR PRINTING	53
54	3160	074135*	5326		JMP	MOREPRM	/KEEP GOING	54
55								55
56								56
57								57
58								58

1	3162		/	COMES HERE IF FIELD DELIMITER FOUND.		1
2	3163					2
3	3164	074136*	2351	FNDDELI, ISZ	NTMP	3
4	3165	074137*	5324	JMP	NEXTREG	4
5	3166	074140*	4501	BOLDLP, JMS I	PCHRPRT/[CHRPRT]	5
6	3167	074141*	1410	TAD I	XR0	6
7	3168	074142*	7421	MQL		7
8	3169	074143*	7701	CLA MQA		8
9	3170	074144*	1130	TAD	Z7741/[-37]	9
10	3171	074145*	7750	SPA SNA	CLA	10
11	3172	074146*	5326	JMP	MOREPRM	11
12	3173	074147*	1116	TAD	Z400/[400]	12
13	3174	074150*	5340	JMP	BOLDLP	13
14	3175					14
15	3176	074151*	0000	NTMP, .-. .		15
16	3177	074152*	3365	PCLIST, CLIST		16
17	3178					17
18	3179			/	SLUSHWARE CODE VALIDATION ROUTINE.	18
19	3180					19
20	3181	074153*	0000	SLCHECK, .-. .		20
21	3182	074154*	6214	RDF		21
22	3183	074155*	1122	TAD	ZCDF/[CDF]	22
23	3184	074156*	3365	DCA	SLUCDF	23
24	3185	074157*	1375	TAD	PSLTDAT/(SLTDAT-1)	24
25	3186	074160*	3011	DCA	XR1	25
26	3187	074161*	6271	CDF	70	26
27	3188	074162*	1411	SLNEXT, TAD I	XR1	27
28	3189	074163*	7450	SNA		28
29	3190	074164*	5373	JMP	SLGOOD	29
30	3191	074165*	6201	SLUCDF, CDF	00+.-. .	30
31	3192	074166*	1410	TAD I	XR0	31
32	3193	074167*	6271	CDF	70	32
33	3194	074170*	7650	SNA CLA		33
34	3195	074171*	5362	JMP	SLNEXT	34
35	3196	074172*	5753	JMP I	SLCHECK	35
36	3197					36
37	3198	074173*	2353	SLGOOD, ISZ	SLCHECK	37
38	3199	074174*	5753	JMP I	SLCHECK	38
39	3200					39
40	3201	074175*	3731	PSLTDAT, SLTDAT-1		40
41	3202					41
42	3203	074176*	0000	ZBLOCK	2	42
43						43
44						44
45						45
46						46
47						47
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57						57
58						58

1	3205	*4200	PAGE		1
2	3206				2
3	3207	/	*****		3
4	3208				4
5	3209	074200*	3661 HIGHROW, ROWADR+1	/POINTER TO ROWTABLE (HIGH-ORDER)	5
6	3210	074201*	2030 L2030, 2030	/CONSTANT 2030	6
7	3211	074202*	0031 L31, 31	/CONSTANT 0031	7
8	3212	074203*	7747 M31, -31	/CONSTANT 7747	8
9	3213				9
10	3214	074204*	0000 CHRPRT, .-. .	/CHARACTER PRINT ROUTINE	10
11	3215	074205*	3254 DCA ATTRVAL	/SAVE PASSED ATTRIBUTE VALUE	11
12	3216	074206*	7701 CLA MQA	/GET THE CHARACTER	12
13	3217	074207*	1126 TAD Z7766/[-"J!300]	/COMPARE TO <LF>	13
14	3218	074210*	7450 SNA	/SKIP IF OTHER	14
15	3219	074211*	5252 JMP JUSTPOS	/JUMP IF IT MATCHES	15
16	3220	074212*	1133 TAD Z7775/[-"M+"J]	/COMPARE TO <CR>	16
17	3221	074213*	7650 SNA CLA	/SKIP IF NO MATCH	17
18	3222	074214*	5252 JMP JUSTPOS	/JUMP IF IT MATCHES	18
19	3223	074215*	1056 TAD ROW	/GET DESIRED ROW NUMBER	19
20	3224	074216*	7104 CLL RAL	/*2	20
21	3225	074217*	1200 TAD HIGHROW/(ROWADR+1)	/POINT TO PROPER HIGH-ORDER ADDRESS	21
22	3226	074220*	3255 DCA CHARADR	/STASH THE POINTER	22
23	3227	074221*	6211 CDF 10	/GOTO ROWTABLE FIELD	23
24	3228	074222*	1655 TAD I CHARADR	/GET THE HIGH-ORDER ROW ADDRESS	24
25	3229	074223*	6271 CDF 70	/BACK TO OUR FIELD	25
26	3230	074224*	3255 DCA CHARADR	/STASH THE POINTER	26
27	3231	074225*	1255 TAD CHARADR	/GET HIGH-ORDER ROW ADDRESS	27
28	3232	074226*	7110 CLL RAR	/MOVE OVER	28
29	3233	074227*	0201 AND L2030/(30)	/JUST FIELD BITS (CLOSE ENOUGH)	29
30	3234	074230*	1122 TAD ZCDF/[CDF]	/FORM CDF INSTRUCTION	30
31	3235	074231*	3247 DCA CHARCDF	/STORE IN-LINE	31
32	3236	074232*	1255 TAD CHARADR	/GET HIGH-ORDER ROW ADDRESS AGAIN	32
33	3237	074233*	0107 AND Z17/[17]	/JUST QUARTER-FIELD BITS	33
34	3238	074234*	7106 CLL RTL;BSW	/*400	34
35	3239	074235*	7002		35
36	3240	074236*	1057 TAD COLUMN	/ADD ON DESIRED COLUMN	36
37	3241	074237*	1153 TAD Z205/[205]	/ADD ON LINE BASE	37
38	3242	074240*	3255 DCA CHARADR	/STASH THE POINTER	38
39	3243	074241*	7701 CLA MQA	/GET THE CHARACTER	39
40	3244	074242*	1256 TAD M40/(-40)	/COMPARE TO <SPACE>	40
41	3245	074243*	7740 SMA SZA CLA	/SKIP IF <SPACE> OR CONTROL CHARACTER	41
42	3246	074244*	7701 CLA MQA	/ELSE GET THE CHARACTER	42
43	3247	074245*	7440 SZA	/SKIP IF NOT DISPLAYABLE	43
44	3248	074246*	1254 TAD ATTRVAL	/ELSE ADD ON PASSED ATTRIBUTES	44
45	3249	074247*	7402 CHARCDF, HLT+.-.	/WILL BE CDF 20 (OR 30) INSTRUCTION	45
46	3250	074250*	3655 DCA I CHARADR	/STORE IN LINE BUFFER	46
47	3251	074251*	6271 CDF 70	/BACK TO OUR FIELD	47
48	3252	074252*	4257 JUSTPOS, JMS SCRPOS	/ADVANCE SCREEN POSITION	48
49	3253	074253*	5604 JMP I CHRPRT	/RETURN	49
50	3254				50
51	3255	074254*	0000 ATTRVAL, .-. .	/PASSED ATTRIBUTE BITS SAVED HERE	51
52	3256	074255*	0000 CHARADR, .-. .	/ADDRESS TEMPORARY	52
53	3257	074256*	7740 M40, -" !200	/CONSTANT 7740	53
54					54
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57					57
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1	3259	/	*****				1
2	3260						2
3	3261	074257* 0000	SCRPOS, .-. .		/SCREEN POSITION ADVANCE ROUTINE		3
4	3262	074260* 7701	CLA MQA		/GET THE CHARACTER		4
5	3263	074261* 1126	TAD	Z7766/[-"J!300]	/COMPARE TO <LF>		5
6	3264	074262* 7440	SZA		/SKIP IF IT MATCHES		6
7	3265	074263* 5266	JMP	SCRCHK	/JUMP IF NOT		7
8	3266	074264* 2056	ISZ	ROW	/BUMP TO NEXT ROW		8
9	3267	074265* 5657	JMP I	SCRPOS	/RETURN		9
10	3268						10
11	3269	074266* 1133	SCRCHK, TAD	Z7775/[-"M+"J]	/COMPARE TO <LF>		11
12	3270	074267* 7640	SZA CLA		/SKIP IF IT MATCHES		12
13	3271	074270* 5273	JMP	SCREGULAR	/JUMP IF NOT		13
14	3272	074271* 3057	DCA	COLUMN	/CLEAR COLUMN		14
15	3273	074272* 5657	JMP I	SCRPOS	/RETURN		15
16	3274						16
17	3275	074273* 2057	SCREGUL, ISZ	COLUMN	/BUMP TO NEXT COLUMN		17
18	3276	074274* 5657	JMP I	SCRPOS	/RETURN		18
19	3277						19
20	3278	/	*****				20
21	3279						21
22	3280	/	HANDLER FOR LEFT ARROW KEY IN SETUP ROUTINE.				22
23	3281						23
24	3282	074275* 0000	MOVLEFT, .-. .		/MOVE BACK ONE POSITION ROUTINE		24
25	3283	074276* 3055	DCA	LCHAR	/SAVE OUR CHARACTER FOR OTHERS		25
26	3284	074277* 1152	TAD	PCALLSUB/[CALLSUB+1-1]	/GET BUFFER POINTER		26
27	3285	074300* 3010	DCA	XR0	/STASH THE POINTER		27
28	3286	074301* 6271	CDF	70	/GOTO OUR FIELD		28
29	3287	074302* 7240	NL7777		/TRY TO BACKUP		29
30	3288	074303* 1060	TAD	CCNT	/COMPARE TO CURRENT SELECT		30
31	3289	074304* 7510	SPA		/SKIP IF IT CAN BACKUP		31
32	3290	074305* 7200	CLA		/ELSE FORGET IT		32
33	3291	074306* 3060	DCA	CCNT	/SAVE NEW POSITION		33
34	3292	074307* 4321	JMS	LOC36	/LOCATE NEW DELIMITER		34
35	3293	074310* 5675	JMP I	MOVLEFT	/RETURN		35
36	3294						36
37	3295	/	*****				37
38	3296						38
39	3297	/	HANDLER FOR RIGHT ARROW KEY IN SETUP ROUTINE.				39
40	3298						40
41	3299	074311* 0000	MOVRITE, .-. .		/MOVE FORWARD ONE POSITION ROUTINE		41
42	3300	074312* 3055	DCA	LCHAR	/SAVE OUR CHARACTER FOR OTHERS		42
43	3301	074313* 1060	TAD	CCNT	/GET CURRENT POSITION		43
44	3302	074314* 1203	TAD	M31/(-31)	/COMPARE TO UPPER LIMIT		44
45	3303	074315* 7710	SPA CLA		/SKIP IF TOO FAR		45
46	3304	074316* 2060	ISZ	CCNT	/ELSE BUMP TO NEXT		46
47	3305	074317* 4321	JMS	LOC36	/LOCATE NEW DELIMITER		47
48	3306	074320* 5711	JMP I	MOVRITE	/RETURN		48
49							49
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1	3308	/	*****				1
2	3309						2
3	3310	/	THIS ROUTINE LOCATES DELIMITER (36 OR 37) CODES IN THE BUFFER. CCNT				3
4	3311	/	DETERMINES HOW MANY.				4
5	3312						5
6	3313	074321* 0000	LOC36, .-. /	DELIMITER SEARCH ROUTINE			6
7	3314	074322* 3056	DCA ROW	/CLEAR ROW			7
8	3315	074323* 3057	DCA COLUMN	/CLEAR COLUMN			8
9	3316	074324* 1152	TAD PCALLSUB/[CALLSUB]	/SETUP THE			9
10	3317	074325* 3010	DCA XR0	/BUFFER POINTER			10
11	3318	074326* 3011	DCA XR1	/CLEAR CHARACTER COUNTER			11
12	3319	074327* 1410	SRCHLUP, TAD I XR0	/GET A CHARACTER			12
13	3320	074330* 7421	MQL	/STASH IT THERE			13
14	3321	074331* 7701	CLA MQA	/GET IT BACK			14
15	3322	074332* 1130	TAD Z7741/[-37]	/COMPARE TO 37 CODE			15
16	3323	074333* 7450	SNA	/SKIP IF OTHER			16
17	3324	074334* 5340	JMP IS37	/JUMP IF IT MATCHES			17
18	3325	074335* 7001	IAC	/COMPARE TO 36 CODE			18
19	3326	074336* 7640	SZA CLA	/SKIP IF IT MATCHES			19
20	3327	074337* 5351	JMP NOTOURS	/JUMP IF NOT			20
21	3328	074340* 1011	IS37, TAD XR1	/GET CHARACTER COUNT			21
22	3329	074341* 7041	CIA	/INVERT FOR TEST			22
23	3330	074342* 1060	TAD CCNT	/COMPARE TO LIMIT			23
24	3331	074343* 7640	SZA CLA	/SKIP IF SEARCHED ENOUGH			24
25	3332	074344* 5350	JMP ISOURS	/JUMP IF NOT			25
26	3333	074345* 2057	ISZ COLUMN	/BUMP TO NEXT COLUMN			26
27	3334	074346* 4353	JMS CUPDATE	/PLACE CURSOR THERE			27
28	3335	074347* 5721	JMP I LOC36	/RETURN			28
29	3336						29
30	3337	074350* 2011	ISOURS, ISZ XR1	/BUMP CHARACTER COUNT			30
31	3338	074351* 4257	NOTOURS, JMS SCRPOS	/ADVANCE SCREEN POSITION			31
32	3339	074352* 5327	JMP SRCHLUP	/KEEP GOING			32
33	3340						33
34	3341	/	*****				34
35	3342						35
36	3343	074353* 0000	CUPDATE, .-. /	CURSOR UPDATE ROUTINE			36
37	3344	074354* 1201	TAD L2030/(2030)	/GET VERTICAL CURSOR REGISTER VALUE			37
38	3345	074355* 6122	LSCREG	/LOAD REGISTER SELECT			38
39	3346	074356* 7200	CLA	/CLEAN UP			39
40	3347	074357* 1056	TAD ROW	/GET CURRENT ROW			40
41	3348	074360* 6124	VLOAD	/LOAD CURSOR VERTICAL POSITION			41
42	3349	074361* 7200	CLA	/CLEAN UP			42
43	3350	074362* 1202	TAD L31/(31)	/GET HORIZONTAL CURSOR REGISTER VALUE			43
44	3351	074363* 6122	LSCREG	/LOAD REGISTER SELECT			44
45	3352	074364* 7200	CLA	/CLEAN UP			45
46	3353	074365* 1057	TAD COLUMN	/GET CURRENT COLUMN			46
47	3354	074366* 6124	VLOAD	/LOAD CURSOR HORIZONTAL POSITION			47
48	3355	074367* 7200	CLA	/CLEAN UP			48
49	3356	074370* 5753	JMP I CUPDATE	/RETURN			49
50	3357						50
51	3358	074371* 0000	ZBLOCK 7	/EMPTY SPACE			51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

	*4400	PAGE		
/		*****		
PCUPDAT,CUPDATE			/POINTER TO CURSOR UPDATE ROUTINE	
HANDLER FOR REMOVE KEY IN SETUP ROUTINE.				
BOOTIT,	.- .		/REMOVE KEY ROUTINE	
CLA			/CLEAN UP	
JMS I AVIDCLR/(VIDCLR)			/CLEAR VIDEO DATA	
CDF 70			/GOTO OUR FIELD	
DCA ROW			/CLEAR ROW	
DCA COLUMN			/CLEAR COLUMN	
JMS I PCUPDATE/(CUPDATE)			/UPDATE CURSOR POSITION	
TAD PCALLSUB/[CALLSUB]			/SETUP THE	
DCA XR0			/DATA POINTER	
NL7775			/SETUP THE	
DCA XR1			/DATA COUNTER	
TAD I XR0			/GET A WORD	
Z7776/[7776]			/COMPARE TO VALIDATION FIELD DELIMITER	
SZA CLA			/SKIP IF IT MATCHES	
JMP LOOKLP			/ELSE KEEP GOING	
ISZ XR1			/FOUND ENOUGH OF THEM?	
JMP LOOKLP			/NO, KEEP GOING	
TAD I XR0			/GET A CHARACTER	
MQL			/LOAD FOR PRINTING ROUTINE	
CLA MQA			/GET IT BACK	
TAD Z7775/[-3]			/COMPARE TO MESSAGE END CHARACTER	
SNA CLA			/SKIP IF NOT THERE YET	
JMP PEND			/JUMP IF AT END	
JMS I PCHRPT/[CHRPRT]			/PRINT THE CHARACTER	
JMP PLUP			/KEEP GOING	
JMS I PKBDIN/[KBDIN]			/GET A KEYBOARD CHARACTER	
L7603/(-175)			/COMPARE TO VALUE FOR DO KEY	
SNA CLA			/SKIP IF IT DOESN'T MATCH	
JMP I PRXOK/(RXOK)			/JUMP IF IT DOES	
DCA CCNT			/CLEAR POSITION COUNT	
JMS I AVIDCLR/(VIDCLR)			/CLEAR VIDEO DATA	
ISZ BOOTIT			/TAKE SKIP RETURN	
JMP I BOOTIT			/RETURN TO COMMAND ROUTINE	
-175			/CONSTANT 7603	
PRXOK,RXOK			/POINTER TO RXOK ROUTINE	
C23, 23			/CONSTANT 0023	
AVIDCLR,VIDCLR			/POINTER TO VIDEO CLEAR ROUTINE	

1	3408	/	*****				1
2	3409						2
3	3410	/	HANDLER FOR DO KEY IN SETUP ROUTINE.				3
4	3411						4
5	3412	074446*	0000	DOKEY,	...	/DO KEY ROUTINE	5
6	3413	074447*	3055	DCA	LCHAR	/SAVE CHARACTER VALUE FOR OTHERS	6
7	3414	074450*	1244	TAD	C23/(23)	/SETUP FOR THE	7
8	3415	074451*	3056	DCA	ROW	/HIGHEST ROW	8
9	3416	074452*	3057	DCA	COLUMN	/CLEAR COLUMN	9
10	3417	074453*	1373	TAD	CM5/(-5)	/SETUP THE	10
11	3418	074454*	3011	DCA	XR1	/ROW COUNT	11
12	3419	074455*	1374	P1LUP, TAD	M120/(-120)	/SETUP THE	12
13	3420	074456*	3012	DCA	XR2	/COLUMN COUNT	13
14	3421	074457*	4501	P2LUP, JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	14
15	3422	074460*	2012	ISZ	XR2	/DONE A LINE YET?	15
16	3423	074461*	5257	JMP	P2LUP	/NO, KEEP GOING	16
17	3424	074462*	3057	DCA	COLUMN	/YES, CLEAR COLUMN	17
18	3425	074463*	2056	ISZ	ROW	/BUMP TO NEXT ROW	18
19	3426	074464*	2011	ISZ	XR1	/DONE ENOUGH ROW?	19
20	3427	074465*	5255	JMP	P1LUP	/NO, KEEP GOING	20
21	3428	074466*	1152	TAD	PCALLSUB/[CALLSUB+1-1]	/SETUP THE	21
22	3429	074467*	3010	DCA	XR0	/DATA POINTER	22
23	3430	074470*	1133	FNDIT, TAD	Z7775/[-3]	/GET DELIMITER VALUE	23
24	3431	074471*	1410	TAD I	XR0	/COMPARE TO LATEST	24
25	3432	074472*	7640	SZA CLA		/SKIP IF IT MATCHES	25
26	3433	074473*	5270	JMP	FNDIT	/ELSE KEEP GOING	26
27	3434	074474*	2010	ISZ	XR0	/BUMP PAST DELIMITER	27
28	3435	074475*	1244	TAD	C23/(23)	/SETUP FOR	28
29	3436	074476*	3056	DCA	ROW	/HIGHEST ROW	29
30	3437	074477*	3057	DCA	COLUMN	/CLEAR COLUMN	30
31	3438	074500*	1410	PNEXT, TAD I	XR0	/GET A CHARACTER	31
32	3439	074501*	7421	MQL		/LOAD FOR PRINTING	32
33	3440	074502*	7701	CLA MQA		/GET IT BACK	33
34	3441	074503*	1133	TAD	Z7775/[-3]	/COMPARE TO DELIMITER	34
35	3442	074504*	7650	SNA CLA		/SKIP IF NOT THERE YET	35
36	3443	074505*	5310	JMP	ATEND	/JUMP IF AT END OF BLURB	36
37	3444	074506*	4501	JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	37
38	3445	074507*	5300	JMP	PNEXT	/KEEP GOING	38
39	3446						39
40	3447	074510*	4551	ATEND, JMS I	PKBDIN/[KBDIN]	/GET A KEYBOARD CHARACTER	40
41	3448	074511*	7041	CIA		/INVERT IT	41
42	3449	074512*	1055	TAD	LCHAR	/COMPARE TO OUR VALUE (DO KEY)	42
43	3450	074513*	7650	SNA CLA		/SKIP IF OTHER	43
44	3451	074514*	5320	JMP	DOIT	/JUMP IF IT MATCHES	44
45	3452	074515*	4645	BMPRET, JMS I	AVIDCLR/(VIDCLR)	/CLEAR VIDEO DATA	45
46	3453	074516*	2246	NOCRET, ISZ	DOKEY	/BUMP RETURN ADDRESS	46
47	3454	074517*	5646	JMP I	DOKEY	/TAKE SKIP RETURN ON USER ABORT	47
48							48
49							49
50							50
51							51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

1	3456	074520*	1123	DOIT,	TAD	PWRIBUFF/[WRIBUFFER+1-1]	/SETUP THE	1
2	3457	074521*	3010		DCA	XR0	/DATA POINTER	2
3	3458	074522*	7305		NL002A		/SETUP 12-BIT EMPTY SILO COMMAND VALUE	3
4	3459	074523*	3053		DCA	RXCMD	/STASH THE COMMAND	4
5	3460	074524*	6231		CDF	30	/GOTO DATA FIELD	5
6	3461			/	CLA		/SET AC T0 1-1 FOR FIRST SECTOR	6
7	3462	074525*	4772		JMS I	LFREAD/(FREAD)	/CALL FLOPPY READ ROUTINE	7
8	3463	074526*	7776		-1-1		/WANT ONE SECTOR	8
9	3464	074527*	0116	L116,	116		/TRACK 78	9
10	3465	074530*	5316		JMP	NOCRET	/COULDN'T DO IT, FORGET IT	10
11	3466	074531*	1123		TAD	PWRIBUFF/[WRIBUFFER+1-1]	/GET BUFFER POINTER	11
12	3467	074532*	7001		IAC		/MOVE TO NEXT POSITION	12
13	3468	074533*	3010		DCA	XR0	/STASH THE POINTER	13
14	3469	074534*	6231		CDF	30	/GOTO BUFFER FIELD	14
15	3470	074535*	4771		JMS I	ASLCHECK/(SLCHECK)	/ENSURE SLUSHWARE IS VALID	15
16	3471	074536*	5366		JMP	SLUBAD	/FORGET IT, BAD SLUSHWARE	16
17	3472	074537*	3010		DCA	XR0	/SETUP POINTER TO 0001	17
18	3473	074540*	4775		JMS I	PPARMOVE/(PARMOVE)	/MOVE IN THE NEW PARAMETERS	18
19	3474			/	CLA		/SET 12-BIT FILL VALUE	19
20	3475	074541*	6751		LCD		/LOAD THE COMMAND FOR FILLING	20
21	3476	074542*	4776	LODLUP,	JMS I	LRXWAIT/(RXWAIT)	/CALL RX50 WAIT ROUTINE	21
22	3477	074543*	5346		JMP	LDTRANS	/TRANSFER FLAG CAME UP	22
23	3478	074544*	5352		JMP	LDDONE	/DONE FLAG CAME UP	23
24	3479	074545*	5777		JMP I	PFERR/(FERR)	/ERROR, GO COMPLAIN THERE	24
25	3480							25
26	3481	074546*	6231	LDTRANS,	CDF	30	/GOTO BUFFER FIELD	26
27	3482	074547*	1410		TAD I	XR0	/GET A WORD	27
28	3483	074550*	6752		XDR		/FILL IT	28
29	3484	074551*	5342		JMP	LODLUP	/KEEP GOING	29
30	3485							30
31	3486	074552*	1327	LDDONE,	TAD	L116/(116)	/SETUP TRACK 78	31
32	3487	074553*	3043		DCA	T1	/STASH IT	32
33	3488	074554*	7301		NL0001	CLL	/SETUP THE	33
34	3489	074555*	3041		DCA	TINC	/SECTOR NUMBER	34
35	3490	074556*	7307		NL0004		/SET WRITE FUNCTION	35
36	3491	074557*	6271		CDF	70	/BACK TO OUR FIELD	36
37	3492	074560*	4545		JMS I	PRXCMD/[RXCMD]	/GO DO THE WRITE COMMAND	37
38	3493	074561*	7410		SKP		/SKIP IF IT WAS OK	38
39	3494	074562*	5315		JMP	BMPRET	/JUMP IF NOT	39
40	3495	074563*	6754		SER		/ANY ERRORS?	40
41	3496	074564*	7410		SKP		/SKIP IF NOT	41
42	3497	074565*	5315		JMP	BMPRET	/JUMP IF SO	42
43	3498	074566*	2246	SLUBAD,	ISZ	DOKEY	/TAKE THE	43
44	3499	074567*	2246		ISZ	DOKEY	/DOUBLE-SKIP	44
45	3500	074570*	5646		JMP I	DOKEY	/RETURN	45
46	3501							46
47	3502	074571*	4153	ASLCHEC,	SLCHECK		/POINTER TO SLUSHWARE VALIDATION ROUTINE	47
48	3503	074572*	1311	LFREAD,	FREAD		/POINTER TO FREAD ROUTINE	48
49	3504	074573*	7773	CM5,	-5		/CONSTANT 7773	49
50	3505	074574*	7660	M120,	-120		/CONSTANT 7660	50
51	3506	074575*	4617	PPARMOV,	PARMOVE		/POINTER TO PARAMETER MOVE ROUTINE	51
52	3507	074576*	1140	LRXWAIT,	RXWAIT		/POINTER TO RX50 WAIT ROUTINE	52
53	3508	074577*	1343	PFERR,	FERR		/POINTER TO ERROR ROUTINE	53
54								54
55								55
56								56
57								57
58								58

1	3510		*4600	PAGE					1
2	3511								2
3	3512		/	*****					3
4	3513								4
5	3514		/	HANDLER FOR METRONOME CHARACTER IN SETUP ROUTINE.					5
6	3515								6
7	3516	074600*	0000	METRONO, .-. .			/METRONOME KEY ROUTINE		7
8	3517	074601*	7200	CLA			/CLEAN UP		8
9	3518	074602*	1055	TAD	LCHAR		/GET PREVIOUS CHARACTER		9
10	3519	074603*	5604	JMP I	.+1/(METRETRY)		/GO THERE		10
11	3520	074604*	3471	METRETRY			/THROUGH HERE		11
12	3521								12
13	3522	074605*	3715	LVIDCLR,VIDCLR			/POINTER TO VIDEO CLEAR ROUTINE		13
14	3523								14
15	3524		/	*****					15
16	3525								16
17	3526		/	HANDLER FOR RETURN CHARACTER IN SETUP ROUTINE.					17
18	3527								18
19	3528	074606*	0000	RETURN, .-. .			/RETURN KEY ROUTINE		19
20	3529	074607*	4605	JMS I	LVIDCLR/(VIDCLR)		/CLEAR VIDEO OUT		20
21	3530	074610*	4217	JMS	PARMOVE		/SETUP THE PARAMETERS		21
22	3531	074611*	2206	ISZ	RETURN		/TAKE THE		22
23	3532	074612*	2206	ISZ	RETURN		/DOUBLE-SKIP		23
24	3533	074613*	5606	JMP I	RETURN		/RETURN		24
25	3534								25
26	3535		/	*****					26
27	3536								27
28	3537	074614*	0024	A14,	24		/POINTER TO 0024		28
29	3538	074615*	0025	A15,	25		/POINTER TO 0025		29
30	3539	074616*	0026	A16,	26		/POINTER TO 0026		30
31	3540								31
32	3541	074617*	0000	PARMOVE, .-. .			/PARAMETER MOVE ROUTINE		32
33	3542	074620*	6201	CDF	00		/SETUP DATA FIELD		33
34	3543	074621*	1025	TAD	TEMP2		/GET SECOND PARAMETER		34
35	3544	074622*	3615	DCA I	A15/(0025)		/STORE IT		35
36	3545	074623*	1024	TAD	TEMP1		/GET FIRST PARAMETER		36
37	3546	074624*	3614	DCA I	A14/(0024)		/STORE IT		37
38	3547	074625*	1026	TAD	TEMP3		/GET THIRD PARAMETER		38
39	3548	074626*	3616	DCA I	A16/(0026)		/STORE IT		39
40	3549	074627*	6271	CDF	70		/BACK TO OUR FIELD		40
41	3550	074630*	3010	DCA	XR0		/SETUP SOURCE POINTER		41
42	3551	074631*	1123	TAD	PWRIBUFF/[WRIBUFFER+1-1]		/SETUP THE		42
43	3552	074632*	3011	DCA	XR1		/DESTINATION POINTER		43
44	3553	074633*	2011	ISZ	XR1		/BUMP PAST HEADER		44
45	3554	074634*	1250	TAD	M377/(-377)		/SETUP THE		45
46	3555	074635*	3247	DCA	MVCNT		/MOVE COUNTER		46
47	3556	074636*	6201	MVLUP, CDF	00		/GOTO DATA FIELD		47
48	3557	074637*	1410	TAD I	XR0		/GET A WORD		48
49	3558	074640*	6231	CDF	30		/GOTO BUFFER FIELD		49
50	3559	074641*	3411	DCA I	XR1		/PUT A WORD		50
51	3560	074642*	2247	ISZ	MVCNT		/DONE ENOUGH?		51
52	3561	074643*	5236	JMP	MVLUP		/NO, GO BACK		52
53	3562	074644*	1123	TAD	PWRIBUFF/[WRIBUFFER+1-1]		/SETUP THE		53
54	3563	074645*	3010	DCA	XR0		/DISK DATA POINTER		54
55	3564	074646*	5617	JMP I	PARMOVE		/RETURN		55
56									56
57									57
58									58

1	3566	074647*	0000	MVCNT, .-. .	/MOVE COUNTER	1
2	3567	074650*	7401	M377, -377	/CONSTANT 7401	2
3	3568	074651*	0017	BAUD17, 17	/CONSTANT 0017	3
4	3569					4
5	3570			/ COMES HERE AFTER TESTING THE KEYBOARD (OR JUST AFTER THE TEST WAS BYPASSED IF		5
6	3571			/ APTEN IS IN EFFECT) TO SETUP THE COMMUNICATIONS CHIP.		6
7	3572					7
8	3573	074652*	1251	COMTEST, TAD BAUD17/(17)	/GET BAUD RATE VALUE FOR 19200 BAUD	8
9	3574	074653*	6363	MSB	/SETUP COMMUNICATIONS CHIP RATE	9
10	3575	074654*	7200	CLA	/CLEAN UP	10
11	3576	074655*	6367	MPRESET	/RESET THE COMMUNICATIONS CHIP	11
12	3577	074656*	6121	APTSKP	/AUTOMATIC PRODUCT TESTING?	12
13	3578	074657*	7410	SKP	/SKIP IF NOT	13
14	3579	074660*	5270	JMP TESTCOM	/JUMP IF SO	14
15	3580	074661*	1104	COMEXIT, TAD Z7/[7]	/GET 1200 BAUD RATE VALUE	15
16	3581	074662*	6363	MSB	/SETUP COMM CHIP FOR 1200 BAUD	16
17	3582	074663*	7200	CLA	/CLEAN UP	17
18	3583	074664*	4667	JMS I PREGINIT/(REGINIT)	/INITIALIZE THE REGISTERS	18
19	3584	074665*	5666	JMP I .+1/(COMDONE)	/GO BACK TO MAIN-LINE CODE	19
20	3585	074666*	1007	COMDONE	/THROUGH HERE	20
21	3586					21
22	3587	074667*	5230	PREGINI, REGINIT	/POINTER TO REGISTER INITIALIZATION ROUTINE	22
23	3588					23
24	3589	074670*	4667	TESTCOM, JMS I PREGINIT/(REGINIT)	/INITIALIZE THE REGISTERS	24
25	3590	074671*	3030	DCA TM0	/CLEAR OUTPUT TEST VALUE	25
26	3591	074672*	3031	DCA COMTM1	/CLEAR INPUT TEST VALUE	26
27	3592	074673*	3050	DCA T6	/CLEAR LOW-ORDER TIME-OUT COUNTER	27
28	3593	074674*	1130	TAD Z7741/[-37]	/SETUP THE	28
29	3594	074675*	3051	DCA T7	/HIGH-ORDER TIME-OUT COUNTER	29
30	3595	074676*	6201	CDF 00	/SET EVEN FIELD TO ACCESS "A" SIDE	30
31	3596	074677*	6316	OLS	/OUTPUT A NULL CHARACTER	31
32	3597	074700*	6271	CDF 70	/BACK TO OUR FIELD	32
33	3598	074701*	2030	ISZ TM0	/BUMP OUTPUT VALUE	33
34	3599	074702*	6301	COMWAIT, ISF	/COMMUNICATIONS PORT FLAG UP?	34
35	3600	074703*	7410	SKP	/SKIP IF NOT	35
36	3601	074704*	5315	JMP COMCHK	/JUMP IF SO	36
37	3602	074705*	2050	ISZ T6	/WAITING TOO LONG?	37
38	3603	074706*	5302	JMP COMWAIT	/NO, KEEP TRYING	38
39	3604	074707*	2051	ISZ T7	/WAITING TOO MANY TIMES?	39
40	3605	074710*	5302	JMP COMWAIT	/NO, KEEP TRYING	40
41	3606	074711*	1372	COMBAD, TAD L100/(100)	/GET COMMUNICATIONS PORT ERROR BIT	41
42	3607	074712*	1054	TAD ESTATUS	/UPDATE CUMULATIVE ERROR STATUS	42
43	3608	074713*	3054	DCA ESTATUS	/STORE BACK	43
44	3609	074714*	5261	JMP COMEXIT	/STOP TESTING	44
45						45
46						46
47						47
48						48
49						49
50						50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	3611	074715*	4543	COMCHK, JMS I	PCOMREAD/[COMREAD]	/READ CONTENTS OF	1
2	3612	074716*	0012	B^10+2		/REGISTER 2B	2
3	3613	074717*	0107	AND	Z17/[17]	/JUST LOW-ORDER BITS AFFECTING "A" HALF	3
4	3614	074720*	1373	TAD	M6/(-6)	/RECEIVE CHARACTER AVAILABLE?	4
5	3615	074721*	7440	SZA		/SKIP IF AS EXPECTED	5
6	3616	074722*	5350	JMP	CCHKMORE	/JUMP IF NOT	6
7	3617	074723*	6201	CDF	00	/NEED EVEN FIELD TO ACCESS "A" SIDE	7
8	3618	074724*	6306	IRB		/GET THE CHARACTER	8
9	3619	074725*	6271	CDF	70	/BACK TO OUR FIELD	9
10	3620	074726*	0115	AND	Z377/[377]	/JUST DATA BITS	10
11	3621	074727*	7041	CIA		/INVERT FOR COMPARISON	11
12	3622	074730*	1031	TAD	COMTM1	/COMPARE TO EXPECTED VALUE	12
13	3623	074731*	7650	SNA CLA		/SKIP IF DIFFERENT	13
14	3624	074732*	5337	JMP	CREADOK	/JUMP IF IT MATCHES	14
15	3625	074733*	4542	JMS I	PCOMLOAD/[COMLOAD]	/CALL REGISTER LOAD ROUTINE	15
16	3626	074734*	0000	A^10+0		/REGISTER 0A	16
17	3627	074735*	0030	030		/CHANNEL RESET	17
18	3628	074736*	5311	JMP	COMBAD	/FORGET IT	18
19	3629						19
20	3630	074737*	2031	CREADOK, ISZ	COMTM1	/BUMP INPUT TEST VALUE	20
21	3631	074740*	1031	TAD	COMTM1	/GET THE VALUE	21
22	3632	074741*	0116	AND	Z400/[400]	/JUST TOO-FAR BIT	22
23	3633	074742*	7640	SZA CLA		/SKIP IF STILL GOING	23
24	3634	074743*	5261	JMP	COMEXIT	/JUMP IF FINISHED	24
25	3635	074744*	4542	CINTCLR, JMS I	PCOMLOAD/[COMLOAD]	/CALL REGISTER LOAD ROUTINE	25
26	3636	074745*	0000	A^10+0		/0A	26
27	3637	074746*	0070	070		/END OF INTERRUPT	27
28	3638	074747*	5302	JMP	COMWAIT	/GO BACK FOR MORE	28
29	3639						29
30	3640	074750*	1374	CCHKMOR, TAD	L2/(-4+6)	/TRANSMITTER BUFFER EMPTY?	30
31	3641	074751*	7640	SZA CLA		/SKIP IF IT MATCHES	31
32	3642	074752*	5311	JMP	COMBAD	/JUMP IF NOT	32
33	3643	074753*	1030	TAD	TM0	/GET OUTPUT TEST VALUE	33
34	3644	074754*	0116	AND	Z400/[400]	/JUST TOO-FAR BIT	34
35	3645	074755*	7650	SNA CLA		/SKIP IF NOW TOO FAR	35
36	3646	074756*	5363	JMP	OUTMORE	/JUMP IF NOT	36
37	3647	074757*	4542	JMS I	PCOMLOAD/[COMLOAD]	/CALL REGISTER LOAD ROUTINE	37
38	3648	074760*	0000	A^10+0		/0A	38
39	3649	074761*	0050	050		/CLEAR TRANSMITTER INTERRUPT	39
40	3650	074762*	5344	JMP	CINTCLR	/CONTINUE THERE	40
41	3651						41
42	3652	074763*	1030	OUTMORE, TAD	TM0	/GET OUTPUT VALUE	42
43	3653	074764*	6201	CDF	00	/SET EVEN FIELD TO ACCESS "A" HALF	43
44	3654	074765*	6316	OLS		/OUTPUT THE CHARACTER	44
45	3655	074766*	6271	CDF	70	/BACK TO OUR FIELD	45
46	3656	074767*	7200	CLA		/CLEAN UP	46
47	3657	074770*	2030	ISZ	TM0	/BUMP TEST VALUE	47
48	3658	074771*	5344	JMP	CINTCLR	/CONTINUE THERE	48
49	3659						49
50	3660	074772*	0100	L100, 100		/CONSTANT 0100	50
51	3661	074773*	7772	M6, -6		/CONSTANT 7772	51
52	3662	074774*	0002	L2, -4+6		/CONSTANT 0002	52
53	3663						53
54	3664	074775*	0000	ZBLOCK 3		/EMPTY SPACE	54
55							55
56							56
57							57
58							58

1	3666	*5000	PAGE			1
2	3667					2
3	3668	/	RD TEST AND SETUP ROUTINE. CHECKS IF RD51 HARDWRE IS PRESENT. IF PRESENT,			3
4	3669	/	WILL WAIT MAY SECONDS UNTIL THE SELF-TEST FINISHES AND RAISES THE DONE FLAG OR			4
5	3670	/	IT TIMES OUT. THEN THE STARTUP SECTOR IS READ IN AND EXECUTED AS A			5
6	3671	/	SUBROUTINE LOADED IN A DESIGNATED PLACE.			6
7	3672					7
8	3673	075000* 0000	RDSETUP, .-. .	/RD TEST AND SETUP ROUTINE		8
9	3674	075001* 7240	NL7777	/SET TEST PATTERN		9
10	3675	075002* 6707	RDTEST	/DO TEST IOT THAT SHOULD CLEAR THE AC		10
11	3676	075003* 7640	SZA CLA	/SKIP IF RD51 HARDWARE ACTUALLY PRESENT		11
12	3677	075004* 5600	JMP I RDSETUP	/ELSE JUST RETURN		12
13	3678	075005* 1256	TAD M12/(-12)	/GET OUTER TIME-OUT CONSTANT		13
14	3679	075006* 3052	DCA T8	/STASH THE OUTER COUNTER		14
15	3680	075007* 1254	RDWLP1, TAD M30/(-30)	/GET MIDDLE TIME-OUT CONSTANT		15
16	3681	075010* 3051	DCA T7	/STASH THE MIDDLE COUNTER		16
17	3682	075011* 6703	RDWLP2, RDSD	/DONE FLAG SET FROM SELF-TEST?		17
18	3683	075012* 7410	SKP	/SKIP IF NOT		18
19	3684	075013* 5223	JMP RDFIN	/JUMP IF SO		19
20	3685	075014* 2050	ISZ T6	/WAIT AWHILE?		20
21	3686	075015* 5211	JMP RDWLP2	/KEEP WAITING		21
22	3687	075016* 2051	ISZ T7	/WAITED ENOUGH?		22
23	3688	075017* 5211	JMP RDWLP2	/KEEP GOING		23
24	3689	075020* 2052	ISZ T8	/WAITED TOO LONG?		24
25	3690	075021* 5207	JMP RDWLP1	/NO, KEEP GOING		25
26	3691	075022* 5600	JMP I RDSETUP	/YES, JUST RETURN		26
27	3692					27
28	3693	075023* 6706	RDFIN, RDSE	/ANY ERRORS?		28
29	3694	075024* 7410	SKP	/SKIP IF NOT		29
30	3695	075025* 5600	JMP I RDSETUP	/JUST RETURN		30
31	3696	075026* 1152	TAD PCALLSUB/[CALLSUB+1-1]	/SETUP THE		31
32	3697	075027* 3010	DCA XR0	/STORAGE POINTER		32
33	3698	075030* 1255	TAD L32/(32)	/GET READ STARTUP SECTOR COMMAND		33
34	3699	075031* 6702	RDSC	/SEND THE COMMAND		34
35	3700	075032* 4270	DTRLUP, JMS RDWAIT	/WAIT FOR FLAGS		35
36	3701	075033* 5600	JMP I RDSETUP	/ERROR OR TIME-OUT, FORGET IT		36
37	3702	075034* 7410	SKP	/TRANSFER REQUEST UP		37
38	3703	075035* 5241	JMP SSDONE	/JUMP THERE IF DONE		38
39	3704	075036* 6704	RDTD	/TRANSFER DATA		39
40	3705	075037* 3410	DCA I XR0	/STORE THE WORD IN THE SUBROUTINE BODY		40
41	3706	075040* 5232	JMP DTRLUP	/KEEP GOING		41
42	3707					42
43	3708	075041* 1152	SSDONE, TAD PCALLSUB/[CALLSUB+1-1]	/SETUP THE POINTER		43
44	3709	075042* 3010	DCA XR0	/TO THE READ-IN DATA		44
45	3710	075043* 1260	TAD PTSTDATA/(TSTDATA-1)	/SETUP THE TEST DATA POINTER		45
46	3711	075044* 3011	DCA XR1	/STASH THE POINTER		46
47	3712	075045* 1411	TSTLUP, TAD I XR1	/GET AN INVERTED TEST VALUE		47
48	3713	075046* 7450	SNA	/SKIP IF NOT AT END OF LIST		48
49	3714	075047* 4552	JMS I PCALLSUB/[CALLSUB]	/CALL THE READ-IN CODE		49
50	3715	075050* 1410	TAD I XR0	/COMPARE TO READ-IN VALUE		50
51	3716	075051* 7640	SZA CLA	/SKIP IF IT MATCHES		51
52	3717	075052* 5600	JMP I RDSETUP	/RETURN IF NOT		52
53	3718	075053* 5245	JMP TSTLUP	/KEEP TESTING		53
54						54
55						55
56						56
57						57
58						58

1	3720	075054*	7750	M30,	-30	/MIDDLE TIME-OUT CONSTANT	1
2	3721	075055*	0032	L32,	32	/CONSTANT 0032	2
3	3722	075056*	7766	M12,	-12	/OUTER TIME-OUT CONSTANT	3
4	3723						4
5	3724	075057*	0006		6	/AN UNUSED CONSTANT 0006	5
6	3725						6
7	3726	075060*	5060	PTSTDAT,	TSTDATA-1	/POINTER TO TEST DATA TABLE (-1)	7
8	3727						8
9	3728			/	TEST CONTENTS OF CALLABLE SUBROUTINE. ALL VALUES ARE INVERTED. THE ACTUAL		9
10	3729			/	READ-IN CODE IS EXECUTABLE AS AND INSTRUCTIONS.		10
11	3730						11
12	3731	075061*	7777	TSTDATA,	-1		12
13	3732	075062*	7656		-"R!200		13
14	3733	075063*	7674		-"D!200		14
15	3734	075064*	7713		-"5!200		15
16	3735	075065*	7717		-"1!200		16
17	3736	075066*	7776		-2		17
18	3737	075067*	0000		0	/THIS ENDS THE LIST	18
19	3738						19
20	3739			/	RD51 WAIT ROUTINE. TAKES IMMEDIATE RETURN ON ERROR OR TIME-OUT; TAKES SKIP		20
21	3740			/	RETURN ON TRANSFER REQUEST FLAG; TAKES DOUBLE-SKIP RETURN ON DONE FLAG.		21
22	3741						22
23	3742	075070*	0000	RDWAIT,	...	/RD51 WAIT ROUTINE	23
24	3743	075071*	1132	TAD	DTIMOUT/[-DTIME]	/SETUP THE	24
25	3744	075072*	3051	DCA	T7	/TIME-OUT COUNTER	25
26	3745	075073*	3050	DCA	T6	/CLEAR INNER TIME-OUT COUNTER	26
27	3746	075074*	6701	RDWATLP,	RDSR	/DATA REQUEST FLAG UP?	27
28	3747	075075*	7410	SKP		/SKIP IF NOT	28
29	3748	075076*	5313	JMP	RDTRAN	/JUMP IF SO	29
30	3749	075077*	6703	RDSD		/DONE FLAG UP?	30
31	3750	075100*	7410	SKP		/SKIP IF NOT	31
32	3751	075101*	5307	JMP	RDDONE	/JUMP IF SO	32
33	3752	075102*	2050	ISZ	T6	/WAITED ENOUGH?	33
34	3753	075103*	5274	JMP	RDWATLP	/NO, KEEP GOING	34
35	3754	075104*	2051	ISZ	T7	/WAITING TOO LONG?	35
36	3755	075105*	5274	JMP	RDWATLP	/NO, KEEP GOING	36
37	3756	075106*	5670	JMP I	RDWAIT	/YES, TAKE IMMEDIATE RETURN	37
38	3757						38
39	3758	075107*	6706	RDDONE,	RDSE	/ANY ERRORS?	39
40	3759	075110*	7410	SKP		/SKIP IF NOT	40
41	3760	075111*	5670	JMP I	RDWAIT	/TAKE IMMEDIATE RETURN IF SO	41
42	3761	075112*	2270	ISZ	RDWAIT	/TAKE DOUBLE-SKIP RETURN ON DONE FLAG RAISE	42
43	3762	075113*	2270	RDTRAN,	ISZ	RDWAIT	43
44	3763	075114*	5670	JMP I	RDWAIT	/RETURN EITHER WAY	44
45							45
46							46
47							47
48							48
49							49
50							50
51							51
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1	3765	/	MPSCC REGISTER READ ROUTINE. CALLED WITH IN-LINE ARGUMENT OF EVEN/ODD FIELD				1
2	3766	/	IN BITS[6-8] AND SPECIFIED REGISTER IN BITS[9-11]. THE CONTENTS OF THE				2
3	3767	/	SPECIFIED REGISTER ARE RETURNED IN THE AC.				3
4	3768						4
5	3769	075115* 0000	COMREAD, .-. .		/MPSCC CHIP REGISTER READ ROUTINE		5
6	3770	075116* 1715	TAD I	COMREAD	/GET THE IN-LINE ARGUMENT		6
7	3771	075117* 2315	ISZ	COMREAD	/BUMP PAST IT		7
8	3772	075120* 3020	DCA	CTMP1	/SAVE IT		8
9	3773	075121* 1020	TAD	CTMP1	/GET IT BACK		9
10	3774	075122* 0366	AND	L70/(70)	/JUST FIELD BITS		10
11	3775	075123* 1122	TAD	ZCDF/[CDF]	/FORM CDF INSTRUCTION		11
12	3776	075124* 3325	DCA	.+1	/STORE IN-LINE		12
13	3777	075125* 7402	HLT+. - .		/WILL BE CDF XX INSTRUCTION		13
14	3778	075126* 1020	TAD	CTMP1	/GET ARGUMENT AGAIN		14
15	3779	075127* 0104	AND	Z7/[7]	/JUST REGISTER BITS		15
16	3780	075130* 1336	TAD	L4000/(4000)	/ADD ON READ ACCESS BIT		16
17	3781	075131* 6366	MPSCC		/SETUP TO READ THE SPECIFIED REGISTER		17
18	3782	075132* 7200	CLA		/CLEAN UP		18
19	3783	075133* 6366	MPSCC		/READ THE REGISTER		19
20	3784	075134* 6271	CDF	70	/BACK TO OUR FIELD		20
21	3785	075135* 5715	JMP I	COMREAD	/RETURN		21
22	3786						22
23	3787	075136* 4000	L4000, 4000		/CONSTANT 4000		23
24	3788						24
25	3789	/	MPSCC REGISTER LOAD ROUTINE. CALLED WITH IN-LINE ARGUMENTS OF EVEN/ODD FIELD				25
26	3790	/	IN BITS[6-8] AND SPECIFIED REGISTER IN BITS[9-11] OF THE FIRST ARGUMENT,				26
27	3791	/	FOLLOWED BY VALUE TO LOAD INTO THE SPECIFIED REGISTER IN THE SECOND ARGUMENT.				27
28	3792						28
29	3793	075137* 0000	COMLOAD, .-. .		/COMMUNICATIONS CHIP REGISTER LOAD ROUTINE		29
30	3794	075140* 7200	CLA		/CLEAN UP		30
31	3795	075141* 1737	TAD I	COMLOAD	/GET THE FIRST ARGUMENT		31
32	3796	075142* 2337	ISZ	COMLOAD	/BUMP PAST IT		32
33	3797	075143* 3020	DCA	CTMP1	/STASH IT		33
34	3798	075144* 1737	TAD I	COMLOAD	/GET THE SECOND ARGUMENT		34
35	3799	075145* 2337	ISZ	COMLOAD	/BUMP PAST IT		35
36	3800	075146* 3021	DCA	CTMP2	/STASH IT		36
37	3801	075147* 1020	TAD	CTMP1	/GET THE FIRST ARGUMENT		37
38	3802	075150* 0366	AND	L70/(70)	/JUST FIELD BITS		38
39	3803	075151* 1122	TAD	ZCDF/[CDF]	/FORM CDF INSTRUCTION		39
40	3804	075152* 3353	DCA	.+1	/STORE IN-LINE		40
41	3805	075153* 7402	HLT+. - .		/WILL BE CDF XX INSTRUCTION		41
42	3806	075154* 1020	TAD	CTMP1	/GET THE FIRST ARGUMENT AGAIN		42
43	3807	075155* 0104	AND	Z7/[7]	/JUST REGISTER BITS		43
44	3808	075156* 7440	SZA		/DON'T NEED TO POINT IF ACCESSING REGISTER 0		44
45	3809	075157* 6366	MPSCC		/POINT TO DESIRED REGISTER NEXT TIME		45
46	3810	075160* 7200	CLA		/CLEAN UP		46
47	3811	075161* 1021	TAD	CTMP2	/GET THE NEW REGISTER CONTENTS		47
48	3812	075162* 6366	MPSCC		/LOAD THE NEW VALUE		48
49	3813	075163* 7200	CLA		/CLEAN UP		49
50	3814	075164* 6271	CDF	70	/BACK TO OUR FIELD		50
51	3815	075165* 5737	JMP I	COMLOAD	/RETURN TO CALLER		51
52	3816						52
53	3817	075166* 0070	L70, 70		/CONSTANT 0070		53
54	3818						54
55	3819	075167* 0000	ZBLOCK 11		/EMPTY SPACE		55
56							56
57							57
58							58

1	3821	*5200	PAGE			1
2	3822					2
3	3823	/	RX50 READ COMMAND ROUTINE. THE COMMAND (TYPICALLY 0006 FOR A READ) IS PASSED			3
4	3824	/	IN THE AC. THE SECTOR IS IN TINC, AND THE TRACK IN T1. TAKE IMMEDIATE RETURN			4
5	3825	/	IF THE COMMAND SUCCEEDS; TAKE SKIP RETURN IF IT TIMES OUT.			5
6	3826					6
7	3827	075200*	0000 RXCOMD, .-. /RX50 READ COMMAND ROUTINE			7
8	3828	075201*	6751 LCD /LOAD PASSED COMMAND			8
9	3829	075202*	1041 TAD TINC /GET THE SECTOR			9
10	3830	075203*	6753 STR /TRANSFER FLAG UP?			10
11	3831	075204*	5203 JMP .-1 /NO, WAIT FOR IT			11
12	3832	075205*	6752 XDR /YES, SEND THE SECTOR			12
13	3833	075206*	7200 CLA /CLEAN UP			13
14	3834	075207*	1043 TAD T1 /GET THE TRACK			14
15	3835	075210*	6753 STR /TRANSFER FLAG UP?			15
16	3836	075211*	5210 JMP .-1 /NO, WAIT FOR IT			16
17	3837	075212*	6752 XDR /YES, SEND THE TRACK			17
18	3838	075213*	7200 CLA /CLEAN UP			18
19	3839	075214*	1132 TAD DTIMOUT/[-DTIME] /SETUP THE			19
20	3840	075215*	3050 DCA T6 /TIME-OUT COUNTER			20
21	3841	075216*	3051 DCA T7 /CLEAR INNER TIME-OUT COUNTER			21
22	3842	075217*	6755 DNWAIT, SDN /DONE FLAG UP?			22
23	3843	075220*	7410 SKP /SKIP IF NOT			23
24	3844	075221*	5600 JMP I RXCOMD /TAKE SUCCESSFUL RETURN IF SO			24
25	3845	075222*	2051 ISZ T7 /WAITING LONG ENOUGH?			25
26	3846	075223*	5217 JMP DNWAIT /NO, KEEP WAITING			26
27	3847	075224*	2050 ISZ T6 /WAITING TOO LONG?			27
28	3848	075225*	5217 JMP DNWAIT /NO, KEEP WAITING			28
29	3849	075226*	2200 ISZ RXCOMD /BUMP TO FAILURE RETURN			29
30	3850	075227*	5600 JMP I RXCOMD /TAKE SKIP RETURN TO CALLER			30
31						31
32						32
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1	3852	075230*	0000	REGINIT, .-. .		/REGISTER INITIALIZE ROUTINE	1
2	3853	075231*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	2
3	3854	075232*	0000	A^10+0		/0A	3
4	3855	075233*	0030	030		/CHANNEL RESET	4
5	3856	075234*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	5
6	3857	075235*	0010	B^10+0		/0B	6
7	3858	075236*	0030	030		/CHANNEL RESET	7
8	3859	075237*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	8
9	3860	075240*	0002	A^10+2		/2A	9
10	3861	075241*	0020	020		/LOW-ORDER BITS ARE AFFECTED, NO DMA	10
11	3862	075242*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	11
12	3863	075243*	0012	B^10+2		/2B	12
13	3864	075244*	0000	000		/INTERRUPT VECTOR IS ALL ZEROES	13
14	3865	075245*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	14
15	3866	075246*	0004	A^10+4		/4A	15
16	3867	075247*	0104	104		/16X CLOCK RATE, 1 STOP BIT ASYNCHRONOUS	16
17	3868	075250*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	17
18	3869	075251*	0001	A^10+1		/1A	18
19	3870	075252*	0022	022		/RECEIVE AND TRANSMIT INTERRUPT ENABLE	19
20	3871	075253*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	20
21	3872	075254*	0011	B^10+1		/1B	21
22	3873	075255*	0004	004		/CONDITION AFFECTS VECTOR, NO B INTERRUPTS	22
23	3874	075256*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	23
24	3875	075257*	0003	A^10+3		/3A	24
25	3876	075260*	0301	301		/8 BITS, ENABLE RECEIVER	25
26	3877	075261*	4542	JMS I	PCOMLOAD/[COMLOAD]	/LOAD REGISTER	26
27	3878	075262*	0005	A^10+5		/5A	27
28	3879	075263*	0150	150		/8 BITS, ENABLE TRANSMITTER	28
29	3880	075264*	5630	JMP I	REGINIT	/RETURN	29
30							30
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1	3882		/	*****			1
2	3883						2
3	3884	075265*	0000	PRINTP,	..	/SIX PARAMETERS PRINT ROUTINE	3
4	3885	075266*	6271	CDF	70	/RESET TO OUR FIELD	4
5	3886	075267*	3056	DCA	ROW	/CLEAR COLUMN	5
6	3887	075270*	3057	DCA	COLUMN	/CLEAR COLUMN	6
7	3888	075271*	1152	TAD	PCALLSUB/[CALLSUB]	/SETUP THE	7
8	3889	075272*	3010	DCA	XR0	/DATA POINTER	8
9	3890	075273*	1351	TAD	L7772/(-6)	/SETUP THE	9
10	3891	075274*	3011	DCA	XR1	/PARAMETER COUNT	10
11	3892	075275*	1355	TAD	PTM0/(TM0-1)	/SETUP THE	11
12	3893	075276*	3353	DCA	PADR	/PARAMETER ADDRESS	12
13	3894	075277*	1410	HDLOOP,	TAD I XR0	/GET A WORD	13
14	3895	075300*	1134	TAD	Z7776/[7776]	/COMPARE TO MESSAGE END	14
15	3896	075301*	7640	SZA	CLA	/SKIP IF IT MATCHES	15
16	3897	075302*	5277	JMP	HDLOOP	/ELSE GO BACK	16
17	3898	075303*	1410	PRLOOP,	TAD I XR0	/GET A CHARACTER	17
18	3899	075304*	7421	MQL		/LOAD IT FOR PRINTING	18
19	3900	075305*	7701	CLA	MQA	/GET IT BACK	19
20	3901	075306*	1130	TAD	Z7741/[-37]	/COMPARE TO DELIMITER	20
21	3902	075307*	7450	SNA		/SKIP IF OTHER	21
22	3903	075310*	5321	JMP	ATDEL	/JUMP IF A MATCH	22
23	3904	075311*	7001	IAC		/COMPARE TO ALTERNATE DELIMITER VALUE	23
24	3905	075312*	7450	SNA		/SKIP IF OTHER	24
25	3906	075313*	5321	JMP	ATDEL	/JUMP IF A MATCH	25
26	3907	075314*	1347	TAD	L4/(4)	/COMPARE TO END OF DATA VALUE	26
27	3908	075315*	7650	SNA	CLA	/SKIP IF OTHER	27
28	3909	075316*	5665	JMP I	PRINTP	/RETURN IF IT MATCHES	28
29	3910	075317*	4501	JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	29
30	3911	075320*	5303	JMP	PRLOOP	/KEEP GOING	30
31	3912						31
32	3913	075321*	4501	ATDEL,	JMS I PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	32
33	3914	075322*	1011	TAD	XR1	/GET PARAMETER COUNT	33
34	3915	075323*	7650	SNA	CLA	/SKIP IF NOT DONE	34
35	3916	075324*	5303	JMP	PRLOOP	/JUST PRINT THE FIELD	35
36	3917	075325*	2011	ISZ	XR1	/COUNT THIS ONE	36
37	3918	075326*	7000	NOP		/IN CASE IT SKIPS	37
38	3919	075327*	2353	ISZ	PADR	/BUMP TO NEXT ADDRESS	38
39	3920	075330*	1350	TAD	M4/(-4)	/SETUP THE	39
40	3921	075331*	3012	DCA	XR2	/DIGIT COUNTER	40
41	3922	075332*	1753	TAD I	PADR	/GET THE VALUE	41
42	3923	075333*	3354	DCA	DIGTMP	/STASH IT	42
43	3924	075334*	1354	DIGLUP,	TAD	DIGTMP	43
44	3925	075335*	7014	R3L		/MOVE UP	44
45	3926	075336*	3354	DCA	DIGTMP	/STORE IT BACK	45
46	3927	075337*	1354	TAD	DIGTMP	/GET IT AGAIN	46
47	3928	075340*	0104	AND	Z7/[7]	/JUST LATEST DIGIT	47
48	3929	075341*	1352	TAD	L60/("0&177)	/MAKE IT ASCII	48
49	3930	075342*	7421	MQL		/STASH IT	49
50	3931	075343*	4501	JMS I	PCHRPRT/[CHRPRT]	/PRINT THE CHARACTER	50
51	3932	075344*	2012	ISZ	XR2	/DONE ALL DIGITS YET?	51
52	3933	075345*	5334	JMP	DIGLUP	/NO, KEEP GOING	52
53	3934	075346*	5303	JMP	PRLOOP	/KEEP GOING	53
54							54
55							55
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1	3936	/	*****				1
2	3937						2
3	3938	075347*	0004	L4,	4	/CONSTANT 0004	3
4	3939	075350*	7774	M4,	-4	/CONSTANT 7774	4
5	3940	075351*	7772	L7772,	-6	/CONSTANT 7772	5
6	3941	075352*	0060	L60,	"0&177	/ASCII NUMERIC CONSTANT 0060	6
7	3942	075353*	0000	PADR,	.-.	/PARAMETER ADDRESS TEMPORARY	7
8	3943	075354*	0000	DIGTMP,	.-.	/OUTPUT VALUE STORED HERE	8
9	3944	075355*	0027	PTM0,	TM0-1	/POINTER TO DATA BLOCK	9
10	3945						10
11	3946	/		RD51 TEST AND WAIT ROUTINE.	IF RD51 IS PRESENT, THEN WAIT AWHILE.		11
12	3947						12
13	3948	075356*	0000	RDTWAIT,	.-.	/RD51 TEST AND WAIT ROUTINE	13
14	3949	075357*	7240	NL7777		/SET TEST PATTERN	14
15	3950	075360*	6707	RDTEST		/AC WILL CLEAR IF RD51 IS PRESENT	15
16	3951	075361*	7640	SZA CLA		/SKIP IF RD51 INSTALLED	16
17	3952	075362*	5756	JMP I	RDTWAIT	/RETURN IF NOT	17
18	3953	075363*	1373	TAD	M2260/(-2260)	/SETUP THE	18
19	3954	075364*	3050	DCA	T6	/OUTER LOOP COUNTER	19
20	3955	075365*	3051	DCA	T7	/CLEAR THE INNER LOOP COUNTER	20
21	3956	075366*	2051	WAITLUP,	ISZ T7	/WAIT FOR AWHILE	21
22	3957	075367*	5366	JMP	WAITLUP	/KEEP WAITING	22
23	3958	075370*	2050	ISZ	T6	/WAITED ENOUGH	23
24	3959	075371*	5366	JMP	WAITLUP	/NO, KEEP GOING	24
25	3960	075372*	5756	JMP I	RDTWAIT	/YES, RETURN TO CALLER	25
26	3961						26
27	3962	075373*	5520	M2260,	-2260	/CONSTANT 5520	27
28	3963						28
29	3964	075374*	0000	ZBLOCK	4	/EMPTY SPACE	29
30							30
31							31
32							32
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1	3966	*5400	PAGE			1
2	3967					2
3	3968	/	THIS ROUTINE CHECKS FOR BYTES LOADED AS 0377 TO BE REPLACED WITH 4000. THE			3
4	3969	/	AFFECTED DATA CONSISTS OF THE FRESHLY READ-IN (IN BYTE MODE) CONTENTS OF TRACK			4
5	3970	/	1, SECTOR 3 PAST THE VALIDATED HEADER BYTES, LOCATIONS 15360-15777 ONLY.			5
6	3971					6
7	3972	075400*	0000 RUBCHK, .-. /377 CODE REPLACE ROUTINE			7
8	3973	075401*	1222 TAD RUBPTR/(T0S3BUFFER+10) /SETUP THE LIST POINTER PAST THE HEADER			8
9	3974	075402*	3223 DCA VPTR /STASH THE POINTER			9
10	3975	075403*	6211 VLUP, CDF 10 /GOTO DATA FIELD			10
11	3976	075404*	1623 TAD I VPTR /GET A WORD			11
12	3977	075405*	7001 IAC /ADD ONE			12
13	3978	075406*	0115 AND Z377/[377] /LOW-ORDER BITS ONLY			13
14	3979	075407*	7640 SZA CLA /SKIP IF THEY WERE ALL SET			14
15	3980	075410*	5213 JMP VAROUND /ELSE RETAIN THIS WORD			15
16	3981	075411*	7330 NL4000 /GET REPLACEMENT VALUE			16
17	3982	075412*	3623 DCA I VPTR /STORE OVER LIST ELEMENT			17
18	3983	075413*	6271 VAROUND, CDF 70 /BACK TO OUR FIELD			18
19	3984	075414*	2223 ISZ VPTR /BUMP TO NEXT			19
20	3985	075415*	7332 NL2000 /GET ADDRESS MASK			20
21	3986	075416*	0223 AND VPTR /JUST QUADRANT BIT			21
22	3987	075417*	7650 SNA CLA /SKIP IF JUST FINISHED			22
23	3988	075420*	5203 JMP VLUP /JUMP IF NOT			23
24	3989	075421*	5600 JMP I RUBCHK /RETURN TO CALLER			24
25	3990					25
26	3991	075422*	5360 RUBPTR, T0S3BUFFER+10 /POINTER TO DATA PAST THE HEADER			26
27	3992	075423*	0000 VPTR, .-. /DATA POINTER			27
28						28
29						29
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1	3994	/	WHILE THERE IS NO MORE ACTUAL CODE, THE FOLLOWING LINES ARE PRESENT TO	1
2	3995	/	CALCULATE THE MOVE LENGTH FOR THE CP-INTERRUPT HANDLER USED DURING THE	2
3	3996	/	LOOPBACK TEST IN MAIN MEMORY. THIS CALCULATION IS PROBABLY INCORRECT AND	3
4	3997	/	MOVES TOO MUCH CODE. THE ONLY POSSIBLE PROBLEM IS INTERACTION WITH APT	4
5	3998	/	DOWNLOADED PROGRAM CONVENTIONS.	5
6	3999			6
7	4000	075424* 0000	ZBLOCK .&7600+200- . /EMPTY SPACE IN THE REST OF THE PAGE	7
8	4001			8
9	4002	/	THIS IS WHERE THE STARTUP SECTOR CODE IS READ IN AS A CALLED SUBROUTINE.	9
10	4003			10
11	4004	075600* 0000	CALLSUB, . . . /TEST SUBROUTINE LOCATION	11
12	4005			12
13	4006	5601	CPEND2= . /MOVED CP INTERRUPT HANDLER CODE ENDS HERE?	13
14	4007			14
15	4008	075601* 0000	TSTBUFF, ZBLOCK 4 /KEYBOARD TEST BUFFER	15
16	4009			16
17	4010	5605	TSTBEND= . /END OF KEYBOARD TEST BUFFER	17
18	4011			18
19	4012	*6000	RELOC 6000 /EMPTY SPACE	19
20	4013			20
21	4014	6000	APTSTRT= . /APT ROUTINE STARTS HERE	21
22	4015			22
23	4016	*6400	RELOC 6400 /EMPTY SPACE	23
24	4017			24
25	4018	6400	WRIBUFF= . /WRITE BUFFER FOR SETUP	25
26	4019			26
27	4020	*6500	RELOC 6500 /EMPTY SPACE	27
28	4021			28
29	4022	6500	APTROUT= . /APT ROUTINE LOADS HERE	29
30	4023			30
31	4024	*6520	RELOC .+20 /EMPTY SPACE	31
32	4025			32
33	4026	6520	APTR02= . /SECONDARY APT ADDRESS HERE	33
34	4027			34
35	4028	*5505	RELOC /UN-FOOL THE ASSEMBLER	35
36	4029			36
37	4030	075505 0000	ZBLOCK ROMGO- . /EMPTY SPACE	37
38	4031			38
39	4032		\$ /THAT'S ALL FOLK!	39
40				40
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1	USER SYMBOL TABLE LISTING														2		
2															3		
3	A	=0000	CLNOP1=6132	DISPAD	3515	INOP2 =6307	LCD	=6751	L72	0250	NEXTRE	4124	PDOCG	1310	4		
4	ADDFIX	2464	CLNOP2=6133	DMLOGO	2403	INTADR	0000	LCHAR	0055	L7603	4442	NL0001=7201	PEND	4432	5		
5	AFREAD	3406	CLNOP3=6134	DNWAIT	5217	INTHND	0250	LDDONE	4552	L77	2764	NL0002=7326	PENDUP	1243	6		
6	APTFL	=6120	CLNOP4=6136	DOCG	3152	INTR	=6756	LDTRAN	4546	L7772	5351	NL0003=7325	PEX	=6004	7		
7	APTIE	=6125	CLNOP5=6137	DOCLK	0277	INTXIT	0275	LFREAD	4572	L7773	3661	NL0004=7307	PFBLIN	1233	8		
8	APTR0U=6500		CLRFIN	1550	DOCOMR	3647	INT1	0001	LISTLU	3126	L7774	0633	NL0006=7327	PFERR	4577	9	
9	APTR02=6520		CLRIT	3213	DOIT	4520	INT2	0002	LISTOK	3137	MCHKLP	1430	NL0010=7215	PFREAD	3142	10	
10	APTSKP=6121		CLRLUP	3201	DOKEY	4446	IRB	=6306	LOADER=	0015	MCLUP	0746	NL002A=7305	PGETPA	3566	11	
11	APTSTR=6000		CLR2	2765	DONEOK	1031	IRS	=6304	LOADIT	0020	MEMCOM	1421	NL0100=7203	PG0	=6003	12	
12	APUCON=0014		CLR23	1513	DONEUP	1067	ISF	=6301	LOADLP	0022	MEMLOA	0311	NL2000=7332	PHLTPC	4110	13	
13	ARXWAI	3700	CLSK	=6131	DONUP	1132	ISOURS	4350	LOC36	4321	MEMTES	0251	NL3777=7350	PINTHN	0244	14	
14	ASLCHE	4571	CMDOK	1104	DONWAI	1017	IS37	4340	LODLUP	4542	MEMTST	7604	NL4000=7330	PKBDIN	0151	15	
15	ATABLE	7745	CMNDOK	1045	DOWNLO	0640	JMPC	=5600	LOOKLP	4414	METREN	3471	NL5777=7352	PKBIPC	4106	16	
16	ATDEL	5321	CMSW	0755	DPTR	1002	JMPIM1	0121	LOW1	2531	METRON	4600	NL6000=7333	PKBOIP	4107	17	
17	ATDELI	4047	CM5	4573	DTIME	=0400	JMPIPM	0756	LOW2	2532	MFL	=6360	NL7775=7346	PKBTEN	0552	18	
18	ATEND	4510	CNTHI	0023	DTIMOU	0132	JMPI2	0245	LPROGM=	7721	MIE	=6365	NL7776=7344	PLBDON	4020	19	
19	ATTRVA	4254	CNTLO	0022	DTRLUP	5032	JUSTEX	4066	LRXWAI	4576	MLC	=6362	NL7777=7240	PL0C36	3410	20	
20	AVIDCL	4445	CODEST	0200	DUMBCF=	6312	JUSTPO	4252	LSCREG=	6122	ML0DLP	0254	NOCRET	4516	PLUP	4422	21
21	A14	4614	CODEXC=	0100	DUMBFL=	6310	J00	2400	LUPDON	0554	MOREPR	4126	NODATA	3200	PMEMCO	0150	22
22	A15	4615	COLUMN	0057	DUMBIE=	6315	J01	2401	LUPIN	0517	MOVCNT	0017	NOERRO	2515	PMTSTL	0464	23
23	A16	4616	COMBAD	4711	DUMBSF=	6311	KAGN	3551	LUPLUP	0504	MOVLEF	4275	NOTAPT	2522	PNEXT	4500	24
24	A7740	0636	COMCHK	4715	DWNAGN	0737	KBDIN	3672	LUPOUT	0534	MOVRIT	4311	NOTDON	1024	PNODAT	3151	25
25	B	=0001	COMDON	1007	DWNENT	0650	KBICLR=	6112	LUPTIM	0553	MPRESE=	6367	NOTOUR	4351	PORTCO=	0036	26
26	BACKEN	0342	COMEXI	4661	EDISPL	2372	KBICON=	0011	LVIDCL	4605	MPSCC	=6366	NPLUP	4113	PORTIN=	0030	27
27	BAUD17	4651	COMFIN	1360	EMPTLU	1106	KBIFL	=6110	L100	4772	MRS	=6364	NPRT	4103	PORTOU=	0031	28
28	BIGFLP	2047	COMLOA	5137	ENDUP	2444	KBIE	=6115	L110	3255	MSB	=6363	NTMP	4151	PPARMO	4575	29
29	BL40	1570	COMREA	5115	ESTATU	0054	KBIPC	0213	L116	4527	MSF	=6361	NXTLIN	3227	PPPTAB	0102	30
30	BMPRET	4515	COMSRC	3474	FBLINK	1554	KBIRB	=6116	L16	0520	MTM1	0032	NXTPRM	4025	PPRINT	3567	31
31	BOLDLP	4140	COMTAB	3736	FERR	1343	KBIRS	=6114	L17	0302	MTM2	0033	OLS	=6316	PPRMPR	3407	32
32	BOOTIT	4401	COMTES	4652	FHERE	1474	KBISF	=6111	L2	4774	MTM5	0036	ONOP1	=6313	PPTABL	1400	33
33	BTABLE	3664	COMTM1	0031	FILLIT	0263	KBOCON=	0005	L200	2574	MTM6	0037	ONOP2	=6317	PRDSET	1235	34
34	BUFLUP	1523	COMWAI	4702	FILLUP	1046	KBOFL	=6050	L202	2442	MTM7	0040	OPC	=6314	PRDTWA	1006	35
35	BUMPIT	4065	COUNTD	0320	FINDIT	3504	KBOIE	=6055	L2030	4201	MTSTLP	0323	OUTLUP	4070	PREGIN	4667	36
36	CALLSU	5600	COUNTU	0334	FLPBAD	1236	KBOIPC	0211	L23	2440	MVCNT	4647	OUTMOR	4763	PREGLI	1447	37
37	CCHKMO	4750	CPD	=6266	FNDDEL	4136	KBOLS	=6056	L247	2441	MVLUP	4636	PADR	5353	PRGDON	0214	38
38	CCNT	0060	CPEND	=4111	FNDIT	4470	KBOPC	=6054	L2525	7717	M100	1163	PAPTRO	0124	PRGLUP	0206	39
39	CDFINS	2024	CPEND2=	5601	FNEXT	1335	KBOSF	=6051	L300	2763	M1000	1004	PAPTST	0637	PRICLR=	6322	40
40	CDFSET	1540	CPIADD	0134	FNXTSC	1325	KBTEND	0630	L31	4202	M12	5056	PAPT2	2535	PRICON=	0032	41
41	CDF20	2045	CPIHND	2536	FRDLUP	1363	KBTIMO	0620	L32	5055	M120	4574	PARMOV	4617	PRIFL	=6320	42
42	CDWNTS	0344	CPINT	4000	FRDOK	1346	KEY	=0003	L320	0635	M16	3660	PASSEN	0377	PRIIE	=6325	43
43	CGDONE	3146	CPLADR	0140	FREAD	1311	KEYLUP	0601	L342	0634	M175	3564	PATTER	0006	PRIN01=	6323	44
44	CGINDX	3257	CPL0AD=	3000	GCF	=6256	KINOP1=	6113	L3705	2761	M2260	5373	PBAUD	0655	PRIN02=	6327	45
45	CGLDLP	3223	CPMCNT	0141	GETPAR	3300	KINOP2=	6117	L3707	2762	M30	5054	PBTABL	3663	PRINTP	5265	46
46	CGLOAD=	6123	CPMTST	0347	GOTKEY	0611	KONOP1=	6052	L375	0550	M31	4203	PCALLS	0152	PRIRB	=6326	47
47	CGPTR	3277	CPMVLP	0444	HALTES	2565	KONOP2=	6053	L377	0246	M377	4650	PCGIND	3256	PRIRS	=6324	48
48	CHARAD	4255	CREADO	4737	HDLOOP	5277	KONOP3=	6057	L4	5347	M4	5350	PCHRPR	0101	PRISKP=	6321	49
49	CHARCD	4247	CSTATU	0005	HEDLUP	4021	LBADR	0463	L40	1175	M40	4256	PCLIST	4152	PRLOOP	5303	50
50	CHRPRT	4204	CTMP1	0020	HIGHRO	4200	LBDONE	0465	L4000	5136	M5	1003	PCLR2	1552	PRMLUP	4037	51
51	CIDF	3405	CTMP2	0021	HIGH1	2533	LBLADR	0136	L4001	0015	M52	0546	PCLR23	0144	PRMPRT	4007	52
52	CINTCL	4744	CUPDAT	4353	HIGH2	2534	LBLOAD=	2600	L45	3565	M6	4773	PCOMLO	0142	PRMTM1	4101	53
53	CLFL	=6130	C23	4444	HLTPC	0203	LBMCNT	0137	L460	2443	M7	4006	PCOMRE	0143	PRMTM2	4102	54
54	CLIE	=6135	DCNT	1001	ICF	=6302	ICF	=6302	L5	3662	NEGTST	0366	PCOMTA	3516	PROCON=	0033	55
55	CLIST	3365	DIGLUP	5334	IFL	=6300	LBTEND=	0347	L5252	7720	NEXTCH	3470	PCOUNT	0547	PROFL	=6330	56
56	CLKCON=	0013	DIGTMP	5354	IIE	=6305	LBTEST	0200	L60	5352	NEXTOU	0314	PCPIHN	0757	PROIE	=6335	57
57	CLKTIC	0004	DINDEX	1000	INOP1	=6303	LB20	0247	L70	5166	NEXTPR	3465	PCUPDA	4400	PROINT	0304	58

1						1
2						2
3	PROLS =6336	RDSE =6706	SELECT 3600	TSTKY0 4056	VPTR 5423	3
4	PROMST 0135	RDSETU 5000	SELSHF 3657	TSTLUP 5045	VREAD =6127	4
5	PRON01=6332	RDSR =6701	SELXIT 3632	TSTTRA 4034	VSF =6061	5
6	PRON02=6337	RDTD =6704	SER =6754	TTNOP1=6073	VWATLP 0224	6
7	PROPC =6334	RDTEST=6707	SETODD 1546	TTNOP2=6074	WAITLU 5366	7
8	PROSKP=6331	RDTRAN 5113	SETOK 3454	TTNOP3=6076	WATLUP 7605	8
9	PROWAD 1450	RDTWAI 5356	SETVLU 3445	TTNOP4=6077	WRIBUF=6400	9
10	PROWLO 0103	RDWAIT 5070	SET2 2747	TTY =0004	WSR =6246	10
11	PROWTA 1446	RDWATL 5074	SHFTLU 3613	TTYCLR=6072	XDR =6752	11
12	PRQEXI 4100	RDWE =6705	SLCHEC 4153	TTYCON=0007	XR0 0010	12
13	PRQ3 =6236	RDWLP1 5007	SLGOOD 4173	TTYFL =6070	XR1 0011	13
14	PRS =6000	RDWLP2 5011	SLNEXT 4162	TTYIE =6075	XR2 0012	14
15	PRSB =6333	REGDON 1511	SLSHAG 1224	TTYPC 0207	XR3 0013	15
16	PRTIT 4051	REGINI 5230	SLSHBA 1240	TTYSF =6071	XR4 0014	16
17	PRTRAT 4064	REGLIS 1664	SLTDAT 3732	T0 0042	XR5 0015	17
18	PRUBCH 3145	REGLUP 1477	SLUBAD 4566	T0S1DA 3755	XR6 0016	18
19	PRXCOM 0145	RETURN 4606	SLUCDF 4165	T0S3BU=5350	XR7 0017	19
20	PRXOK 4443	REVIT 4062	SPACSW 2046	T0S3DA 3775	ZCDF 0122	20
21	PRXRDY 1234	RIMADR 0000	SPD =6276	T0S3RD 3111	ZERLUP 1525	21
22	PRXWAI 1242	RIMEND 0705	SP1SAV 0034	T0S4DA 0760	ZERNXT 1516	22
23	PSET2 2530	RIMLUP 0662	SP2SAV 0035	T0S7DA 3764	Z0301 0114	23
24	PSLCHE 1306	RIMST 0734	SRCHLU 4327	T1 0043	Z10 0105	24
25	PSLTDA 4175	ROMADR=0000	SSADR 2032	T2 0044	Z102 0111	25
26	PSSTRI 0146	ROMGO 7600	SSDONE 5041	T3 0045	Z12 0106	26
27	PSTATU 0061	ROMSP 7776	SSLUP 2007	T4 0046	Z17 0107	27
28	PTABLE=1401	ROMST 7777	SSTRIN 2000	T5 0047	Z20 0110	28
29	PTIMOU 1553	ROW 0056	STR =6753	T6 0050	Z200 0112	29
30	PTMO 5355	ROWADR=3660	SUBAC 3514	T7 0051	Z205 0153	30
31	PTPATT 1005	ROWLOA 1451	SUBCDF 3512	T8 0052	Z237 0113	31
32	PTSTBE 0632	ROWLUP 1461	SUBLUP 2457	USEDAT 3205	Z2525 0117	32
33	PTSTBU 0551	ROWTAB 1747	SUBR 3412	VAGAIN 0223	Z377 0115	33
34	PTSTDA 5060	RSP1 =6207	SUB176 0176	VALAGN 1624	Z3777 0120	34
35	PTTYPC 4105	RSP2 =6227	TEMP1 0024	VALALT 1657	Z400 0116	35
36	PT0S1D 3411	RTRK1 1211	TEMP2 0025	VALID8 1600	Z7 0104	36
37	PT0S3B 3144	RUBCHK 5400	TEMP3 0026	VAROUN 5413	Z7400 0131	37
38	PT0S3D 3143	RUBPTR 5422	TESTCO 4670	VCL =6062	Z7741 0130	38
39	PT0S3R 1307	RXCMD 0053	TESTCP 4021	VDUMM1=6064	Z7760 0127	39
40	PT0S4D 3570	RXCOMD 5200	TESTHL 4070	VDUMM2=6066	Z7766 0126	40
41	PT0S7D 3147	RXCON =0075	TIMCHK 0321	VDUMM3=6067	Z7775 0133	41
42	PUSEDA 3150	RXERRO 1176	TIMEOU 0003	VERHLT 3520	Z7776 0134	42
43	PVALID 1232	RXINIT=6757	TIMLUP 1170	VERLUP 3540		43
44	PVIDCL 3517	RXLOAD 1244	TIMOUT 1164	VERMAT 3547		44
45	PVIDPC 4104	RXOK 1201	TINC 0041	VERPOK 3562		45
46	PWRIBU 0123	RXRDY 3701	TM0 0030	VFL =6060		46
47	P1LUP 4455	RXSKIP 1161	TPATTR 0771	VIDCLR 3715		47
48	P2LUP 4457	RXSKP2 1160	TRNCDF 1361	VIDCON=6126		48
49	P24 3415	RXWAIT 1140	TRNSUP 1052	VIDDEV=0012		49
50	P25 3421	RXWATI 1154	TRYDON 1367	VIDINT=0006		50
51	P26 3417	RXWATL 1144	TRYPRA 3634	VIDPC 0205		51
52	RCPLUP 0725	R3L =7014	TSTATU 0007	VIDWAI 3724		52
53	RDDONE 5107	SCRCHK 4266	TSTBEN=5605	VIE =6065		53
54	RDFIN 5023	SCREGU 4273	TSTBUF 5601	VLOAD =6124		54
55	RDNOP =6700	SCRPOS 4257	TSTDAT 5061	VLUP 5403		55
56	RDSC =6702	SDN =6755	TSTDLU 3167	VNOP1 =6063		56
57	RDSO =6703	SEL =6750	TSTKYI 4045	VPAST 0232		57
58						58

1		1
2		2
3	NO ERRORS DETECTED	3
4		4
5	NO LINKS GENERATED	5
6		6
7	8K MEMORY UTILIZED	7
8		8
9	3 FILES CREATED	9
10		10
11	700 SYMBOLS	11
12		12
13		13
14		14
15		15
16		16
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18		18
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